



D4.2: Second wave of DEDALUS Digital Twins and data-driven services for residential DR

D4.2: Second wave' of DEDALUS Digital Twins and data-driven services for residential DR.

WP4



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List of Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BEMS	Building Energy Management System
BIM	Building Information Modelling
CHI	Calinski-Harabasz Index
CO	Combinatorial Optimization
CSV	Comma-Separated Values
DAE	Denoising Autoencoder
DBI	Davies-Bouldin Index
DDTW	Derivative Dynamic Time Warping
DH	District Heating
DR	Demand Response
DTW	Dynamic Time Warping
DRAG	DR Action Generation
DSO	Distribution System Operator
DTs	Digital Twins
ED	Euclidean Distance
EDA	Energy Data Access
ELD	Electricity Load Disaggregation
EV	Electric Vehicle
EVCS	Electric vehicle charging stations
FHMM	Factorial Hidden Markov Model
FR	Functional Requirements
GUI	Graphical User Interface
HAC	Hierarchical Agglomerative clustering
HHO	Harris Hawks Optimisation



HLUC	High Level Use Case
HVAC	Heating, Ventilation, and Air Conditioning
IEQ	Indoor Environmental Quality
IET	Institution of Engineering and Technology
IoT	Internet of Things
ISP	Integrated Scheduling Programming
KPI	Key Performance Indicators
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MIP	Mixed Integer Program
ML	Machine Learning
MLP	Multi-Layer Perceptron
MQTT	Message Queuing Telemetry Transport
NILM	Non-Intrusive Load Monitoring
NN	Neural Network
P2P	Peer-to-Peer
PM	Particulate Matter
PMV	Predicted Mean Vote
PV	Photovoltaic
RC	Resistance-Capacitance
REDD	Reference Energy Disaggregation Data
RES	Renewable Energy Share
REST	Representational State Transfer
RH	Relative Humidity
SIL	Silhouette score
sPMV	Simplified Predicted Mean Vote
SVM	Support Vector Machine
TC	TotalConsumption
TE	Technological Enabler



ToM	Time-of-Maximum Analysis
TRL	Technology Readiness Level
TSO	Transmission System Operator
TSV	Thermal Sensation Vote
UI	User Interface
USG	Universal Secure Gateway
WP	Work Package



Executive summary

This document outlines the second wave of advancements in Digital Twins (DTs) and data-driven services for residential Demand Response (DR) within the Horizon Europe-funded DEDALUS project. Deliverable D4.2 represents the culmination of the second technological development cycle (MS7) under Work Package 4 (WP4), which focuses on the creation and deployment of value-added energy and non-energy services for energy-efficient residential buildings. It builds upon the foundation laid in D4.1, integrating lessons learned to refine and enhance the functionalities of Digital Twins and associated services. The progress made between Deliverables 4.1 and 4.2 marks a significant milestone in the DEDALUS project, reflecting the collective effort to bridge conceptual frameworks with practical implementation across the seven pilots.

Significant progress was made in Task 4.1: Digital Twins for optimal assets vs. consumers flexibility planning. Notably, for Pilots 2 and 4, the project transitioned from a theoretical definition of interfaces and basic requirements to the development of an initial version of the digital twin. This first iteration provides a simplified representation of data and service outputs, marking a crucial step forward. For Pilot 3, a detailed mock-up of the digital twin interface was developed, establishing the foundation for significant advancements in the next phase. Substantial progress was also achieved in building simulation models designed for Building Managers and defining their integration into the digital twin. Additionally, for Pilot 2, a navigable 3D model of the apartments was created. The source code was made available in the project repository.

The Apartment Building Digital Twin for blockchain integration has entered the development stage. It will be integrated with the DLT/Blockchain/Smart Contracts solution (TE-5) to facilitate the simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts. The DT has been refined in terms of components and models, and two algorithms have been integrated: an Electricity Load Disaggregation Algorithm and a DR Action Generation algorithm. Simultaneously, the mock-ups for the DT dashboards have been designed. In the next iteration of the deliverable, the mock-ups will be implemented, and the optimization algorithms will be finalized.

Significant progress was also made in Task 4.2: Optimal trade-off among comfort, energy, and building performance. Efforts were primarily focused on defining and implementing the architecture of the comfort flexibility model. For the Italian and Danish pilots, data collection and processing from installed sensors were initiated, including the calculation of results such as thermal comfort. Specific questionnaires on thermal sensation votes were distributed to tenants in the Danish and Italian pilots, with particular emphasis on simplifying the questionnaire for the Italian pilot.

In Task 4.3, Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers, eight different services have been defined, developed by the involved partners, and are soon to be applied in the respective pilots. Significant progress has already been made in all services. DOMX prepares a service for nudging the end-users of the Greek pilot towards improving their energy behaviour through specific simulation and clustering algorithms. NTUA employs clustering techniques and develops energy



disaggregation mechanisms using data from HERON's energy meters. HERON develops a tariff system to encourage energy-efficient operation and a multi-value DR approach for household energy savings. ENG integrates multi-value DR services into the Digital Twin in Task 4.1. OURPOWER, in collaboration with BLUEPRINT, enhances a mobile application for user engagement and optimized energy consumption.

In Task 4.4, Local flexibility management for energy communities and self-consumption/cost optimization, the project transitioned from foundational conceptualization to practical implementation and integration of key components. This included advancements in flexibility management, optimization algorithms, secure data transactions, and enhanced system interfaces across the Austrian and Spanish pilots. These developments reflect a robust evolution toward operational readiness.

In Task 4.5: Optimal Aggregation of flexible energy resources for heat vs. electrical flexibility valorisation, significant progress has been made in scoping, developing, and implementing the required functionality. The platform general framework has been extended from the present setup focusing on the building to interacting with tenants in apartments. Backend and GUI functionality have been developed to allow tenants to see their own apartment data and provide input on set points and flexibility preferences. Progress has been made on developing and implementing a controller taking feedback from smart thermostats.

The deliverable D4.2 marks a step forward in realizing the vision of the DEDALUS project. Many significant advancements were achieved, showcasing progress in refining Digital Twin technologies, expanding their applicability across diverse pilots, and integrating innovative tools to enhance energy management and consumer participation. By combining technological innovation with user-centred design, the project continues to demonstrate the transformative potential of Digital Twins in driving sustainable, participatory residential Demand Response across Europe.

DEDALUS will continue to build on these achievements by refining its technologies and expanding their applicability. These initiatives aim to further demonstrate the value of Digital Twins and data-driven services in fostering sustainability.



1. Introduction

1.1. Purpose and audience of the document

The main purpose of this document is to provide information about the second wave of the DEDALUS Digital Twins and data-driven services for residential DR, as main outcome of development activity of WP4 Value added energy and non-energy data-driven services for DR in energy-efficient residential buildings.

Moreover, this report provides relevant information with regards of the functional and technical specification of each service and details about the architectural aspects and the designed algorithms.

1.2. Relation to other activities

This deliverable builds on the functional specifications formally established in WP2. Use cases and functional requirements outlined in D2.1 [1] and subsequently updated in D2.3 [2] served as the foundation to define the DTs and data-driven services from a technical perspective. This specification has guided the development of the first software release. Moreover, WP4 has a strong relationship with D3.1 [3] and D3.2 [4] focus on the development of DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR. These deliverables provide a foundation for the interoperability, data governance, and decentralization features.

Notably, this deliverable is directly linked to D4.1 [5], which represents the initial wave of Digital Twins development in the DEDALUS project. D4.1 detailed the early-stage creation and implementation of Digital Twins and data-driven services for residential DR. The outcomes of this deliverable will significantly support WP5 activities.

1.3. Structure of the document

This document maintains the structure of D4.1 and is organized into the following sections:

- Chapter 2 provides information about the development activities under the scope of Task 4.1 Digital Twins for optimal assets vs consumers flexibility planning. It is mainly related to TE-8 Digital Twins for consumer-aware optimal flexibility planning and to the Apartment building Digital Twin for blockchain integration that is delivered as part of TE-5 blockchain and smart contracts integrated DT
- Chapter 3 provides information about software expected to be developed as outcome of Task 4.2. This development is linked to the TE-9 Comfort-based flexibility model
- Chapter 4 gives a detailed description of software delivered as outcome of Task 4.3 corresponding to several services of optimal DR strategies.
- Chapter 5 is about the local flexibility management for energy communities and self-consumption/cost optimisation service as outcome of Task 4.4. The task is involved in the development of the TE-12 "Peer-to-Peer (P2P) digital platform for trading pre-aggregated building flexibility" and TE-13 "District-level DR for self-consumption and cost optimization".



- Chapter 6 provide information about the software expected to be developed as an outcome of Task 4.5 and corresponding to the TE-14 Aggregation of flexible energy resources for district heating management.
- Chapter 7 serves as the conclusion, where the project progress is highlighted.

As outlined above, each section corresponds to a specific task defined in the grant agreement. The Figure 1 highlights the pilots involved in each task, providing a clearer understanding of their roles and contributions within the project.

Table 1: Task and pilot involvement

Task	Pilot 1 Austria	Pilot 2 Denmark	Pilot 3 Greece	Pilot 4 Italy	Pilot 5 Spain	Pilot 6 Ireland	Pilot 7 Romania
Task4.1							
Task4.2							
Task4.3							
Task4.4							
Task4.5							



2. Digital Twins for optimal assets vs consumers flexibility planning

This section provides information about the development activities under the scope of Task 4.1 Digital Twins for optimal assets vs consumers flexibility planning. The Technological Enablers (TE) that are involved are:

- TE-8: Digital Twins for consumer-aware optimal flexibility planning.
- TE-5: Blockchain and smart contracts integrated DT.

TE-8 focuses in developing cross-commodity DTs centred on DER/DR flexibility. The approach involves integrating multiple methodologies to create a robust and interpretable framework. This objective is pursued employing clustering algorithms to analyse residential user loads, incorporating thermal models leveraging real-world assets connected through building energy management systems, and integrating Human-In-The-Loop models assessing and modelling community-level flexibility.

Focusing on the second one, the development is related to the Apartment building Digital Twin for blockchain integration. Both TEs are extended in the context of the project according to the functional requirements introduced in D2.1: DEDALUS gap analysis and use case capitalization and further detailed in D2.3: High-level requirements for the DEDALUS social, technological and business framework.

2.1. Digital Twins for consumer-aware optimal flexibility planning

As powerful tool designed to optimise the management of various assets, DTs are commonly known for facilitating the oversight of buildings, district heating networks, dispatching grids, and similar infrastructures. In literature [6], the term of DT is used to identify the virtual representation of real system (e.g., building, district heating) and it is characterized by a two-way interaction (manual or automatic) between the physical and digital entity. The DT is fed by historical and real-time data and any changes in the real system is mirrored in the DT and vice versa. Moreover, the DT can offer a simulation environment to support the decision-making process enhanced by machine learning and optimisation techniques. The possibility of controlling the real system through the DT depends on the level of maturity of the DT itself spanning from 0 to 5 according to the level of integration between the physical system and its virtual representation. According to the report presented by the IET [7] and shown in Figure 2, the level 3 defines the minimum requirements for having a DT since it encompasses the real-time interaction with sensors, connected devices and Internet of Things (IoT) technologies.



Maturity element (logarithmic scale of complexity and connectedness)	Defining principle	Outline usage
0	– Reality capture (e.g. point cloud, drones, photogrammetry, or drawings/sketches)	– Brownfield (existing) as-built survey
1	– 2D map/system or 3D model (e.g. object-based, with no metadata or BIM)	– Design/asset optimisation and coordination
2	– Connect model to persistent (static) data, metadata and BIM Stage 2 (e.g. documents, drawings, asset management systems)	– 4D/5D simulation – Design/asset management – BIM Stage 2
3	– Enrich with real-time data (e.g. from IoT, sensors)	– Operational efficiency
4	– Two-way data integration and interaction	– Remote and immersive operations – Control the physical from the digital
5	– Autonomous operations and maintenance	– Complete self-governance with total oversight and transparency

Figure 1: Digital Twin maturity level according to IET

2.1.1. Overview of the services

In the context of the project and with regards to TE-8, Digital Twins are addressed to residential consumers to improve their awareness and active participation in Demand Response services. The primary objective is to empower District Heating Network Managers, Building Managers, Facility Managers, and similar roles, to fully harness the potential of Digital Twins by supporting the management of diverse infrastructure types, such as heat districts or power distribution networks. This support extends to highlighting flexibility opportunities and assisting operators in their day-to-day management tasks. Within the context of the DEDALUS project, Digital Twins will be applied across different scopes in pilots 2, 3, and 4 as Front Runners, as well as in pilot 6 and 7 as Multiplier. The development process is guided by High Level Use Case 4 (HLUC4) Digital Twins prediction of user energy profile and flexibility for consumer-aware optimal flexibility planning, defined in D2.1 [1] and specific use cases defined in D2.3 [2], including UC6, UC7, UC8, UC9, UC10, and UC11. Table 1 reports the complete list of Functional Requirements defined on top of the previously mentioned Use Cases.



Table 2: Task4.1 - Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
Digital twin with an interface that will visualize the different cluster of consumers and the flexibility assessment for each cluster.	TE8-FR1	UC10	TE-1, 4, 8	Pilot 3 – GR	Not Started
Digital twin will deliver analytics (environmental/energy) to the energy manager.	TE8-FR2	UC9	TE-8	Pilot 4 - IT	Started
Digital Twin will provide recommendations/alerts (not direct control) to the energy manager based on energy consumption, while considering their comfort preferences.	TE8-FR3	UC10	TE-8	Pilot 4 - IT Pilot 3 - GR	Started
Digital Twins will enhance the district visualization by providing a 3D visualization basing on IFC Files, enabling direct connection with monitoring real time data of different assets.	TE8-FR4	UC7	TE-8	Pilot 2 - DK	Started
Digital Twin will provide a simulation environment in order to optimize the management of district heating networks and enable the possibility to develop and integrate Digital Models of assets.	TE8-FR5	UC11, UC8	TE-8	Pilot 2 - DK Pilot 3 - GR	Started



2.1.2. Architectural Diagram

Figure 3 illustrates the high-level architecture of the service, highlighting its components and their mapping to the DEDALUS Architectural Diagram presented in D2.3 [2]. Each component will be customized to meet the specific needs of the pilot projects, following the previously defined Functional Requirements (FRs). However, the development will strictly adhere to the DEDALUS Architecture and comply with the interoperability guidelines established in Task 3.2. Specifically, data provided by each pilot will be utilized by the Digital Twin via the FIWARE Orion Context Broker. The architecture remains consistent with the one outlined in Deliverable D4.1 [5].

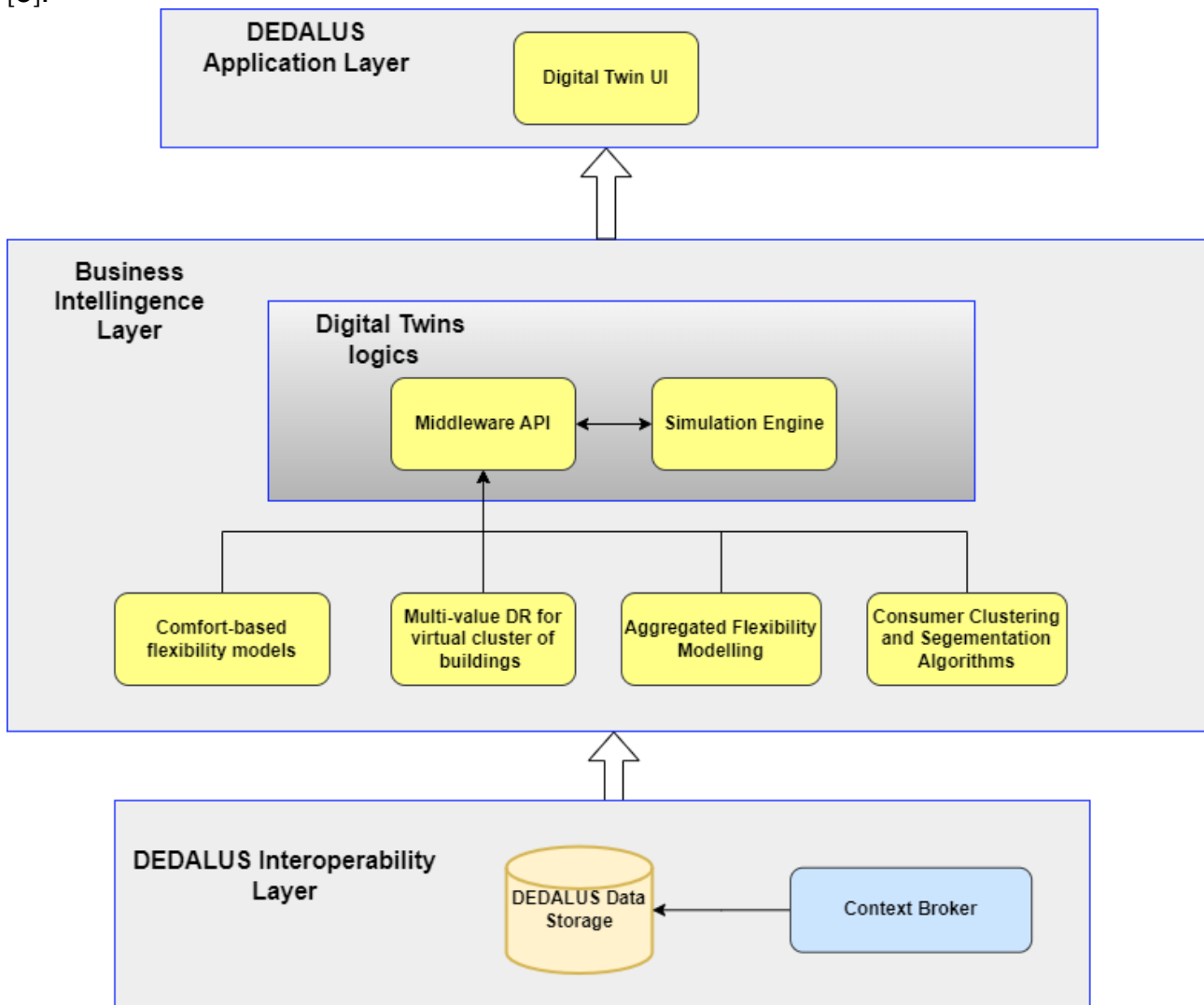


Figure 2: Digital Twins for consumer-aware optimal flexibility planning – Architectural diagram

The Digital Twin is composed by the following components:

- User Interface (UI): it serves as the primary access point for the Digital Twin (DT).
- Middleware API: it is responsible for receiving all the communications coming into the DT, formalizing them into proper data structures to allow the communication and data sharing between different components. For instance, the results of the simulation



engine will be provided to the middleware layer and subsequently to the UI, only once the Middleware has formalized the data. The co-existence of this layer with the DEDALUS Interoperability Layer allows a stronger and deeper integration between data and services and between service and service.

- Simulation Engine: it oversees performing different simulation scenarios. It can be utilized to evaluate 'what-if' scenarios by altering certain parameters to assess their impact on the infrastructure. Additionally, it can simulate Digital Models of assets, thereby enabling the Facility Manager to gauge the potential flexibility offered by the addition of specific assets.
- Business Intelligence Layer components: The other components of the Business Intelligence Layer are developed in the context of the task 4.3, reported in section 4 and will be integrated in the Digital Twin through the Middleware API (Application Programming Interface).
- Dedalus Interoperability Layer: It is responsible for taking the raw data and making it available to all developed services, it is developed under WP3.

2.1.3. Data for service execution

Data requirements for each service can be defined based on pilots and related use cases. The data to be employed for the development of the DT are determined based on the needs of the pilots and are appropriately utilized across different components. Table 3, Table 4 and Table 5 provide the data requirements for each pilot in which the DT will be applied, that are: Pilot 2 Denmark, Pilot 3 Greece and Pilot 4 Italy. For each identified data set, the related data format and protocol is defined; if the dataset is already available details are provided in the Notes column. Moreover, for each data set is specified the involved component.

Table 3: Data Requirements for Pilot 2 - DK

Dataset Name	Protocol	Format	Used By	Available	Notes
Energy Meter DHW	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
DHW Circuit	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Energy Meter Main	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Heat Circuit	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September



Dataset Name	Protocol	Format	Used By	Available	Notes
Indoor Climate	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Weather Data	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Radiator thermostat	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September

Table 4: Data Requirements for Pilot 3 - GR

Dataset Name	Protocol	Format	Used By	Available	Notes
Smart Meter consumption data: energy, power, voltage, power factor	API	JSON, CSV	Context Broker	Yes	Access from HERON Energy Control (smart metering platform)
Smart plug data: energy, power, operation (on/off)	API	JSON, CSV	Context Broker	Yes	Access from HERON Energy Control (smart metering platform)
GR Day-ahead market scheduling and prices	API	CSV, JSON	Simulation Engine	Yes	EnEx data filtered through a HERON API
RES, HYDRO and Load from GR ISP1 results for calculation of the Green %	API	CSV, JSON	Simulation Engine	Yes	EnEx data filtered through a HERON API
HERON 500 KW PV	API	CSV, JSON	Simulation Engine	Yes	HERON data for owned PV
brl_mod_level: Boiler Modulation Level	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2



Dataset Name	Protocol	Format	Used By	Available	Notes
blr_t: Water Temperature Inside the Boiler	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Heat: Heat Indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Flame: Flame indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Water: Domestic Hot Water	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_out: Outdoor Temperature	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_ret: Inlet Temperature	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_r: Indoor Temperature	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_r_set: Indoor Stepoint	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2



Dataset Name	Protocol	Format	Used By	Available	Notes
otc_cur: Heating Balance	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_set: Boiler Setpoint	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
otc_max: Mx Adaptive Setpoint	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Bypass: Bypass Indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Nodata: Nodata Indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2

Table 5: Data Requirements for Pilot 4 - IT

Dataset Name	Protocol	Format	Used By	Available	Notes
Building Consumptions	MQTT, API	CSV	UI	Yes	Sensors installed. Data available through energy consumption monitoring platform since June 2024
IAQ Data	MQTT, API	CSV	UI	Yes	Sensors installed. Data available



Dataset Name	Protocol	Format	Used By	Available	Notes
					through monitoring platform since June 2024
Comfort Data	API	CSV	UI	Yes	TSV (thermal sensation vote) questionnaire provided to users and feedback are going to be collected.

2.1.4. Service output

The outcome of TE-8 is multiple, achieved through the implementation of various components tailored to different objectives. The primary access point is a user-friendly interface, presented as a web page, designed to assist the involved actors like the Building and the Facility Manager in addressing flexibility proposals and DR programs.

The Digital Twin Logic models will produce specific outcomes for each pilot.

- Pilot 2 – Denmark: the digital twin will display information to give a representative picture of the status of the entire building. Furthermore, the model will provide a set of optimised temperatures, enabling the Building/Facility Manager to assess the impact of temperature changes within the building in terms of consumption and/or comfort status.
- Pilot 3 – Greece: the digital twin will display various information on device usage and consumption and usage habits, giving the Building/Facility Manager the opportunity when and how to implement DR suggestions also shown directly in the Digital Twin. In addition, a digital model of a storage system will be developed based on specifications provided by the pilot. This model will be used to develop other potential utilization strategies to reduce peak energy consumption and minimize the carbon footprint, providing a new set of recommendations on how to optimally utilize the storage system.
- Pilot 4 – Italy: the digital twin will show different metrics for estimating the comfort values found in the building environments as well as consumption values. A recommendation system will also be made available to the Building/Facility Manager with simple and clear instructions for making comfort values optimal by also considering consumption values along with temperature and humidity.

2.1.5. Main benefits and innovation

The main innovation brought about by the Digital Twin is the ability to facilitate the management of different types of infrastructure in an immediate and intuitive manner. The DT ensures that the Building Manager, rather than the Facility Manager always can have a clear and detailed view of the infrastructure he must manage, as flexible asset, via the User Interface (UI), and can assess the planning of the resources at his disposal, facilitating DR and flexibility programs. A



further strength of this technology, in the context of DEDALUS, is the integration of comfort models, which allow DR programs to be refined by also considering the well-being of the facilities' occupants. In addition, the use and integration of simulation, optimisation and Digital Models allow the near real-time evaluation of hypothetical conditions (what-if scenarios), future energy balances and the presence of assets that do not actually exist (Storage Systems' Digital Model), without any detrimental effect on the real infrastructure, adding value to energy usage planning and optimisation.

In some cases, it has already been possible to visualize in the Digital Twin the representations of buildings either by sensor values or as a 3D representation model.

2.1.6. Development activity

The result of the development activity as achievement of MS7 2st technology development cycle completion is mainly related to Pilot 2 and Pilot 4. Regarding the Pilot 3, the work has been mostly focused on the technical specifications of the service and the Digital Twin UI definition. More details are provided in the following sections:

2.1.6.1. Pilot 2 - Denmark

Significant progress has been achieved in the development of the DT within the scope of Pilot 2. A functional initial version of the digital twin has been implemented (Figure 3), showcasing key features such as the visualization of apartment temperature and humidity data. Currently, historical data is presented; however, ongoing integration with the interoperability layer will soon enable real-time data visualization, enhancing the tool's functionality and responsiveness.

Additionally, the digital twin includes a navigable 3D model of the apartments managed by the social housing company Fællesbo, located in Denmark. This feature allows users to explore the spatial representation of the apartments, further enriching the user experience by providing a tangible, immersive view of the environment.

An interactive form has also been integrated into the interface, allowing users to query the simulation model. By insert specific parameters, such as indoor and outdoor temperature and humidity, the building manager can simulate various scenarios and analyse potential outcomes. This feature serves as a vital tool for understanding and optimizing the environmental conditions of the apartments.

These advancements mark a pivotal step forward in the development of the digital twin, laying a strong foundation for future enhancements and integrations.

The digital twin has been developed using React with TypeScript and styled with Bootstrap 5 to ensure a modern and responsive user interface. For the 3D model visualization, the library That Open was employed, specifically designed for creating Building Information Modelling (BIM) software, and it leverages Three.js for managing 3D models on the web.

The 3D model, however, has proven to be too large, causing a noticeable slowdown in navigation and interaction within the system. To address this issue, we plan to optimize the model by



dividing it into separate segments corresponding to each individual building. This approach will allow us to reduce the computational load and improve performance.

Additionally, we intend to integrate a selector within the digital twin interface, enabling users to choose which specific building they wish to visualize. This enhancement will not only improve navigation speed but also provide a more user-friendly and efficient experience by focusing on smaller, more manageable portions of the 3D environment.

The source code is available in the project's GitHub repository¹.

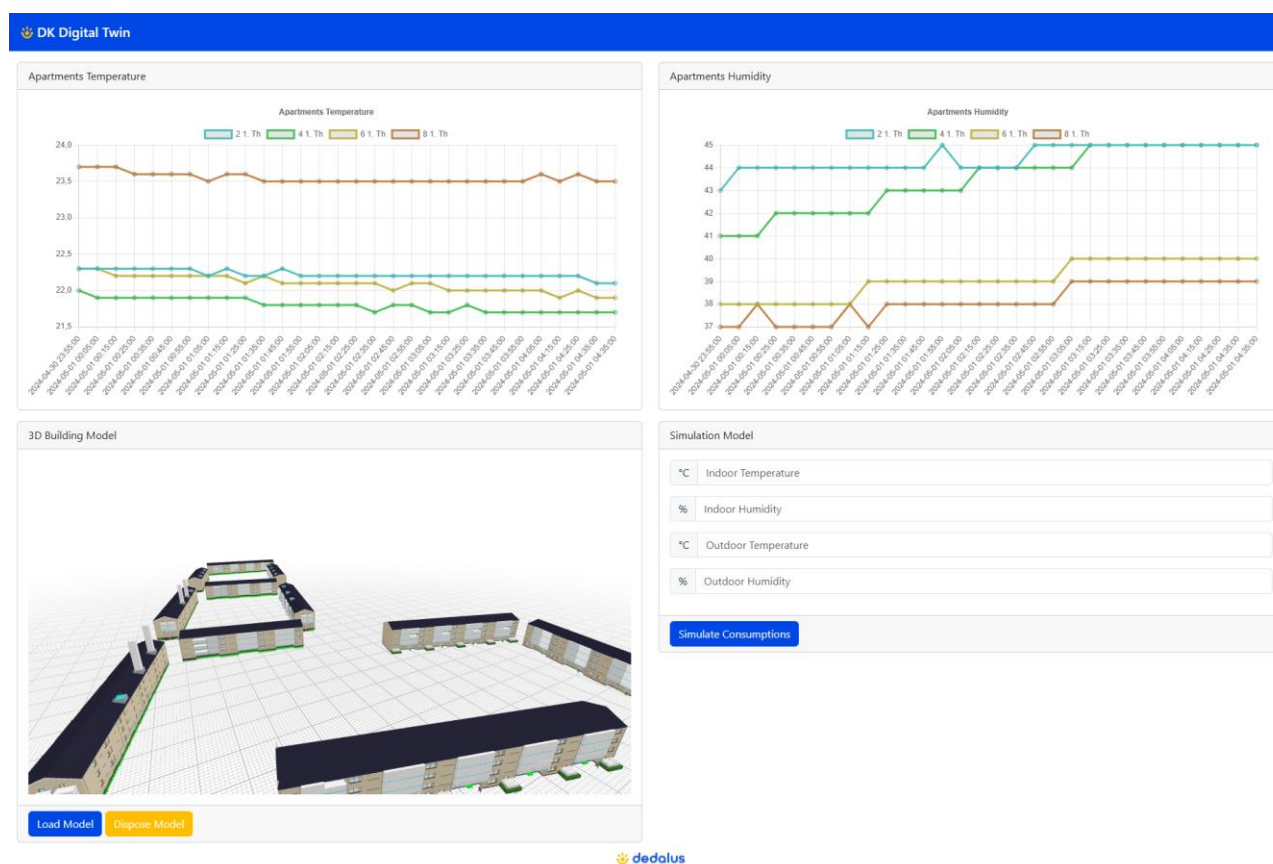


Figure 3: Pilot 2: Denmark Digital Twin UI

Regarding the simulation model, significant progress was made. The following is the architecture of the simulation engine already presented in the previous deliverable D4.1.

¹ <https://github.com/dedalus-technology-release/T4.1-Pilot-2-Digital-Twin>.



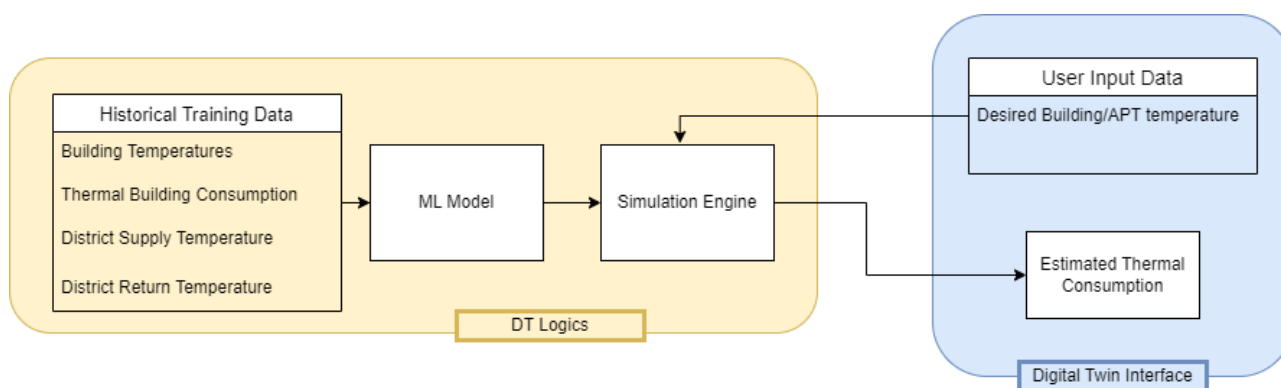


Figure 4: Simulation Engine - Architectural model

The simulation engine was designed to estimate electricity consumption based on the temperature and humidity values set by the building manager, supporting better energy efficiency and management practices.

To achieve this goal, an extensive training phase was conducted in which various machine learning models were implemented and evaluated for their predictive accuracy. The models tested included:

- Gradient Boost Regressor (GB)
- Random Forest Regressor (RF)
- Support Vector Machine (SVM)
- K-Nearest Neighbors Regressor (KNN)
- Bayesian Ridge Regressor (BR)
- Multi-Layer Perceptron (MLP) Regressor
- A custom-built Neural Network with 6 layers (NN)

These models were trained using a comprehensive dataset derived from multiple sources, including apartment sensor measurements, external environmental data, in particular, the indoor and outdoor temperature, the percentage of cloudiness, the incidence of the direct sun and heat meter readings. To ensure the reliability and accuracy of the training data, the raw input from these sources was thoroughly unified, cleaned, and harmonized. This preprocessing step was critical in creating a high-quality dataset that could effectively capture the complex relationships between input variables and electricity consumption.

After rigorous testing, the neural network with six layers emerged as the most promising model together with the Random Forest Regressor and the Support Vector Machine which obtained a Mean Absolute Error (MAE) of: 1295, 1186 and 1298 on the evaluation set. The results were compared with a naive model that uses the mean value to predict consumption and obtains an MAE value of 1496. The MAE, Mean Absolute Percentage Error (MAPE) and the coefficient of determination (R^2) values for all models tested are given in the Table 6. The results showed a strong predictive potential in estimating electricity consumption based on the parameters provided. However, these results are only partial and will need to be challenged again using the totality of the available data. Therefore, a further refinement and model selection phase will have to follow. This model takes advantage of both internal and external environmental factors, making it a powerful tool for data-driven decision making in energy management.



Table 6: Simulation Engine - Model evaluation in comparison

	MAE	MAPE	R ²
Baseline	1496.16	2.54	-0.01
GB	1238.86	1.69	0.20
RF	1186.70	1.84	0.28
SVM	1298.42	2.17	0.15
KNN	1290.21	1.84	0.16
BR	1419.36	2.40	0.07
MLP	1420.21	2.30	0.07
NN	1295.14	2.25	0.19

While the current version of the model has shown promising results, further improvements are planned to enhance its performance and adaptability. The model will be refined by incorporating new data as it becomes available, followed by a fresh round of data analysis to ensure the updated dataset captures any emerging patterns or trends. This iterative process is expected to yield a model with greater accuracy and robustness, ensuring its applicability across a broader range of scenarios.

Figure 5 illustrates the predictions made by three models compared to actual observed values, highlighting their current predictive capabilities. Additionally, Figure 6 presents a comparison between the MAE calculated on the training set and the MAE calculated on the validation set for the custom neural network. This comparison demonstrates that the model, as well as the other, does not suffer from issues of overfitting or underfitting, confirming its ability to generalize effectively to unseen data.

The source code is available in the project's GitHub repository².

² https://github.com/dedalus-technology-release/T4.1_pilot2_simulation-engine/tree/main/simulation_engine



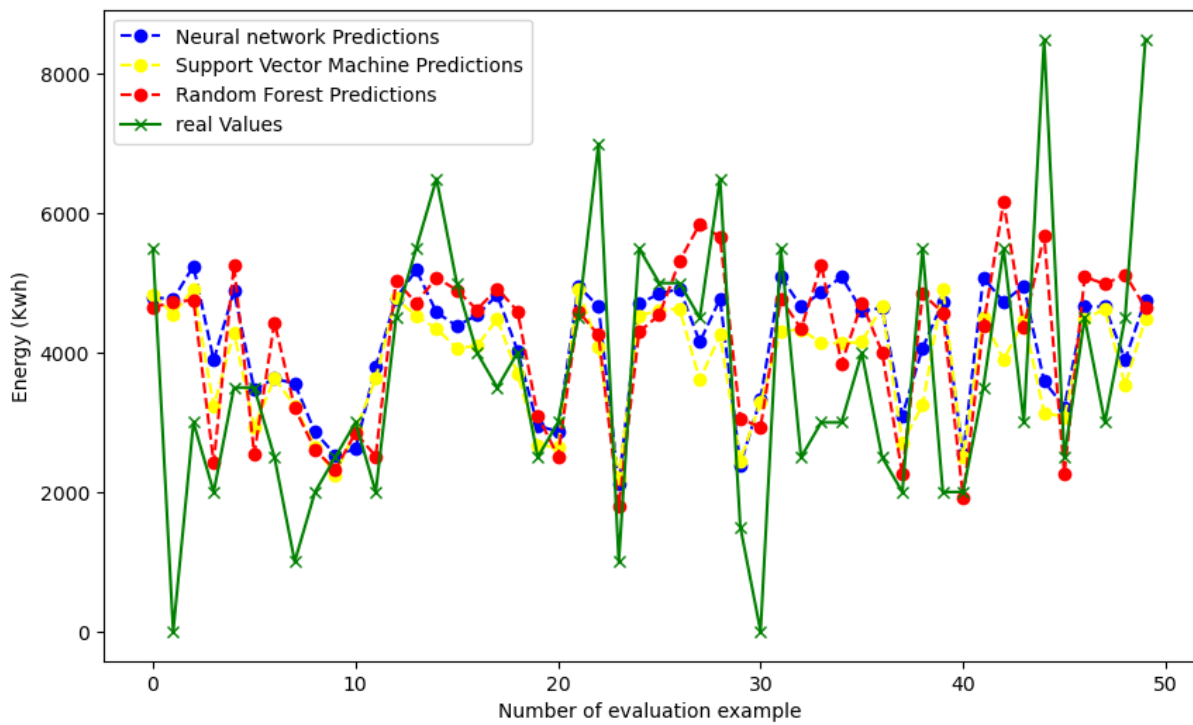


Figure 5: Simulation Engine - Comparison of predicted and actual values

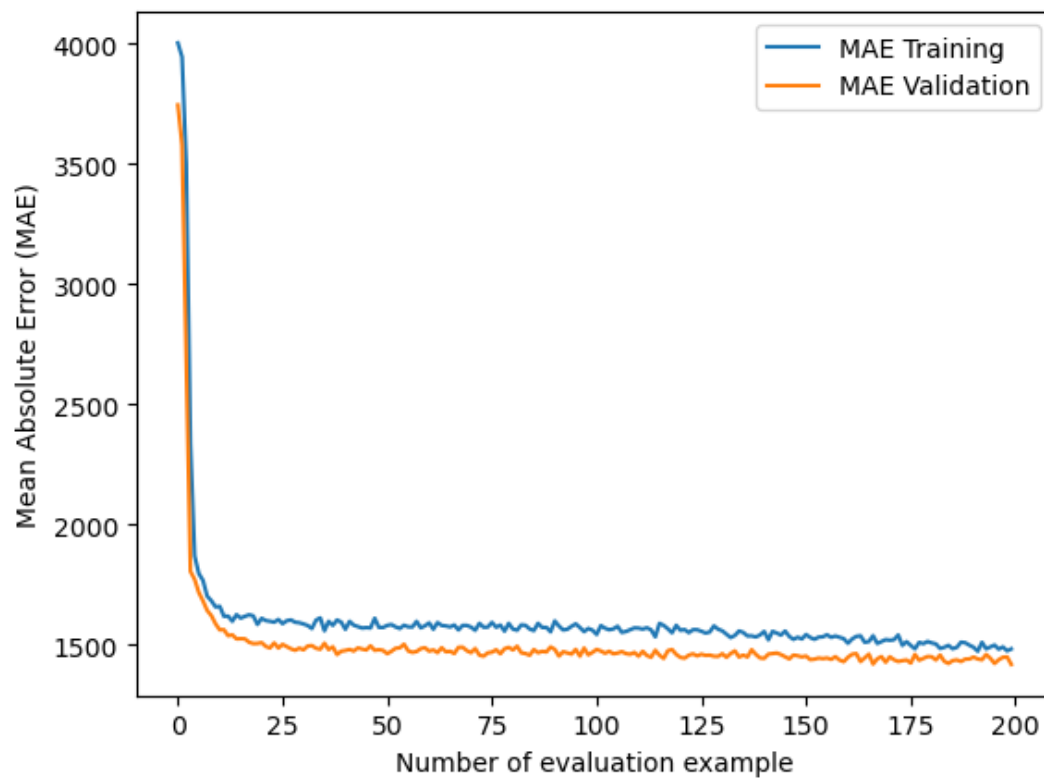


Figure 6: Simulation Engine - Comparison of MAE values in the training set and validation set.



2.1.6.2. Pilot 3 – Greece

Regarding the Pilot 3, on the second wave of development, a first version of mock ups for the Digital Twin was designed. The mock ups were produced with a high level of detail based on feedback from HERON acting also as a Demand Response / Renewable Energy Share (RES) Aggregator, in addition to electricity supplier, in the Greek pilot. The mock ups show a first clear organization of the data to be shown and the services to be implemented/integrated into the Digital Twin. the Digital Twin is organized into three sections. The first section will show general/aggregated data also with extrapolation data that will evaluate the impact of DR strategies at a large share of HERON's customer base. The second will show the household-level information. Finally, the third section will show the results of storage simulations based on real-time market data.

The three sections as follows:

1. Overview/Aggregated data section: this section reports high-level Key Performance Indicators (KPIs) such as flexibility potential or energy saving. It also reports measures to comprehensively evaluate all households, such as comparing consumption to RES or consumption of households categorized into clusters to better manage their visualization and protect their privacy. Still measures consumption by device type, flexible assets, and RES shares.
2. Household level section: this section reports many of the measures already presented in the first section but at a greater level of detail; in fact, this section allows you to obtain, search, and visualize data from an individual household. KPIs, consumption, and devices are reported referring to the household of interest. This section also includes a table showing the DR recommendations with the relative Energy saving/RES increment and CO2 footprint reduction that can be achieved.
3. Storage Simulation section: The last section provides the results of simulations performed considering a storage system. On this page the consumption and RES curves can be viewed along with that of the simulated storage as well. KPIs considering storage are also given, as well as recommendations and impact in terms of energy flexibility.

Mock-ups for each section follow with a detailed description for each.



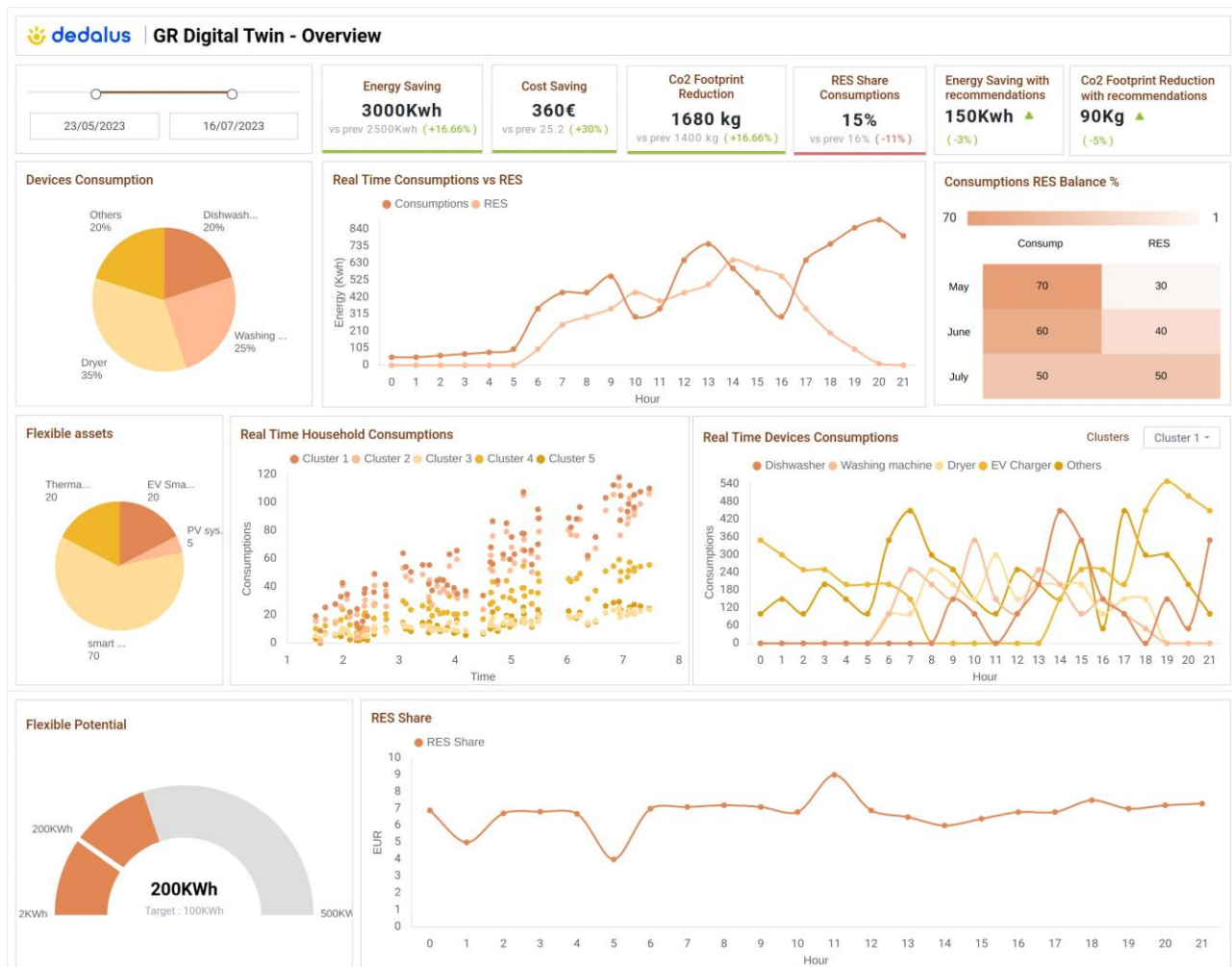


Figure 7: Pilot 3 Digital Twin User Interface first section – Overview

- **Energy Saving:** The KPI shows the energy savings compared to a previous period (e.g., last year)
- **Cost Saving:** The KPI shows the cost savings compared to a previous period (e.g., last year)
- **CO₂ Footprint Reduction:** The KPI shows the CO₂ Footprint reduction compared to a previous period (e.g., last year).
- **Energy Saving with recommendations:** The KPI shows the potential energy saving following the recommendations provided by the DR recommendation system, the number of recommendations to take in consideration will be selectable from the UI.
- **CO₂ Footprint Reduction with recommendations:** The KPI shows the potential CO₂ Footprint Reduction following the recommendations provided by the DR recommendation system, the number of recommendations to take in consideration will be selectable from the UI.
- **Devices Consumptions:** The pie chart shows the consumptions based on device types for a specific time frame (e.g. current year).
- **Real Time consumptions vs RES:** The line chart shows the consumptions and RES % real time values compared



- **Consumptions RES balance %:** The heatmap shows the consumptions and RES % values compared for the last 3 months.
- **Flexible Assets:** The pie chart shows the amount and types of flexible assets available in all households.
- **Real Time Household Consumptions:** The scatter plot shows the households consumptions considering the cluster for easier and more useful visualization.
- **Real Time Devices Consumptions:** The line chart shows the devices consumptions for a given household cluster for easier and more useful visualization.
- **Flexibility Potential:** The tab shows the energy flexibility potential for all households.
- **RES Share:** The line chart shows the RES Share values from the HERON's dynamic tariff (Service D).

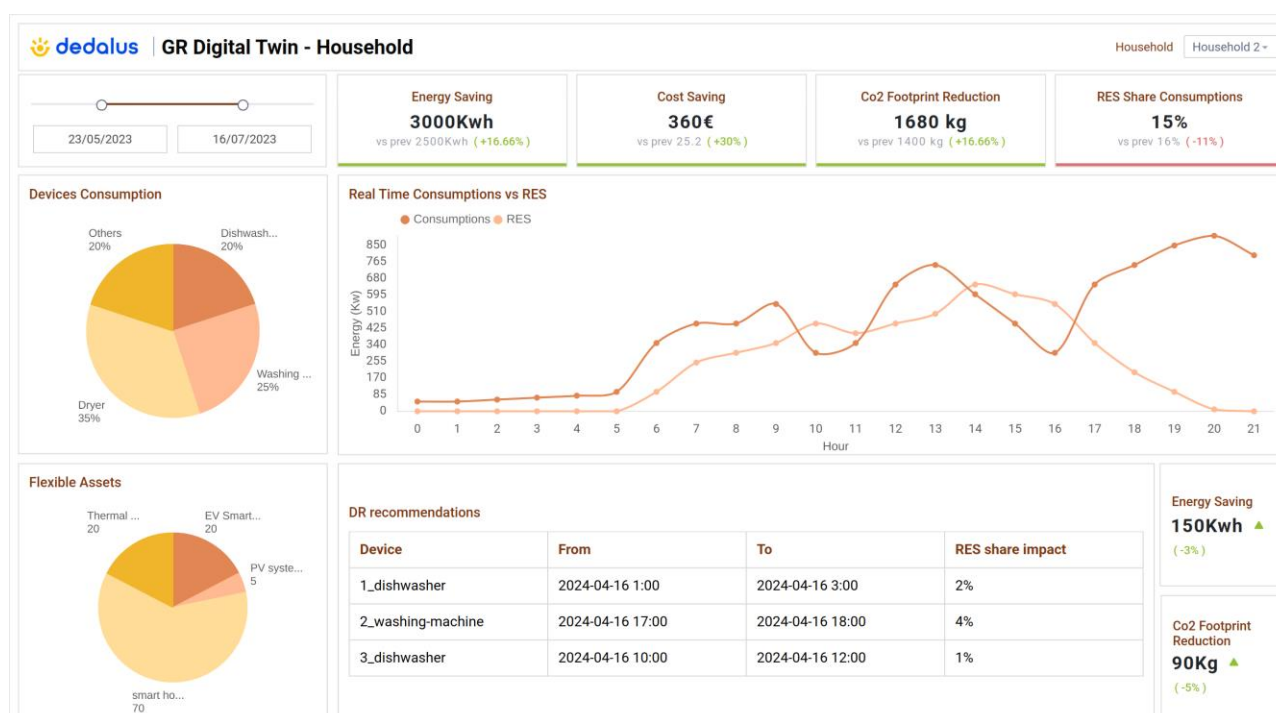


Figure 8: Pilot 3 Digital Twin User Interface second section – Household

- **Energy Saving:** The KPI shows the energy savings for the selected household compared to a previous period (e.g., last year).
- **Cost Saving:** The KPI shows the cost savings for the selected household compared to a previous period (e.g., last year).
- **CO₂ Footprint Reduction:** The KPI shows the CO₂ Footprint reduction for the selected household compared to a previous period (e.g., last year).
- **Devices Consumptions:** The pie chart shows the consumptions of the selected household based on device types for a specific time frame (e.g. current year).
- **Real Time consumptions vs RES:** The line chart shows the consumptions and RES % real time values, for the current household, compared
- **Flexible Assets:** The pie chart shows the amount and types of flexible assets available in the selected household.



- **DR Recommendations:** The tab shows the DR recommendations generated by the service F and provided by the service G showing the preferred time frame operation with the corresponding RES share impact.
- **Energy Saving:** The KPI shows the potential energy saving following the recommendations provided by the DR recommendation system.
- **CO₂ Footprint Reduction:** The KPI shows the potential CO₂ Footprint Reduction following the recommendations provided by the DR recommendation system.

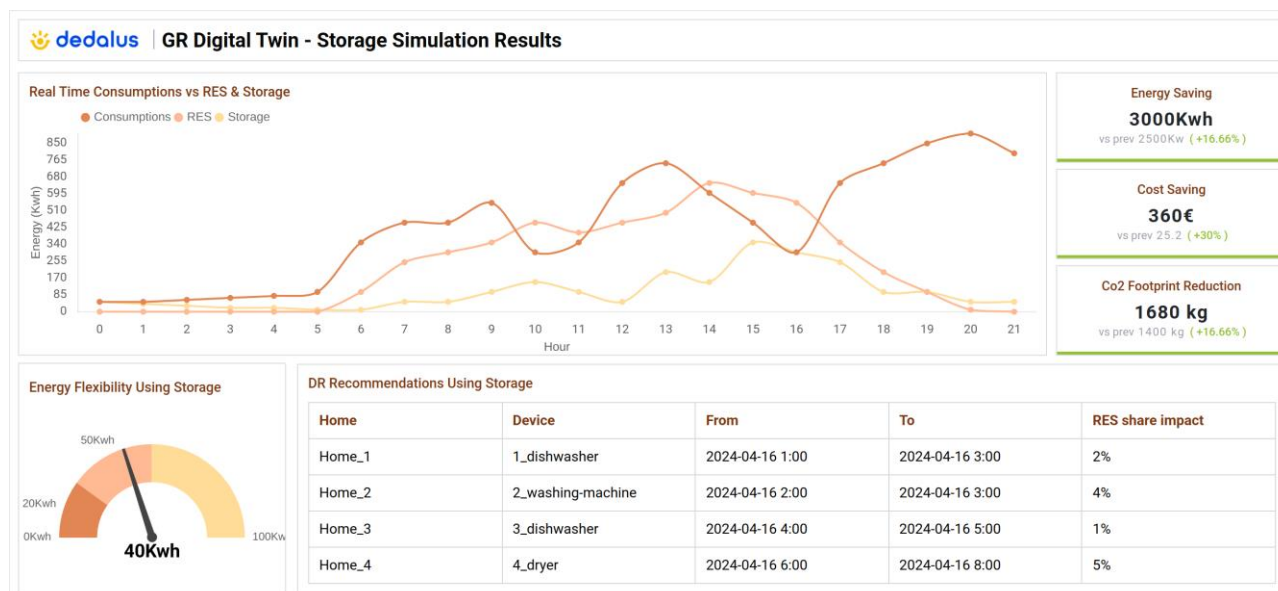


Figure 9: Pilot 3 Digital Twin User Interface third section - Storage Simulation Results

- **Real Time consumptions vs RES & Storage:** The line chart shows the consumptions and RES real time values, considering also the storage system simulated.
- **Energy Saving:** The KPI shows the energy savings compared to a previous period (e.g., last year) taking into consideration also the storage system simulated.
- **Cost Saving:** The KPI shows the cost savings compared to a previous period (e.g., last year) taking into consideration also the storage system simulated.
- **CO₂ Footprint Reduction:** The KPI shows the CO₂ Footprint reduction compared to a previous period (e.g., last year) taking into consideration also the storage system simulated.
- **Energy Flexibility Using Storage:** The tab shows the increase in energy flexibility considering the usage of the storage system simulated.
- **DR Recommendation Using Storage:** The tab shows the DR recommendations generated by the service F and provided by the service G considering also the storage system simulated, the table show the preferred time frame operation with the corresponding RES share impact.

2.1.6.3. Pilot 4 - Italy

The development activity in the Italian pilots mainly consist in the first version of the Digital Twin UI. The activity began with the mock-up analysis (Figure 10) and the development of an initial version of the dashboard for the Building Manager (Figure 11).



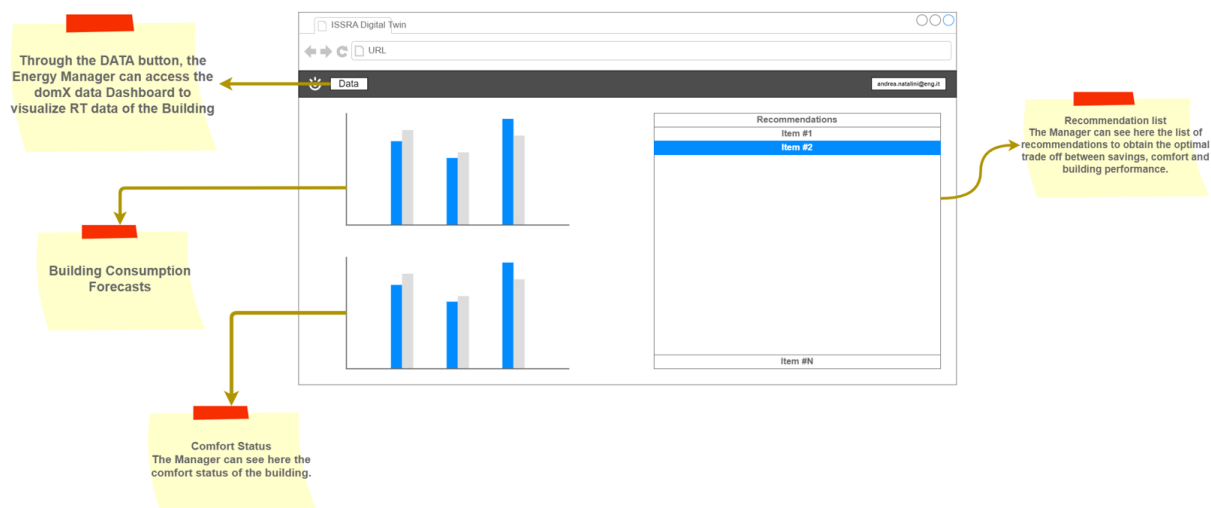


Figure 10: Pilot 4 – Digital Twin User Interface Mock-up



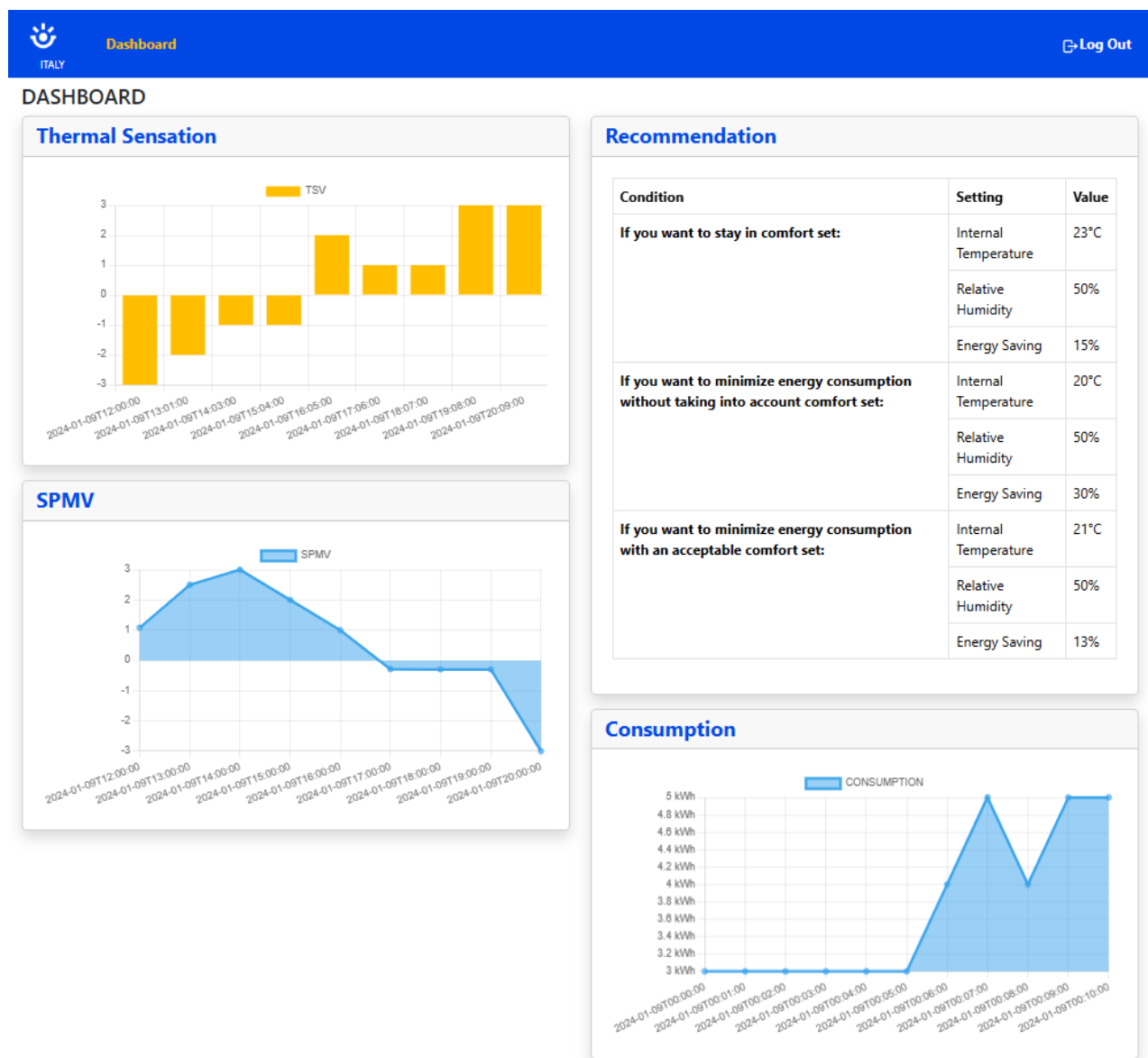


Figure 11: Pilot 4 – Digital twin User Interface Dashboard

The digital twin shows three different charts. The first, thermal sensation vote, shows the comfort data for the people in the building. The second chart, simplified Predicted Mean Vote (sPMV), presents the calculated values of sPMV, while the last chart shows the estimated consumption values of the building. Lastly, on the right is a panel showing DR recommendations for the building manager. These recommendations are presented in textual form, of the type “if you want to stay in comfort set...” and contain information on the degrees at which to set the temperature, humidity, and the percentage of energy saved.

More details about how these measures are calculated can be found in section 3.

At this stage, example data were shown in the digital twin for the purpose of testing their visualization, in future developments the charts will be populated with real data.



The digital twin has been developed using React with TypeScript and styled with Bootstrap 5 to ensure a modern and responsive user interface.

The source code is available in the project's GitHub repository³.

2.1.7. Outlook for the future expected release

The work planned for the next expected release, Deliverable D4.3, is specified per pilot in the following subsections.

2.1.7.1. Pilot 2 – Denmark

Following the data analysis process, the creation of the first simulation model and the development of the first version of the Digital twin UI carried out in this development cycle, the following steps will be developed:

- Improve the rendering speed of 3D models of apartments.
- Investigate and test the possibility of divide 3D model in multiple models of apartments.
- Improve the machine learning model that can estimate consumption based on the environmental variables set by the building manager.
- Development of the Middleware API.
- Integration of the Simulation Model with DEDALUS Interoperability Layer and with the Digital Twin UI.
- Integration of flexibility and clustering models
- Integration of the UI with the interoperability Layer to display real time data.
- Deployment of Digital Twin UI and services.

2.1.7.2. Pilot 3 – Greece

Following the construction of the mock up and a clearer definition of the requirements of the digital twin, the next steps are:

- Analysis of available data from the Pilot Site.
- Storage System specification implementation: starting from the specification provided by the Pilot Leader, a preliminary version of the Storage System simulator will be constructed.
- Development of the Digital Twin UI and it's integration with existing data and services. Integration with flexibility recommendation provided in the context of Task 4.3.
- Deployment of the Digital Twin UI and services.

2.1.7.3. Pilot 4 - Italy

The work planned for the next expected releases is specified as follows:

³ https://github.com/dedalus-technology-release/T4.1_pilot4_UI



- Refinement of the Middleware API Layer to integrate Comfort Services.
- Integration with the Middleware API layer and Comfort Services.
- Integration of available data and services with the Digital Twin UI.
- Deployment of the Digital Twin UI and services,

2.2. Apartment building Digital Twin for blockchain integration

2.2.1. Overview of the service

In the context of Romanian replication pilot 7, a DT model for the students' apartments building will be developed incorporating data from heterogeneous sources with the purpose of supporting decision making in DR programs and simulating the execution of apartment level DR actions. The DT model is interacting with the DLT/Blockchain/Smart contracts solution (TE-5) to facilitate the simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts. This integration allows for communication and data exchange between the DT components and the TE-5 smart contracts that are described in deliverables D3.1 [3] and D3.2 [4]. The DT model is leveraged to determine which appliances should be activated or adjusted to meet the specified constraints. The development process is guided by HLUC10 - Building-level decentralized flexibility aggregation and tokenization for DR, with specific use case UC21 defined in D2.3 [2] (see Table 7).

Table 7: DT Functional Requirements for TE-5

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated site	pilot	Development Status
Digital twin of the building to simulate the potential DR actions that can be taken within the apartments and calculate the expected impact of these actions on energy usage levels	TE5-FR5	UC21	TE-5	Pilot 7 - RO		Under development

2.2.2. Architectural diagram

Figure 12 shows the architectural diagram of the DT and its interaction with TE-5 that consists of several interconnected components that facilitate building-level DR by decentralized aggregation of apartment flexibility. A blockchain-based distributed ledger is set up at the building level to store and digitally fingerprint energy data. Smart contracts are defined to manage flexibility using decentralized market trading mechanisms. A building manager smart contract enables the publishing of flexibility bids, while apartments-associated smart contracts allow the publishing of flexibility offers. Several apartments' flexibility offers are matched with the building manager's bid, and flexibility transactions are generated. During actual delivery,



smart contracts are used for settlement and token distribution. More details about the latest version of TE-5 are presented in deliverable D3.2 (2nd technology release) [4].

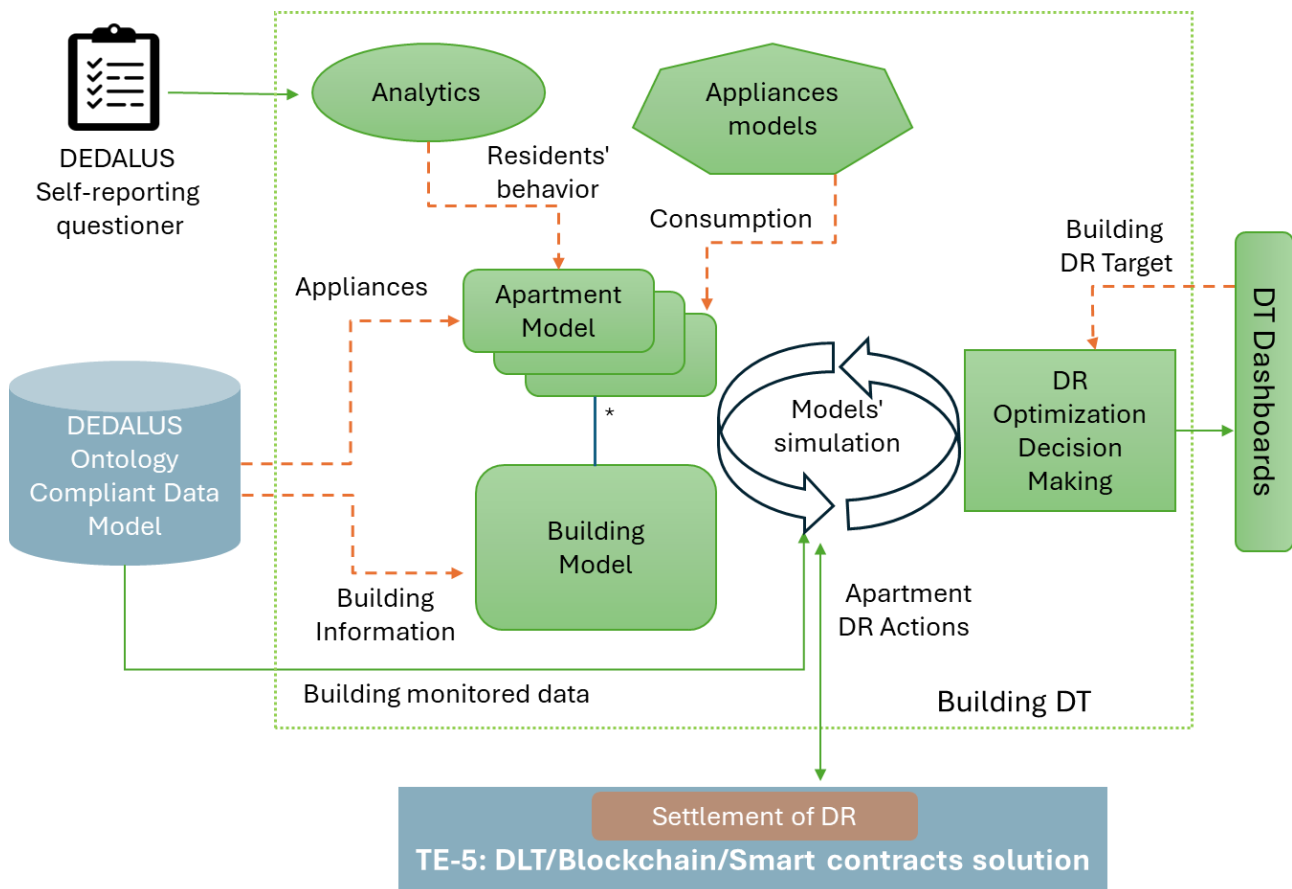


Figure 12: Apartments building DT, TE-5 integration – Architectural Diagram

The building DT model integrates a diverse range of data sources, such as the DEDALUS self-reporting questionnaire and DEDALUS Ontology Compliant Data Model. The DEDALUS self-reporting questionnaire was filled in by the students with the purpose of capturing energy-related behaviours such as the time intervals of apartments occupancy and appliances' usage (i.e. when and how often appliances are used). Analytics are applied on top of this information to identify patterns of energy related behaviours and to construct models of the energy demand profiles of the apartments. The objective is to simulate the daily energy consumption patterns of each apartment starting from the residents' behaviour by applying clustering algorithms to create initial consumption patterns that can be randomly varied, or by employing other machine learning algorithms such as neural networks.

The DEDALUS Ontology Compliant Data Model provides the technical information about the building (e.g. the number of apartments, the number of floors and the organization of apartments on each floor), information about the apartments (e.g., number of residents living in the apartment, the number and type of appliances used by the residents), and monitored data about the total energy consumed at the level of the building acquired at a given rate. Also, the data model provides information about the appliances used in each apartment and their models that contain specific information about the energy consumption characteristics of each individual



appliance. By integrating these data sources and models, the apartment's model simulates and determines energy consumption profiles that accurately represent how appliances contribute to the overall energy consumption within the apartment. The building model uses the information provided by the DEDALUS Ontology Compliant Data Model to represent the building as resulted from the technical information as well as the daily hourly total energy consumption obtained after processing the raw monitored consumption data. The apartment and building models, along with the building's target energy specifications provided by the building manager through the digital twin's front-end, are processed by the DR Optimisation Decision Making component. This component aims to identify apartments that should adjust their energy consumption profile to meet the building's target energy profile while optimising appliance usage patterns within those apartments. The target energy can be expressed either as an hourly consumption profile or as a percentage of the building's total energy usage. The problems of identifying these apartments and optimising appliance schedules are combinatorial optimisation problems complex to solve due to the complex interactions and large search space involved.

To address these challenges, the DR Optimisation Decision Making component uses bio-inspired meta-heuristics, such as the Harris Hawks Optimisation (HHO) algorithm. This approach leverages principles from nature to efficiently explore and optimise the solution space, ensuring that the building's energy consumption is adjusted effectively while meeting specified targets. The use of bio-inspired meta-heuristics is particularly advantageous in handling the complexities of real-world energy optimisation tasks within a dynamic building environment.

2.2.3. Data for service execution

The data employed for the development of the DT are derived from Pilot 7 as shown in Table 8.

Table 8: Data Requirements for Pilot 7 - RO

Dataset Name	Protocol	Format	Used By	Available	Notes
Pilot7_energy_consumption	FTP/ MQTT	CSV	Building model and DR optimisation decision making	Yes	
Pilot7_questionnaire_data	-	CSV	Analytics	Yes	

2.2.4. Service output

The output of the service consists of insights on which apartments should adjust their energy consumption profiles to help the building meet DR goals. Additionally, it includes information on potential appliance activations in the context of DR settlement smart contracts. The output



profiles will be displayed on the DT dashboard, while the appliance activation details will be integrated with the smart contracts of TE-5.

2.2.5. Main benefit and innovation

Essentially, the innovation lies in combining DT models with blockchain and smart contracts to simulate DR actions regarding appliance activation within the context of DR settlement smart contracts. Additionally, the use of nature-inspired optimization heuristics helps in finding the most effective solutions to meet the specified constraints, such as matching a demand curve, while considering various factors like occupancy patterns, appliance usage, and external environmental conditions. Moreover, combinatorial simulations with nature-inspired heuristics allow for the exploration of diverse solution possibilities within the DT, often leading to solutions that are close to optimal in a reasonable amount of time.

2.2.6. Development activity

The development activities within the reported timeframe have focused on the following:

- The refinement of the Analytics component.
- The refinement of the Apartment models.
- The development of two algorithms integrated in the DR Optimization Decision Making component:
 - an Electricity Load Disaggregation (ELD) Algorithm which integrates the HHO algorithm [8] to estimate the individual energy consumption of each apartment starting from the total energy consumption of the building.
 - a DR Action Generation (DRAG) algorithm that combines HHO with genetic algorithms to identify the minimum number of apartments and how these apartments should adjust their energy consumption in response to a DR action.
- The development of the DT Dashboards mock-ups.

The DT components that will be implemented will be uploaded into the DEDALUS repository when ready, at the next deliverable iteration.

In what follows, we present: the architecture of the Apartments building DT highlighting the technologies that will be used for development, the formal definition of the DT followed by a description of the two DR Optimization Decision Making algorithms, and DT Dashboards mock-ups.

2.2.6.1. Apartments building DT Technological View

The technical architecture of the DT is illustrated in Figure 13 and consists of the following components:

- Analytics component – simulates energy demand profiles for the apartments of a building using data collected from the self-reporting questionnaires, the Ontology Compliant Data Model and appliances models. Implemented in Pytho⁴, this component applies clustering

⁴ <https://www.python.org/>



algorithms to generate initial consumption patterns which are afterwards adjusted based on specific constraints are varied considering some constraints. For instance, the fridge must remain continuously connected to electricity, while the usage of light bulbs is restricted to specific periods, depending on seasonal variations.

- **Model API** – this Representational State Transfer (REST) API⁵, developed using Java Spring Boot⁶, is used to persist and retrieve information specific to the DT in/from the Data Model database, such as: the building's daily energy consumption, consumption patterns simulated by the Analytics component, information about the devices from each apartment, the decisions of the DR Optimization Decision Making component.
- **DR Optimization Decision Making component** – implemented as a REST API, using Django⁷ framework, the component integrates the Load Disaggregation and DR Action algorithms. It interacts with the Model API component to: (i) retrieve information about the devices from each apartment and the residents' time and comfort constraints for using the devices; (ii) store the estimated energy consumption profiles resulted after the execution of the load disaggregation algorithm; (iii) store the energy consumption profiles recommended for a minimum number of apartments that should be involved in DR, resulted after the execution of the DR Action algorithm.
- **DT Dashboards API** – implemented as a REST API using Java Spring Boot, this component is the back end of the DT Dashboards front-end, connecting it to the DR Optimization Decision Making and Model API components.
- **DT Dashboards** – represents the DT's front-end component, implemented using React⁸. This component can be accessed by the (i) building manager to view the apartments from the building, their devices and estimated energy consumption, and to insert the target energy consumption curve for the DR program, and (ii) apartments' owners to view their devices and estimated energy consumption together with the recommended energy consumption profile if their apartments have been selected in the DR program.

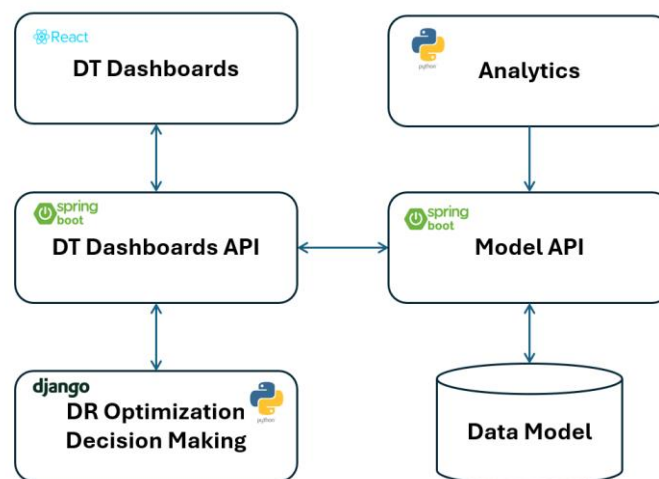


Figure 13: Pilot 7 - Technical architecture of the DT Prototype

⁵ <https://www.ibm.com/topics/rest-apis>

⁶ <https://spring.io/projects/spring-boot>

⁷ <https://www.djangoproject.com/>

⁸ <https://react.dev/>



2.2.6.2. Formal definition of the DT

In our approach an apartment building, B , is modelled as follows:

$$B = \{ap_i | i \in \{0, n - 1\}\} \quad (1)$$

where n represents the total number of apartments ap . An apartment ap_i , is defined as a tuple consisting of the apartment's identification number (id), and a set of devices Did :

$$ap_i = (id, D_{id}) \quad (2)$$

The set of devices is defined as:

$$D_{id} = \{d_j | j \in [1, dn_{id}]\} \quad (3)$$

where dn_{id} is the number of devices from the apartment with the identification number id .

Each device d_j from an apartment has a set of basic information that describes it (BI_{d_j}), a comfort vector (CV_{d_j}), and a current consumption pattern vector (CPV_{d_j}):

$$d_j = \langle BI_{d_j}, CV_{d_j}, CPV_{d_j} \rangle \quad (4)$$

The basic information, BI_{d_j} , that describes a device d_j includes the identification number of the device, id , its name ($name_{d_j}$), its consumption rating expressed in kWh ($consumption_{d_j}$), a flag indicating whether the device is deferrable or not ($isDeferrable$), and the minimum number of hours the device must be turned on during a day ($minUsageHours_{d_j}$).

$$BI_{d_j} = \langle id, name_{d_j}, consumption_{d_j}, isDeferrable, minUsageHours_{d_j} \rangle \quad (5)$$

The comfort vector (CV_{d_j}) of a device d_j from an apartment is a vector of 24 elements, where each value corresponds to an hour of a day (see Formula 6).

$$CV_{d_j} = \{comfortValue_h | h \in \{0, 1, \dots, 23\}\} \quad (6)$$

The comfort vector represents how uncomfortable the apartment's residents are to use a certain device at a given hour. The elements of the comfort vector can take the following values: 0 - the resident is comfortable using the device at a certain hour, 1 - the resident agrees to make a compromise and use the device at a certain hour; 2 - there is a great discomfort for the resident to use the device at that hour.

The current consumption pattern vector CPV_{d_j} is a binary vector that shows in which hours a device d_j is usually used by the resident. A vector element contains a value that indicates whether the device is used or not at a certain hour (see Formulas 7 and 8).

$$CPV_{d_j} = \{isUsed_h | h \in \{0, 1, \dots, 23\}\} \quad (7)$$

$$isUsed_h = \begin{cases} 0, & \text{if the device is not used at hour } h \\ 1, & \text{otherwise} \end{cases} \quad (8)$$

The schedule for a device d_j from an apartment is a binary vector, $Schedule_{d_j}$, of 24 elements, like the current consumption vector. The value 0 indicates that the device should be turned off at hour h , conversely, the value 1 means that the device should be turned on.



$$Schedule_{d_j} = \{isUsed_h | h \in \{0, 1, \dots, 23\}\} \quad (9)$$

Each type of device has its own constraint that must be considered when rescheduling them. Non-deferrable devices must maintain their initial consumption pattern throughout the scheduling procedure as defined in the formula below:

$$Schedule_{d_j}[h] = CPV_{d_j}[h], \forall h \in \{0, 1, \dots, 23\} \quad (10)$$

The individual device schedules are aggregated in a matrix representing the schedules of the devices from all the apartments from the building:

$$Schedule_{BT} = \begin{bmatrix} Schedule_{d_0} \\ \dots \\ Schedule_{d_m} \end{bmatrix} \quad (11)$$

where each row of the matrix corresponds to the schedule $Schedule_{d_i}$ of a device from an apartment, m is the total number of devices from all the building's apartments, and T is the timeframe for which the schedule of the devices has been done.

The number of hours interruptible devices are turned on over a day must be greater than or equal to the minimum usage hours specified for the device:

$$\sum_{h=0}^{23} Schedule_{d_j}[h] \geq minUsageHours_{d_j} \quad (12)$$

The total hourly amount of energy consumed at the level of the building that will be disaggregated is defined as a vector of 24 elements, where each element represents the amount of kWh consumed at the hour equal to the index of the element:

$$TC = \{consumption_h \in R | \forall h \in \{0, 1, \dots, 23\}\} \quad (13)$$

The overall comfort penalization (CVP) for each device for the day will be computed based on the element-wise multiplication of the values from its schedule definition and comfort definition.

$$CVP_{d_j} = \sum_{h=0}^{23} Schedule_{d_j}[h] * CV_{d_j}[h] \quad (14)$$

2.2.6.3. DR Optimization Decision Making Algorithms

This section presents the algorithms used for load disaggregation and DR actions.

The ELD Algorithm (see algorithm below) estimates the schedules of individual devices across the building's apartments based on the building's hourly energy consumption data. This estimation process is a minimization problem, seeking to derive energy consumption curves that align closely with the observed hourly energy usage of the building. To achieve this, the ELD Algorithm integrates the HHO algorithm, optimizing device schedules to fit the known building consumption.

Algorithm: Electricity Load Disaggregation

Input: B – building; TC – building's total hourly amount of energy that needs to be disaggregated; T – the timeframe for which load disaggregation will be performed; popSize – population size; maxIter – maximum number of iterations



Output: ScheduleBT – the schedule of all the devices from the building's apartment for a given timeframe T for which the best fitness value has been obtained for each hour

Comments: Ph – population of agents corresponding to the

Begin

```

1. ScheduleBT = []; it = 0
2. foreach h in T do
3.   Ph = GENERATE_INITIAL_POPULATION(B, popSize)
4.   while it < maxIter do
5.      $X_{rabbit_h}$  = SELECT_BEST_AGENT_POSITION(Ph)
6.     foreach  $X(i)_h$  in Ph do
7.       Compute the values of E0, E and J
8.       if  $|E| \geq 1$  then update  $X(i)_h$  using Formula 23
9.       else
10.        if  $|E| \geq 0.5$  and  $r \geq 0.5$  then update  $X(i)_h$  using Formula 25
11.        if  $|E| < 0.5$  and  $r \geq 0.5$  then update  $X(i)_h$  using Formula 28
12.        if  $|E| < 0.5$  and  $r < 0.5$  then update  $X(i)_h$  using Formula 29
13.        if  $|E| \geq 0.5$  and  $r < 0.5$  then update  $X(i)_h$  using Formula 31
14.      end if
15.    end foreach
16.     $X_{rabbit_h}$  = SELECT_BEST_AGENT_POSITION(Ph)
17.    it++
18.  end while
19.  Append  $X_{rabbit_h}$  to ScheduleBT
20. end foreach
21. return ScheduleBT
End

```

HHO is inspired by the collaborative hunting of Harris Hawks and belongs to the swarm-based algorithms category. The hunting method of Harris Hawks is interesting due to their ability to gather the members of the same family as a swarm to hunt for food. The ritual starts by dispersing onto high trees within the realm and occasionally diving close to ground level to search for hiding prey (e.g. rabbit). When the rabbit is localized they will coordinate to exhaust and confuse it by performing "surprise pounces". That means that hunters will cooperate to attack the rabbit from different angles

In our approach, the population of Harris Hawks is modelled as a set of agents with each agent having a position associated which may change from one iteration to another. The position of the agent is defined as a candidate solution to our optimization problem, i.e., a schedule of the devices from all the building's apartments. Initially, we have defined the schedule for the whole day, as defined by Formula 10, however, this representation of the solution, proved to include a number of parameters equal to the total number of devices from all apartments multiplied by the total number of hours, and with the increase in the number of apartments and devices, it would explode, leading to an extremely large search space whose processing is time consuming. To address this issue, we have reduced the search space by applying the HHO for each hour of the day. In this way, the number of parameters is reduced by a factor equal to the timeframe on which the scheduling is done. In this case the number of parameters for an hour h is equal to the number of devices from all the building's apartments. Additionally, the usage and comfort



matrices are processed by extracting the column corresponding to each hour one-by-one and integrating this information in the HHO. After the schedules for all hours are obtained, the whole schedule for the day is reconstructed. Consequently, the position of a Harris Hawk is defined as a matrix with one column and a number of rows equal to the number of devices from the building's apartments, dn:

$$X(i)_h = \begin{bmatrix} \text{Schedule}_{d_0}[h] \\ \vdots \\ \text{Schedule}_{d_m}[h] \end{bmatrix} \quad (15)$$

where m is the number of devices and $\text{Schedule}_{d_i}[h]$ encodes whether the device d_i has been scheduled to be used at hour h:

$$\text{Schedule}_{d_i}[h] = \begin{cases} 1, & \text{if the device } d_i \text{ is scheduled to be used at hour } h \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

In other words, the position of a Harris Hawk represents the schedule of all devices from the building's apartments at a given hour h.

The objective function used for evaluating the position of a Harris Hawk is defined in Formula 17 and evaluates how well the corresponding schedule approximates the total energy consumption of the building at a given hour on one hand and the comfort impact ($\text{Fitness}_{\text{comfort}}$) of the solution on the other hand. The influence of each component in the fitness function can be modified through the weight parameter, w.

$$\text{Fitness}(X(i)_h) = (1 - w) * \text{Euclidean Distance}(X(i)_h) + w * \text{Fitness}_{\text{comfort}}(X(i)_h) \quad (17)$$

The fitness component *EuclideanDistance* is a linear combination between the normalized Euclidean distance between the total consumption (TC) of the building and the total consumption of the scheduled devices at a given hour h:

$$\text{EuclideanDistance}(X(i)_h) = \frac{|\text{TotalConsumption}(X(i)_h) - \text{TC}[h]|}{\text{baseline_consumption}} * 100 \quad (18)$$

where TotalConsumption computes the total energy consumed by the devices scheduled as working at hour h in $X(i)_h$, TC[h] is the building's total consumption at hour h, while baseline_consumption represents the maximum (or close to maximum) value that the Euclidean distance fitness function can take. This constant value is dataset dependent and is obtained by running the HHO with the weight parameter set to 1. Meaning that the schedule's deviation from the target curve will peak when the optimization algorithm focuses only on the comfort objective.

The fitness component $\text{Fitness}_{\text{comfort}}$ is defined as the sum of the values given by an exponential function for each time devices are scheduled outside the comfort intervals as shown in Formula 19.

$$\text{Fitness}_{\text{comfort}}(X(i)_h) = \frac{\sum_{i=0}^n \sum_{d=0}^{dn_i} P^{CVP_{d_i}[h]-1}}{\text{baseline_comfort}} * 100 \quad (19)$$

Where n is the number of apartments in the building, dn_i is the number of devices from the apartment with the id=i, P is a configuration parameter representing the base of the exponential function, The greater the base is, the more penalty will be given for even a slight violation of the comfort interval. The baseline_comfort represents the maximum (or close to maximum) discomfort tenants can experience. The value is computed for a dataset by running the HHO with w set to 0, thus taking into consideration only the approximation objective.



The ELD algorithm generates the hourly schedules of the devices from the building's apartments, by running the HHO algorithm for each hour of the desired timeframe. For a given hour h , the HHO algorithm starts by randomly initializing the position of each agent in the population (see line 3). The random initialization of an agent's position implies that each device considered in the schedule will be randomly set as working (i.e. value 1 in the corresponding element from the position) or stopped (i.e. value 0) in the corresponding element from the position (see Formula 16).

Next, in the iterative stage (lines 4-18), each agent will be evaluated using the fitness function defined in Formula 17, and the best position will be identified and set as the position of the rabbit X_{rabbit_h} (line 5). Afterwards, the agents will be repositioned in the search space through exploration and exploitation strategies aiming to minimize the fitness of their position (i.e. solution). The transition from exploration to exploitation is assured by the variable E , called the energy of the rabbit. Initially, when the energy variable is large, exploration is done. Over time, as iterations pass, the energy will decrease, and a transition toward exploitation is obtained. The transition is modelled by the following function:

$$E = 2 * E_0 * (1 - \frac{iter}{maxIter}) \quad (20)$$

where $maxIter$ is the total number of iterations, $iter$ is the current iteration number, while E_0 randomly changes at each iteration in the range $(-1, 1)$ according to Formula 21. One can observe, that as time passes, the random energy is multiplied with smaller and smaller values, which will lead to the gradual decrease of the energy.

$$E_0 = 2 * randomNr - 1 \quad (21)$$

The jump strength of the rabbit is another parameter that is computed in each HHO iteration, using the formula below, where $randomNr$ is a randomly chosen number in the interval $(0, 1)$:

$$J = 2 * (1 - randomNr) \quad (22)$$

In the exploration phase (line 8), the search agents discover new positions within the search space hoping to find the global optimum. Two strategies are applied depending on a random variable with equal chances. The two strategies consist of the agents moving toward another agent selected at random or moving toward the best solution found yet. This is done to explore the search space as much as possible and to avoid being stuck in a local optimum. Through the extensive search, they ensure that when the transition to the next phase is made some of the agents will discover or are close to the ideal solution. This behaviour can be described by the equation:

$$X(i+1)_h = \begin{cases} X_{rand}(i) - r_1 * |X_{rand}(i) - 2 * r_2 * X(i)_h|, & q \gg 0.5 \\ (X_{rabbit}(i) - X_m(i)) - r_3 * (LB + r_4 * (UB - LB)), & q < 0.5 \end{cases} \quad (23)$$

where X_{rand} is the position of the randomly selected agent, X_{rabbit} is the position of the agent with a solution resulting in the best fitness, $X(i)_h$ is the position of the current agent in the current step and X_m is the average position of the search agents. Random variables improve the diversity of the model. Despite that the current agent is moved into the vicinity of the selected random agent, the randomness ensures that new space is explored around them. In case it moves towards the best solution found yet, the algorithm first computes the mean of the population. The mean will be the average position of the agents (the average of their optimization parameters at each position). This way the new position of the agent will be around the best



position but shifted in the direction where most of the agents are. This way the search agents will move as a conglomerate towards the best agent.

As the number of epochs increases the energy value decreases. This triggers the exploitation phase (lines 10-13), in which the best solutions found are refined, rather than generating new solutions. Several cases are considered in this phase: the first category of cases is given by the “behaviour” of the target solution. Depending on a random number (r) the best solution can be approached directly by other search agents or by simulating a random movement. The other set of cases is given by the energy value ($|E|$): the values of the optimization parameters of an agent can be computed based only on the position of the agent containing the best solution found so far (target agent), or by moving the target agent in a random direction. As a result of these scenarios, we will have 4 cases:

1. Soft besiege: $|E| > 0.5$, $r > 0.5$ (line 10). In this case, the so-called target still has plenty of energy but is not able to dodge the attacks. The new solution of the current agent is computed as the parameter-wise difference between the best and current agent, from which is subtracted a value modelling a sudden random movement, damped by the energy factor, see Formula 24, where the difference between the solutions is computed as shown in Formula 25.

$$\Delta X(t) = X_{rabbit}(t) - X(t) \quad (24)$$

$$X(t+1) = \Delta X(t) - E * |J * X_{rabbit}(t) - X(t)| \quad (25)$$

where the strength of the random movement J is computed based on Formula 22. On one hand, the random movement ensures that a larger space is covered during the search. On the other hand, it also guarantees the agent to escape from local optima.

2. Soft besiege with progressive rapid dives: $|E| \geq 0.5$, $r < 0.5$ (line 13). In this strategy, the agents will get close to the best solution found so far by doing one or two types of random movements, depending on which yields a solution with better fitness. To model the movement of the agent trying to reduce the distance from the target agent the levy flight concept is used (random walk, where the steps are based on lengths that follow a probability distribution). The next steps of the attackers are made by deciding to follow a path described by the Formula 27 or adding to that movement the LF-based leapfrog movement. The decision is taken based on which next step will reduce the fitness of the current position as seen in Formula 28. The function Z will give the next step containing a zig-zag movement as seen in Formula 26, where S is a random vector of dimension $1 \times D$ (where D is the number of optimization parameters), and LF is the levy flight function.

$$Z = Y + S * X * LF(D) \quad (26)$$

$$Y = X_{rabbit}(t) - E * |J * X_{rabbit}(t) - X(t)| \quad (27)$$

$$X(t+1) = \begin{cases} Y, & \text{if } F(Y) < F(X(t)) \\ Z, & \text{if } F(Z) < F(X(t)) \end{cases} \quad (28)$$

3. Hard besiege with progressive rapid dives: $|E| < 0.5$, $r < 0.5$ (line 12). This is almost the last stage of approaching the best solution. In this phase, search agents will reduce their average distance from the target agent. This ensures that the population will start forming a cluster by moving towards each other and moving towards the agent having the best solution. Formula 29 is used to compute the new position, as in the case of the soft besiege with progressive rapid dives, however the Y component is computed as an average position of the whole population in this case – see Formula 30.



$$X(t+1) = \begin{cases} Y, & \text{if } F(Y) < F(X(t)) \\ Z, & \text{if } F(Z) < F(X(t)) \end{cases} \quad (29)$$

$$Y = X_{rabbit}(t) - E|JX_{rabbit}(t) - X_m(t)| \quad (30)$$

4. Hard besiege: $|E| < 0.5$, $r \geq 0.5$ (line 11). This strategy is employed when the energy level is low. The new solution is computed only based on the distance of the parameters of the best solution found yet and a new position resulting from the distance of the current solution and the best solution damped by the energy factor. This solution will give a solution very close to the best solution. Thus, exploring the search space near to the best solution. Formula 31 describes this movement.

$$X(t+1) = X_{rabbit}(t) - E|\Delta X(t)| \quad (31)$$

Using the disaggregated energy consumption profiles for each apartment, provided by the ELD Algorithm, the DRAG Algorithm identifies the minimum number of apartments that need to adjust their consumption in response to a DR action, such as reducing the building's total energy usage by a specified percentage. We model this task as a bi-level optimization problem, where the objective is to find the optimal solution to an upper-level problem, constrained by the optimal solution of a lower-level problem. Solving such problems is particularly challenging, as they are NP-hard and computationally intensive; each evaluation of an upper-level solution requires solving an optimization problem at the lower level. The DRAG Algorithm integrates the genetic algorithm to solve the upper-level problem and the HHO algorithm to solve the lower-level problem, like the ELD algorithm. Given these considerations, an individual of the genetic algorithm is mapped to a candidate solution of the upper-level optimization problem, has a compound structure including a binary array encoding the apartments selected to be involved in the DR action and the schedule of their devices:

$$sol_{upper} = \langle x, Schedule_{BT} \rangle \quad (32)$$

where x is a binary array with n elements, each element in the binary array corresponding to an apartment from the building:

$$x = [x_1, x_2, \dots, x_n] \quad (33)$$

$$x_i = \begin{cases} 1, & \text{if the apartment with the index } i \text{ has been selected for demand response} \\ 0, & \text{otherwise} \end{cases} \quad (34)$$

In Formula 32, $Schedule_{BT}$ (see Formula 11) represents the lower-level solution identified for the apartments encoded in x as being selected for the DR action.

The fitness function of the upper-level problem is defined as:

$$Fitness_{upper}(sol_{upper}) = w_1 * F_{subs}(sol_{upper}) + w_2 * F_{oversampling}(sol_{upper}) + w_3 * EuclideanDistance(sol_{upper}.Schedule_{BT}) + w_4 * Fitness_{Comfort}(sol_{upper}.Schedule_{BT}) \quad (35)$$

where:

- F_{subs} is the subset fitness computing the percentage of apartments included in the optimization:

$$F_{subs} = \frac{\sum_{i=1}^n x[i]}{n} \quad (36)$$



- $F_{over_sampling}$ gives the percentage of hours in a day when the target consumption (TC) is lower than the consumption (CNI) of devices not selected to be part of the DR program (see Formula 37). Since devices from the apartments that were not selected still contribute to the total consumption of the building according to their initial usage data, this consumption must be subtracted from the target curve. This means that the lower-level optimization algorithm must obtain a solution that fills the gap between the consumption given by devices that will not be scheduled, and the target curve by using the devices from the selected apartments. In case the gap is negative (i.e. with the current subset there is no way of achieving the target consumption curve) the subset algorithm has to be penalized. This penalization is quantified by the over-sampling fitness.

$$F_{over_sampling}(sol_{upper}) = 100 * \frac{\sum_{h=0}^T isOverConsumption(sol_{upper})[h]}{24} \quad (37)$$

where:

$$isOverConsumption(sol_{upper})[h] = \begin{cases} 0, & TC[h] \geq CNI[h] \\ 1, & otherwise \end{cases} \quad (38)$$

- *EuclideanDistance* is defined similar to Formula 18, with the difference that it evaluates the Euclidean distance considering the devices from the apartments selected in x over the given timeframe T.
- *FitnessComfort* is defined similar to Formula 19, with the difference that it evaluates the discomfort generated by the devices from the apartments selected in x over the given timeframe T.

Besides the building information, target energy consumption as computed based on the DR action, timeframe and genetic algorithm and HHO algorithm specific parameters, the DRAG Algorithm takes as input the schedules (i.e. estimated consumption profiles) of all devices from the building's apartments returned by the ELD algorithm. These profiles will represent the residents' preferences for using their devices and the constraints guiding the devices' rescheduling.

2.2.6.4. DT Dashboard mock-ups

This section presents the mock-ups of the DT Dashboards. The DT Dashboards can be accessed by the building manager and the apartment owners through a login interface as illustrated in Figure 14.



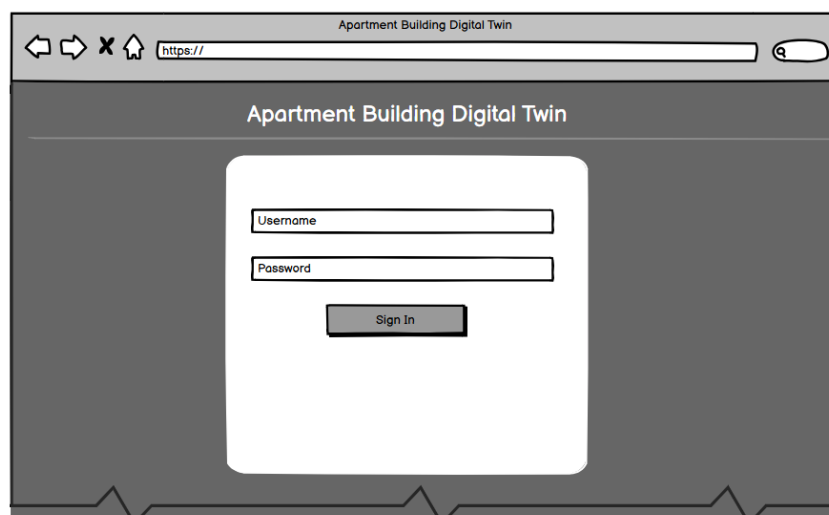


Figure 14: Pilot 7 - DT Dashboards login

a) Building manager dashboards

Once the building manager logs in the DT Dashboards, it can access the following functionalities:

- View the building's overview information (see Figure 15): number of apartments, types of apartments, the total hourly energy consumption of the building for the previous days (by default the previous day is selected in the calendar).

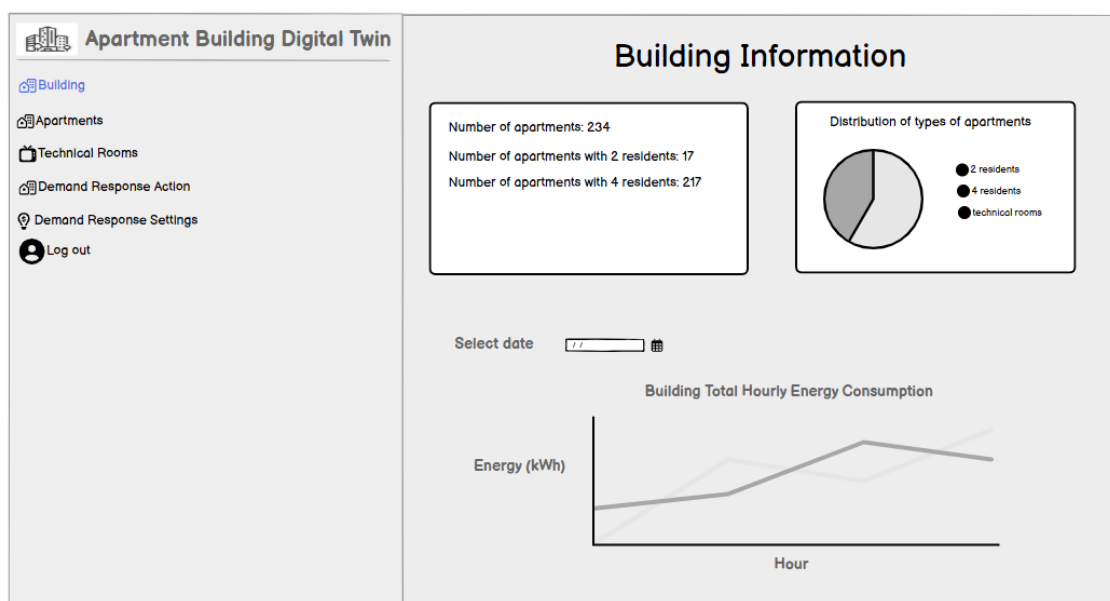


Figure 15: Pilot 7 - Building owner - Building overview dashboard

- View information about the apartments (see Figure 16): the ID and number of residents is displayed for each apartment. The building owner can view the history of estimated consumption profiles of an apartment as resulted from load disaggregation, and the



adjusted energy consumption curves proposed for the apartment when it was involved in the DR program.

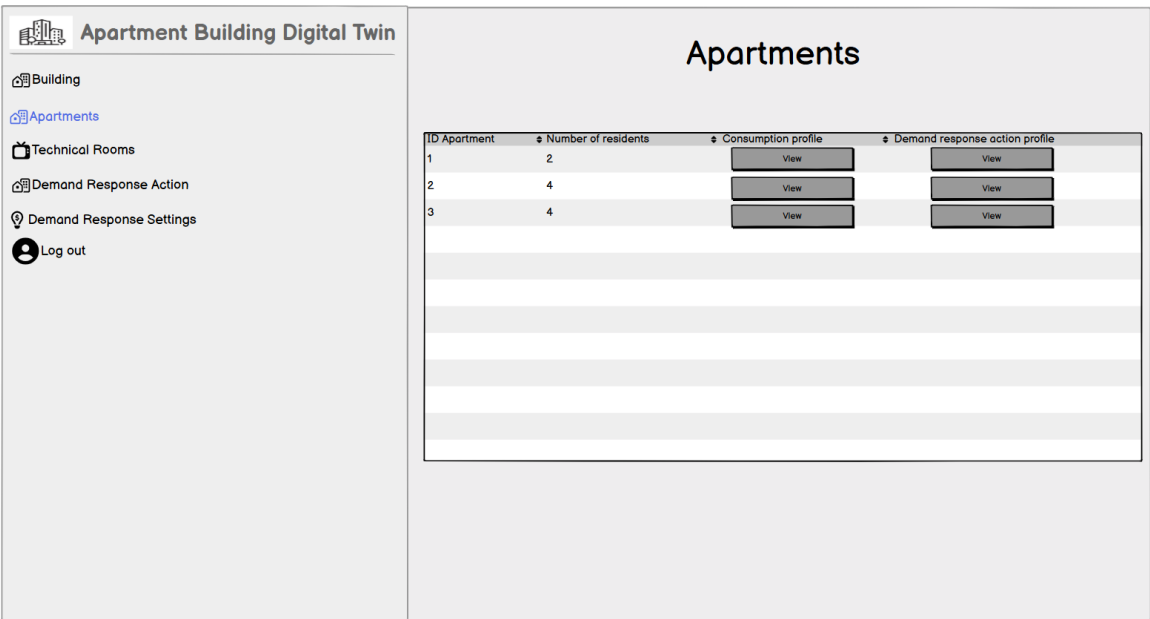


Figure 16: Pilot 7 - Building owner - Apartments dashboard

- View information about the technical rooms from the building (see Figure 17): the building owner can view the devices from the technical rooms and their energy consumption profiles.

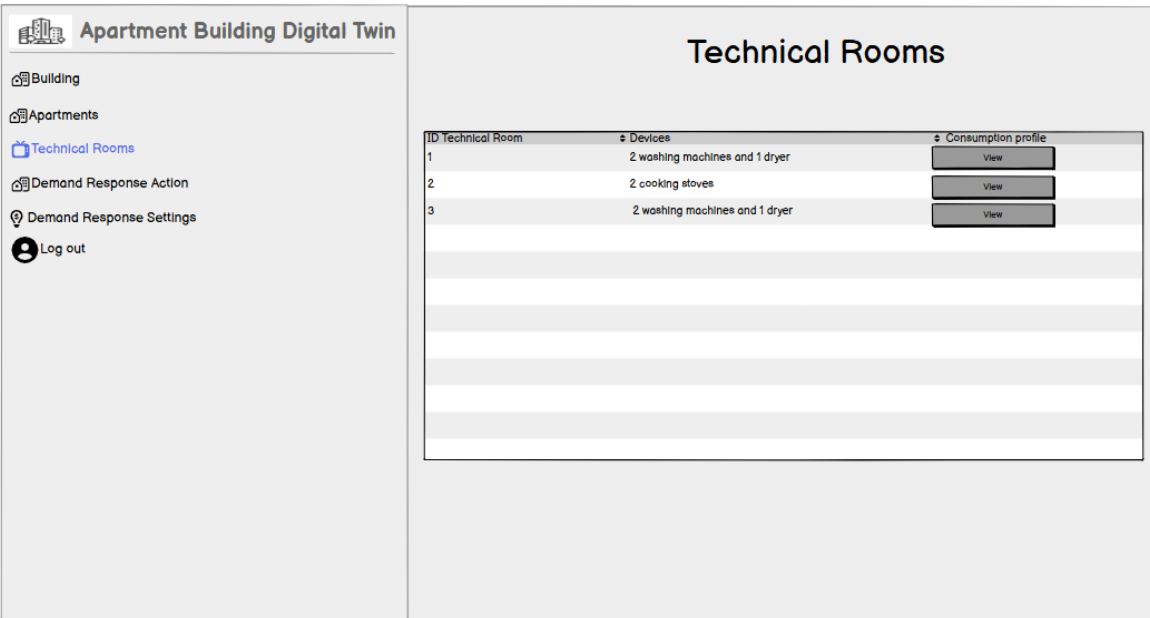


Figure 17: Pilot 7 - Building owner - Technical rooms



- View DR Action (see Figure 18): the building owner can view the target energy consumption curve for the next day and how it can be approximated by a minimum number of apartments. Additionally, the building owner can view the impact on the comfort of the selected apartments' residents.

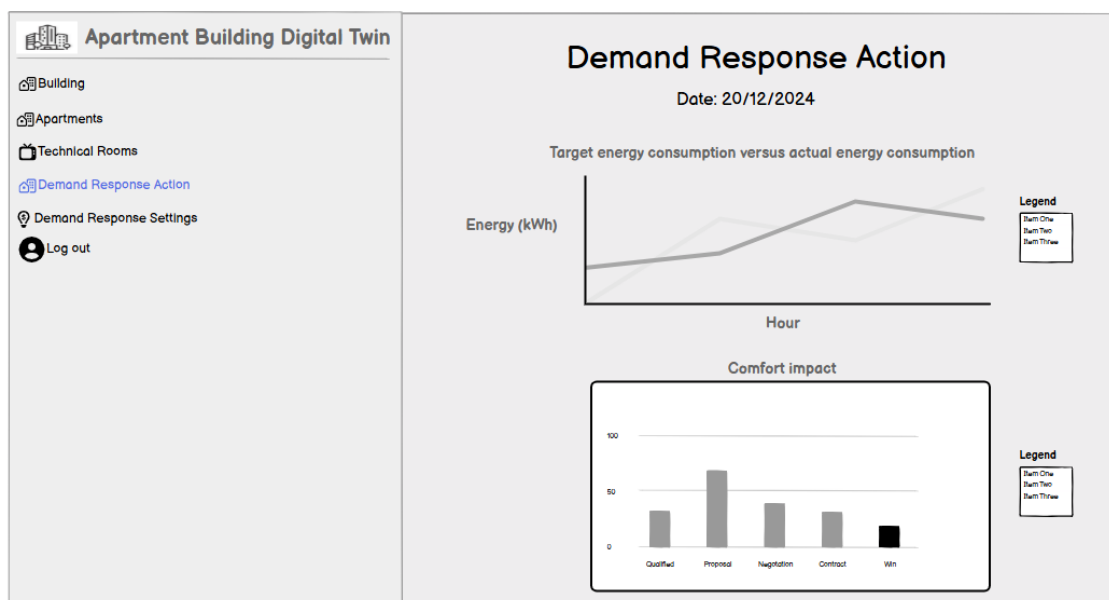


Figure 18: Pilot 7 - Building owner - DR Action

- Configure target curve for DR action (see Figure 19): the building owner can set the hourly energy consumption values for the target consumption curve as requested by the DR action.

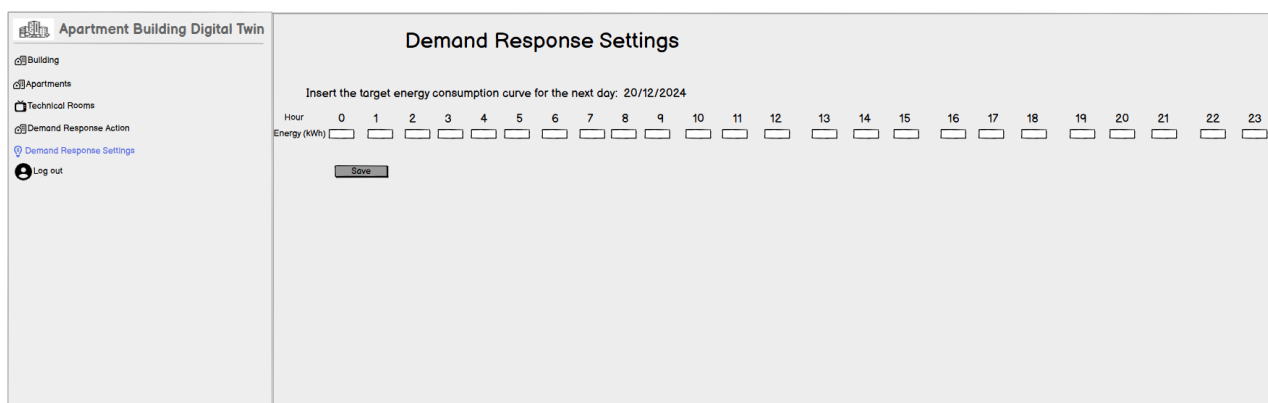


Figure 19: Pilot 7 - Building owner - DR settings

b) Apartment owner dashboards

An apartment owner has access to the following dashboards once it logs in the system:



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- View the load disaggregation for the apartment (see Figure 20): the apartment owner can view the estimated energy consumption of the apartment for the current day implicitly, however it can view for previous days as well by selecting a date from the calendar. Additionally, the apartment owner can view the load disaggregation per device, both for the current day and for previous days.

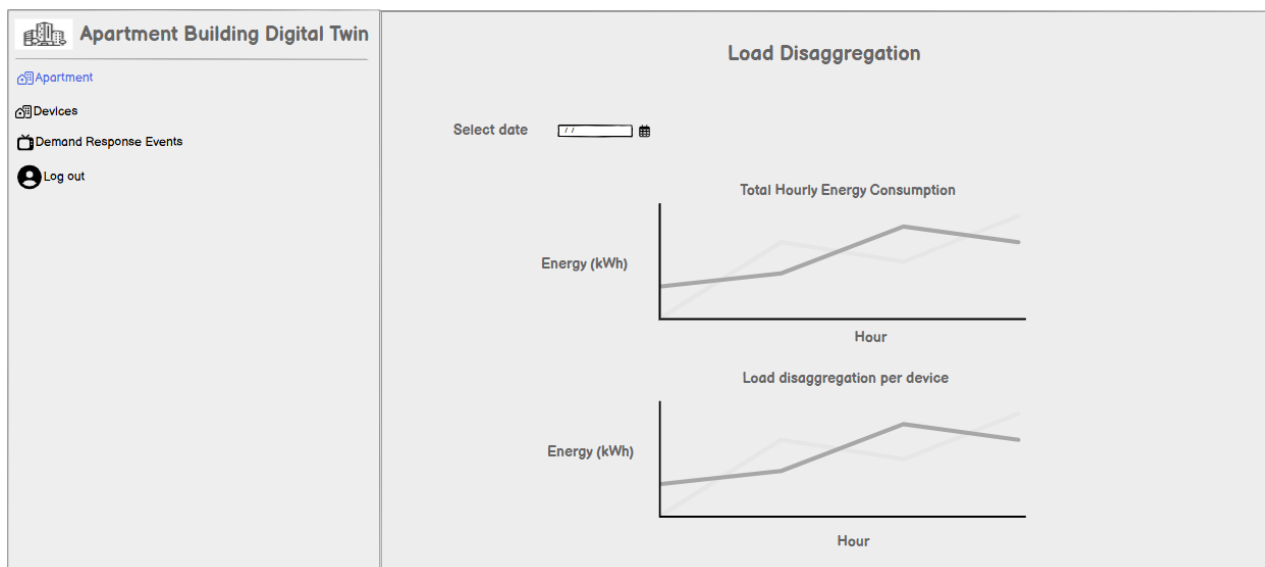


Figure 20: Pilot 7 - Apartment owner - Apartment load disaggregation

- View devices (see Figure 21): the apartment owner can view the list of the devices from its apartment. Additionally, for each device, the user can configure the preferred time intervals for using the device and the comfort of using the device during other hours except the ones in the preferred time intervals – this can be done by clicking on the Configure button corresponding to each device. Consequently, the popup in Figure 22 will be displayed. For the usage preferences the apartment owner can use checkboxes; if a checkbox is marked it means that the apartment owner prefers to use the device at that hour, otherwise it means that the apartment owner does not prefer to use the device at that hour. For the comfort the drop down lists can be used with 3 values (0, 1, 2) reflecting the level of comfort; for example, if the owner specifies that it prefers to use the device at a given time, then the comfort for that time will be set to 0, otherwise it will be set to 2; or if the apartment owner agrees to use the device an hour or two hours earlier or later than the preferred time then the comfort will be set to 1.



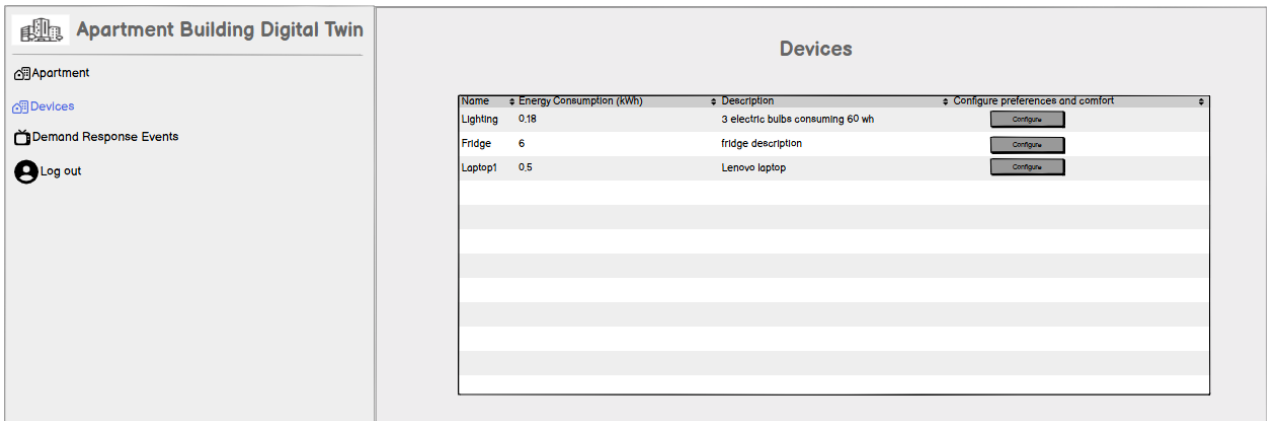


Figure 21: Pilot 7 - Apartment owner - View of devices

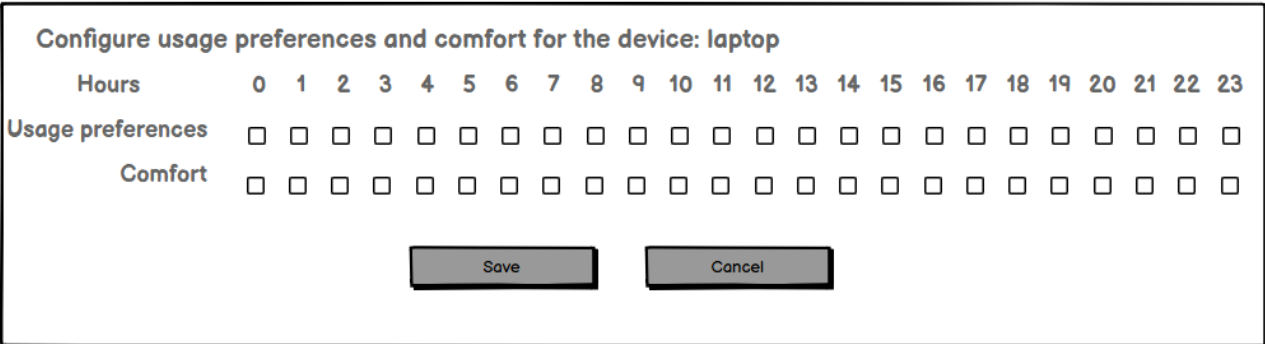


Figure 22: Pilot 7 - Apartment owner - Configure a device's usage preferences and comfort

- View DR event (see Figure 23): if the apartment has been selected to be part in the DR action, the apartment owner can view the recommended schedule for its devices for the current day.

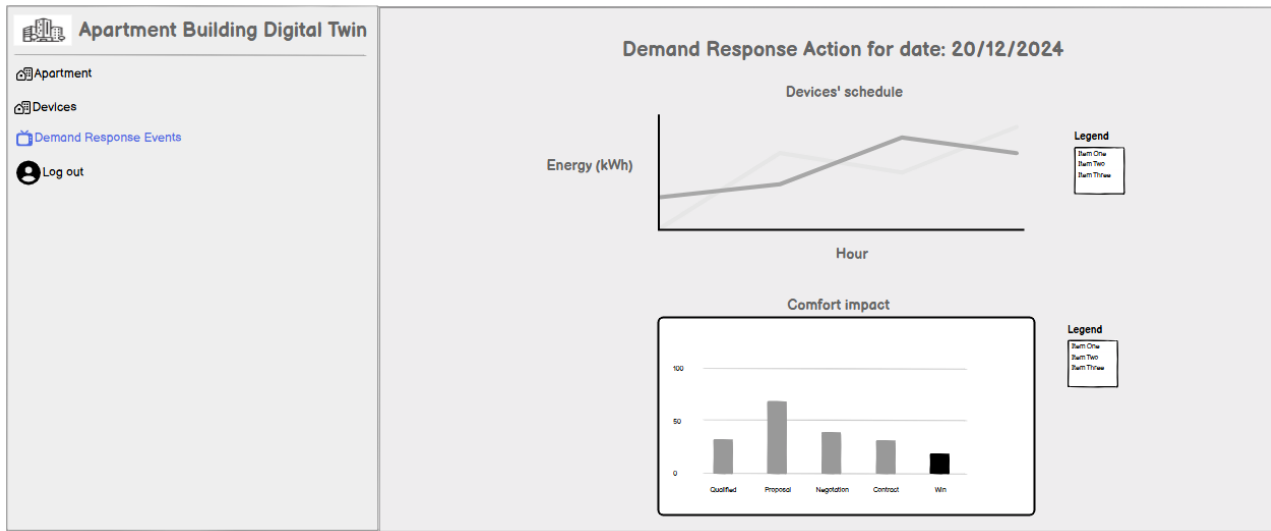


Figure 23: Pilot 7 - View of DR event



2.2.7. Outlook for future expected release

For the next steps we intend to perform the following:

- Implement and integrate the DT's dashboards.
- Finalize the implementation of the DR Optimisation Decision Making component.
- Integrate the DT with the smart contracts for DR settlement to simulate the actions enforcement.



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3. Optimal trade-off among comfort, energy and building performance

This section provides information about the development activities under the scope of task 4.2: Optimal trade-off among comfort, energy and building performance.

The TE that is involved is:

- TE-9: comfort-based flexibility model.

The technology develops a comfort-based flexibility model for DR, combining sensors to monitor indoor parameters (thermal, acoustic, visual, air quality) and weather data with advanced analysis to balance comfort, energy consumption, and building performance.

Supported by the DEDALUS IoT platform, the service integrates data on indoor conditions, energy consumption, and renewable production, enabling dynamic actions to adjust energy demand. A web tool collects user preferences and facilitates seamless communication of DR actions.

3.1. Overview of the service

The service aims to ensure comfort in indoor environments by harmonizing Indoor Environmental Quality (IEQ) factors such as thermal comfort with energy consumption and costs. Additionally, it facilitates DR services while emphasizing the preservation of health and well-being during their implementation.

This goal is achieved through the utilization of data collected from sensors installed in various buildings. To accomplish this, the service develops and applies Artificial Intelligence (AI)-based algorithms capable of quantifying and predicting comfort-related factors, such as the Thermal Sensation Vote (TSV), using building data. This prediction relies on historical data collected from the involved pilots, as well as real-time data acquired from the sensor network provided by the pilot buildings. By employing this AI-based approach, the service assists stakeholders in managing buildings more efficiently and sustainably while ensuring occupants' comfort levels. Additionally, alongside maintaining acceptable comfort levels, reducing energy consumption and costs is a primary objective of the service. Furthermore, the service provides three different solutions for DR, based on end-users preferences, aimed at facilitating the proper management of indoor building conditions. The comfort prediction generated by the developed AI model serves as feedback for end-users to adjust indoor environmental conditions based on his needs and then restore optimal comfort status. Moreover, the service collects feedback directly from end-users regarding indoor environmental and comfort conditions using survey-based approaches. This feedback is utilized to refine the AI model implementation and ensure that comfort predictions align with occupants' preferences.

The development process involves several steps:

- Definition of user requirements and technical specifications (step 0): this initial phase serves to map user requirements and technical specifications. Technical actions required



for service application in the building are determined. Specifications related to installing missing data sources (e.g., sensors) are provided.

- Assessment model for comfort-based flexibility (step 1): this phase involves analysing the status of the building to benchmark pilot building features and verify the availability of necessary data sources and technologies. This assessment informs the strategy and methodologies for service implementation, considering factors such as energy efficiency, digitalization level, structural characteristics, and existing technologies/sensors.
- Interoperability and quality (step 2): upon gathering necessary information about the building and available data sources, the collected data is integrated into the service platform to ensure seamless compatibility and high data quality.
- Baseline monitoring (step 3): this phase involves tracking energy consumption and comfort parameters over historical data to establish a baseline for evaluating building performance. Data analysis prepares and harmonizes input datasets for AI model training.
- Profiling and identification of anomalies (step 4): analysed data is utilized to create profiles of energy consumption and indoor environmental conditions, enabling the identification of anomalies. This step enhances understanding of building dynamics and supports targeted interventions for improved performance.
- Integration of service during DR events (step 5): during DR events, the service seamlessly integrates with building operations to implement comfort-driven actions. This ensures that energy demand is dynamically reshaped while preserving occupant comfort and well-being.
- Delivery of coaching solution for comfort-based flexibility (step 6): the final step involves delivering three different solutions for DR aimed at enhancing comfort and well-being. Leveraging insights from previous steps, an AI-based model is trained and tested with real-time data to provide personalized recommendations. Additionally, user feedback gathered through surveys informs model refinement, ensuring ongoing optimization and user satisfaction.

A dedicated interface within the Digital Twin will be provided to allow the visualization of the AI-model results concerning comfort quantification and to monitor the consumption condition of the building. Additionally, a dedicated tool will be used to collect end-users' preferences and their opinions on indoor comfort conditions. The input datasets will include building's occupants feedback, environmental parameters, and energy consumption data. The service will be implemented through the Python programming language. Furthermore, the mentioned dashboard will also present KPIs related to the service. Hence, all the necessary data for their calculation will be collected.



Table 9 shows the functional specification of the service mapped with FR and Use cases defined in D2.1 and D2.3.

Table 9: Task4.2 - Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
The service provides customised energy-saving recommendations for the building, prioritising occupants' comfort preferences.	TE9-FR1	UC4	TE-9	Pilots 2, 3, 4, 5	Ongoing

UNIVPM outlined the necessary software components and visualization tools essential for implementing the service within pilot scenarios. The comfort model considered is the sPMV [9].

The sPMV utilize environmental parameters such as indoor temperature, relative humidity, and occupants feedback as key inputs. UNIVPM chose to employ a combination of advanced algorithms to address the problem. A Long Short-Term Memory (LSTM) network is utilized to forecast temperature and relative humidity for the following day by leveraging historical data. These predicted values are then used to calculate thermal comfort. Machine Learning (ML) techniques are employed to predict user feedback in terms of TSV. The ML model is trained on historical data, including temperature, relative humidity, alongside user-provided TSV ratings. Once trained, this model allows for the automatic prediction of TSV without requiring further user feedback.

In addition to these software components, the service implementation includes the development of a tool to collect occupants feedback on TSV, and a dashboard serving as a graphical user interface (GUI).

This dashboard is seamlessly integrated into the Digital Twin (Task 4.1), providing an overview of indoor environmental conditions and energy consumption, incorporating outputs from the implemented AI/ML-based models for enhanced monitoring and analysis.



3.2. Architectural diagram

The comfort-based flexibility model aims to achieve an optimal trade-off among comfort, energy efficiency, and building performance. Figure 24 provides the conceptualised service architecture that will be integrated in the DEDALUS architecture.

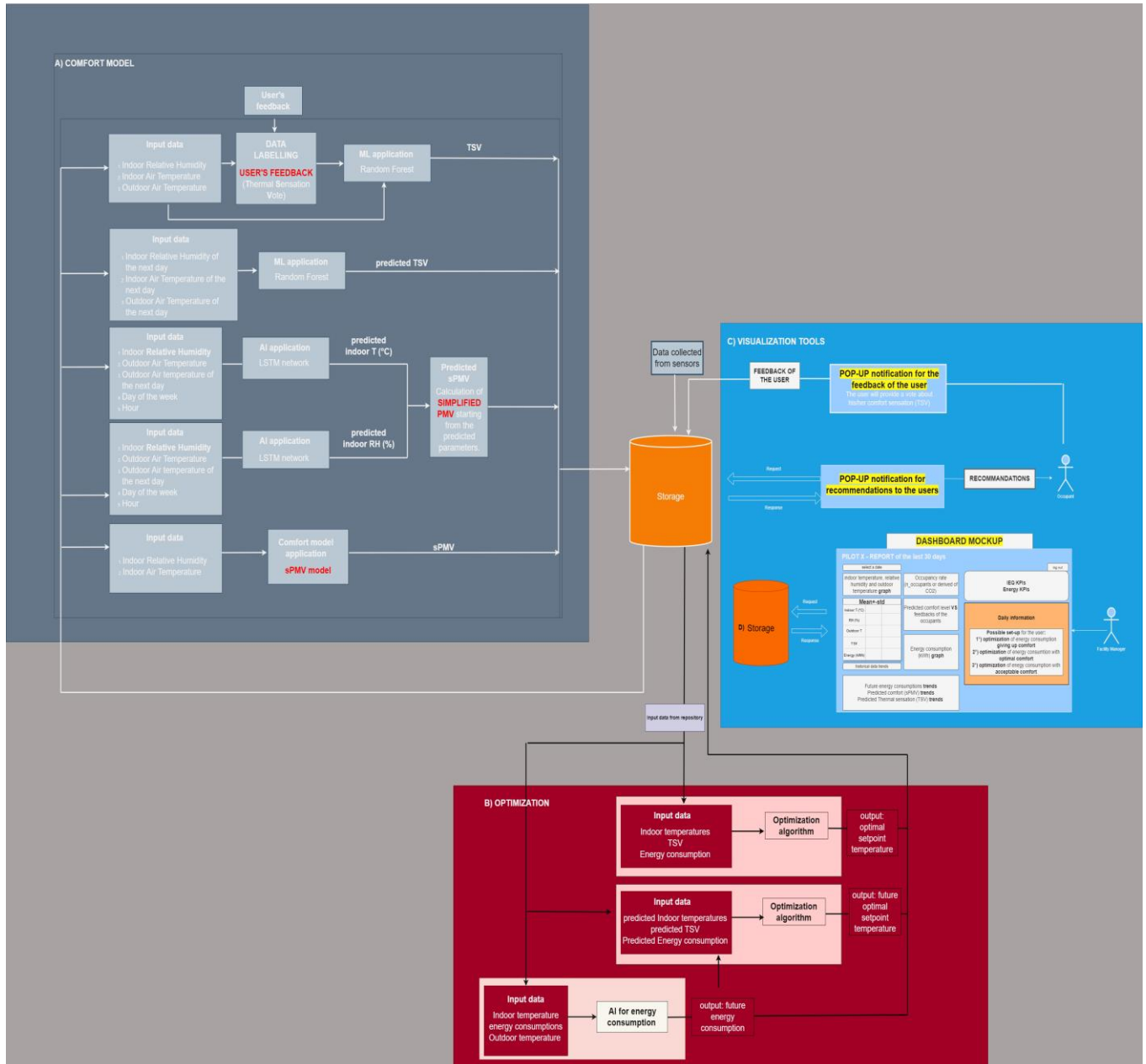


Figure 24: Comfort model - Architectural diagram of the service

The proposed "Comfort model", section A, retrieves the necessary information from a local business repository, identified as 'Storage,' to ensure its proper functioning. The storage contains all the environmental parameters collected from sensors installed in various pilots. Additionally, the model takes as input users feedback TSV obtained from an online repository. Finally, within



the model, there is a prediction component for both TSV and sPMV, using an LSTM model, which is essential for defining possible setups for the following day.

The section C provides an overview of the visualization interface integrated into the technical architecture of the service.

The first interface is designed to collect user's feedback on their thermal sensation vote. This input is crucial as it will later serve as a data input for the comfort model, aiding refine the comfort model.

The second interface is a dashboard accessible to the user, displaying all relevant information, such as indoor temperature, outdoor temperature, relative humidity, and energy consumption, along with their respective trends. Similarly, it also shows forecasted values for the next day.

The third interface presents three possible setups that users can choose to adopt based on their needs. The proposed setups are as follows:

1. Minimization of energy consumption at the expense of comfort.
2. Minimization of energy consumption with optimal comfort.
3. Minimization of energy consumption with acceptable comfort.

Users can select the solution that best suits their preferences and requirements. This dashboard will be integrated within the Digital Twin, provided by Task 4.1.

The final component highlighted in section B Figure 24 focuses on developing an optimization algorithm that considers forecasted energy consumption and comfort levels. Specifically, the goal is to determine three optimal temperatures and relative humidity levels that minimize energy consumption while maintaining a certain level of comfort, allowing the user to choose the one that best suits their needs. This component includes sections dedicated to energy consumption and energy forecasting that are provided by NTUA.

3.3. Data for service execution

The successful execution of the comfort-based flexibility model relies on the input data collected from the buildings involved in different pilots. These inputs play a crucial role in shaping the model's understanding of IEQ and energy consumption dynamics. The primary data sources include:

- IEQ-related parameters: sensors and systems are integrated to measure various indoor comfort parameters, including thermal, visual, and air quality factors. Air temperature, relative humidity, CO₂ concentration, and lux levels are collected from the buildings. Additionally, data from a weather station is collected to provide contextual information and thermal sensation vote is collected from tenants.
- Energy consumption metrics: data collection involves monitoring energy consumption factors such as metering, sub-metering, and smart plugs. This encompasses information on heating systems, lighting systems, and electrical systems within the buildings. Real-time data on on-site renewables production is also gathered. Understanding energy consumption patterns is essential for optimizing comfort while minimizing energy usage. These data are provided by NTUA.



3.3.1. Questionnaire for TSV collection

The TSV represents the subjective evaluation expressed by an individual regarding their perception of the surrounding environment based on personal preference [10]. Similar to the Predicted Mean Vote (PMV), the TSV is measured on a scale ranging from -3 (very cold) to +3 (very hot). To collect user feedback, a questionnaire is sent at hourly intervals via a notification on the Telegram app (Figure 25). By accessing the provided link, users are redirected to a web page where they can complete the questionnaire by entering their location information and providing their vote.

In the Italian pilot, specific measures were adopted to simplify the questionnaire for the target audience. Considering the tenants were primarily older adults, the questionnaire was designed to be more intuitive, with efforts made to minimize ambiguity and ensure ease of use during the filling process. In Figure 26 is reported the simplified questionnaire.

In the other pilots, the questionnaire provided is reported in Figure 27.

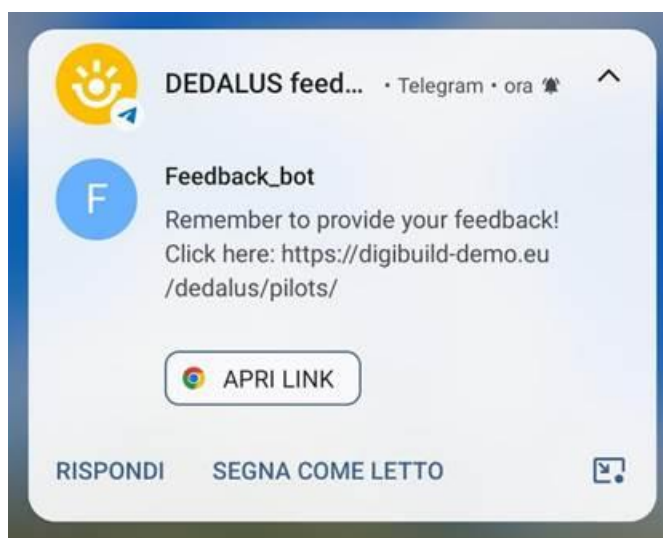


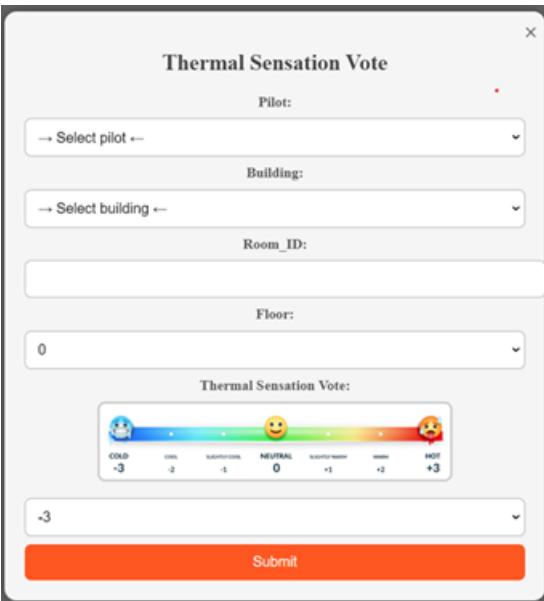
Figure 25: Notification from Telegram





The image shows a web-based questionnaire interface for the Italian pilot. At the top is the 'dedalus' logo. Below it, there are two dropdown menus: 'EDIFICIO:' with 'Casa Maddalena' selected, and 'APPARTAMENTO:' with 'E145' selected. Underneath these is the instruction 'INSERISCI LA TUA SENSAZIONE TERMICA:'. This is followed by seven blue buttons representing thermal sensation levels: 'MOLTO FREDDO', 'FREDDO', 'POCO FREDDO', 'ACCETTABILE', 'POCO CALDO', 'CALDO', and 'MOLTO CALDO'. At the bottom is a green 'INVIA' button.

Figure 26: Simplified questionnaire for the Italian pilot.



The image shows a 'Thermal Sensation Vote' questionnaire. It includes dropdown menus for 'Pilot:' (with a placeholder '→ Select pilot ←'), 'Building:' (with a placeholder '→ Select building ←'), 'Room_ID:', and 'Floor:' (with '0' selected). Below these is a 'Thermal Sensation Vote:' section featuring a horizontal scale from -3 to +3. The scale uses icons: a blue face for -3 (COLD), a blue face for -2 (COOL), a neutral face for 0 (NEUTRAL), a yellow face for +1 (SLIGHTLY WARM), a red face for +2 (WARM), and a red face for +3 (HOT). Below the scale is a dropdown menu showing '-3'. At the bottom is an orange 'Submit' button.

Figure 27: Questionnaire provided to pilots

3.4. Service output

The output of the service encompasses multifaceted contributions to both building management and grid operation. Through meticulous data analysis and leveraging insights from a range of



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sensors including environmental parameters and energy metrics, these services formulate actionable DR strategies. These strategies are tailored to dynamically adjust building energy demand while upholding optimal comfort levels for occupants. The systematic orchestration of DR decisions within the service framework ensures the efficient dispatch of flexibility actions to meet the requirements of grid operators or aggregators. This seamless integration enables buildings to actively participate in DR programs, thereby bolstering grid stability and sustainability while safeguarding occupant comfort and well-being. Furthermore, the outcomes of these service seamlessly integrate into the Digital Twin, providing a comprehensive platform for monitoring indoor environmental conditions, comfort levels, energy consumption, and relevant KPIs.

3.4.1. Concept for demand response strategy

This paragraph explains the preliminary draft of the concept adopted to facilitate the selection of the optimal trade-off solution between comfort and energy, in the context of DR.

For the correct functioning of the service, two components are required: forecasted indoor environmental data for the next day and predicted energy consumption data for the same period.

The predicted energy consumption data is directly obtained from the service provided by Task 3.2, which supplies hourly ranges of predicted energy consumption, considering flexibility.

By using these data as input to the comfort model and to the optimization algorithm, the output will consist of three solutions, each representing an internal setup (temperature, relative humidity) to achieve the optimal trade-off between comfort and energy. These solutions are



provided in hourly intervals, allowing the manager/the user to select the optimal configuration based on user needs and requirements from the grid or the Distribution System Operator (DSO). This service, therefore, outputs three distinct internal setups aimed at enhancing indoor comfort while considering energy flexibility, thereby achieving the optimal trade-off between energy and comfort.

Figure 28 illustrates the concept of how the service will operate from the initial acquisition of data till the choice of the best solution.

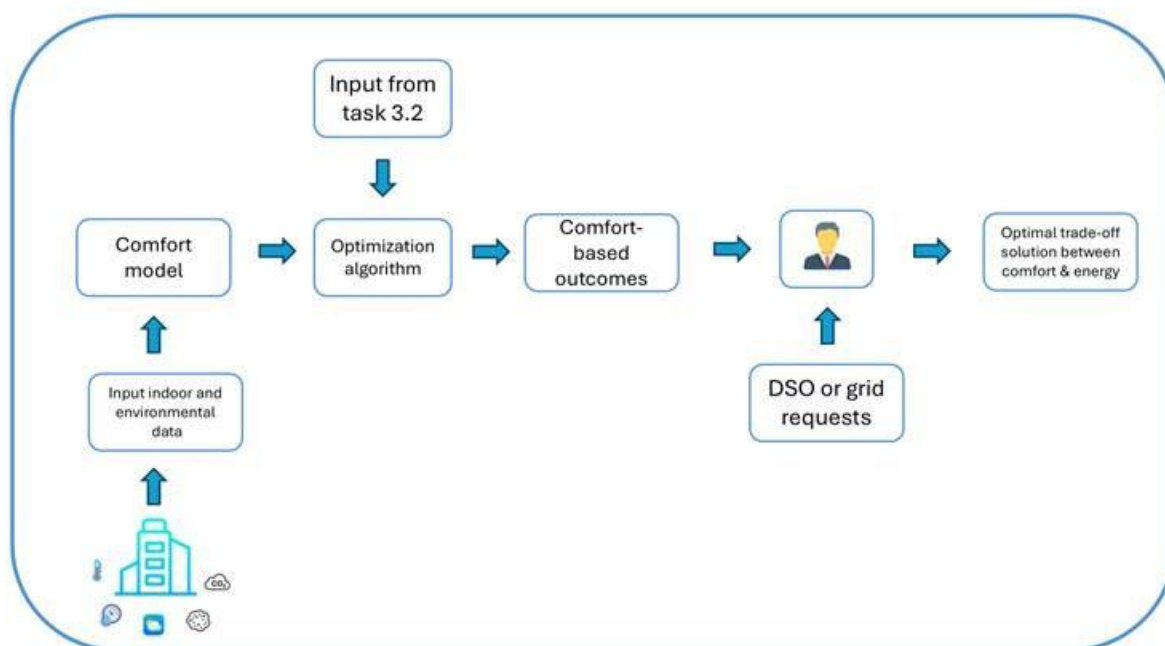


Figure 28: Comfort model - Concept for demand and response strategy

3.5. Main benefits and innovation

The main benefits of the service can be summarised as follows:

- Enhanced comfort: the service aims to provide a personalized comfort-based model, leveraging advanced learning techniques and considering user behaviour and profiles.



This ensures that comfort levels are optimized while balancing energy consumption and overall building performance.

- **Energy efficiency:** by dynamically reshaping energy demand based on comfort-driven actions, the service contributes to improved energy efficiency in buildings.
- **Health and well-being:** the emphasis on preserving health and well-being during the application of DR services ensures that occupants remain comfortable and healthy throughout the process.
- **Robust data collection:** the service ensures robust data collection, including indoor conditions monitoring, energy consumption metrics, and real-time data on on-site renewables production.
- **Seamless communication:** the Digital Twin developed in Task 4.1 facilitates seamless communication of flexibility actions from the DSO/Aggregator to end-users, enhancing the user experience and ensuring effective implementation of DR measures.

The innovative aspects enclosed in the service touch different dimensions of digital service development. From a technical perspective, leveraging ML and AI techniques to classify and predict comfort, specifically in terms of sPMV, distinguishes the service from the state of the art. Additionally, the pivotal role of the user in its development and application represents another important novelty. In many cases of comfort assessment, user feedback is sought for a direct but subjective evaluation of thermal sensation, yet usually without considering a coaching solution. In this context, users' feedback is crucial for implementing ML and AI algorithms effectively. Furthermore, the significant role of the end-user is based on the recommendations that should be delivered to actively adjust the environmental surroundings and maintain or guarantee a proper comfort level.

- **Personalized comfort-based model:** the service innovatively crafts a personalized comfort-based model, considering user behaviour and profiles, to optimize comfort levels while minimizing energy consumption.
- **Integration of sensors and systems:** the seamless integration of sensors and systems designed to measure various indoor comfort parameters ensures comprehensive data collection and analysis.
- **Utilization of advanced learning techniques:** advanced learning techniques are employed to create the personalized comfort-based model, contributing to its effectiveness and adaptability.
- **Comprehensive approach:** the service takes a comprehensive approach, considering various factors such as user preferences, indoor conditions, energy consumption, and real-time data, to orchestrate the optimal dispatch of flexibility actions for DR.

Overall, the service resulting from Task 4.2 represents an innovative approach to integrating comfort considerations into DR services, with a focus on enhancing energy efficiency and overall well-being.

3.6. Development activity

This paragraph outlines the progress achieved through activities conducted from M13 to M18. During these months, data from the Italian and Danish pilot were extracted. Subsequently,



following a data pre-processing phase, the initial parameters necessary for the proper functioning of the comfort flexibility model were calculated. This process resulted in the calculation of two key parameters: sPMV and sPMV predicted. Most of the efforts during this period, however, were dedicated to designing the architecture of the service.

3.6.1. Pilot 2 – Denmark

The data collected from the Danish pilot focus on environmental variables from a single building: Thyrasvej. Figure 29 illustrates the sensors installed in each apartment, the room IDs within the building, and the data availability status. Specifically, the data pertain to the Danish pilot, 'Building Thyrasvej,' with green sections indicating data currently available and red sections representing data that is either unavailable or yet to be provided.

ROOM ID	DATA AVAILABLE				
STARTING ACQUISITION FROM 01/07/2024					27 APARTMENTS
SENSOR	T	RH	CO2	T OUTDOOR	TSV
VALUE	°C	%	ppm	°C	
24929					
24930					
24931					
24932					
24933					
24934					
24935					
24936					
24937					
24938					
24939					
24940					
24941					
24942					
24943					
24947					
24948					
24950					
24951					
24953					
24956					
24982					
24989					
24997					
24998					
25016					
25475					

Figure 29: Pilot 2 - Data available "Building Thyrasvej"



The data were pre-processed by creating a data frame in which all available environmental parameters required by the comfort flexibility model converge.

Figure 30, Figure 31 and Figure 32 show the environmental parameters of temperature (T), Relative Humidity (RH), CO₂ concentration in a sample room. The same procedure was applied to the rooms of the other buildings.

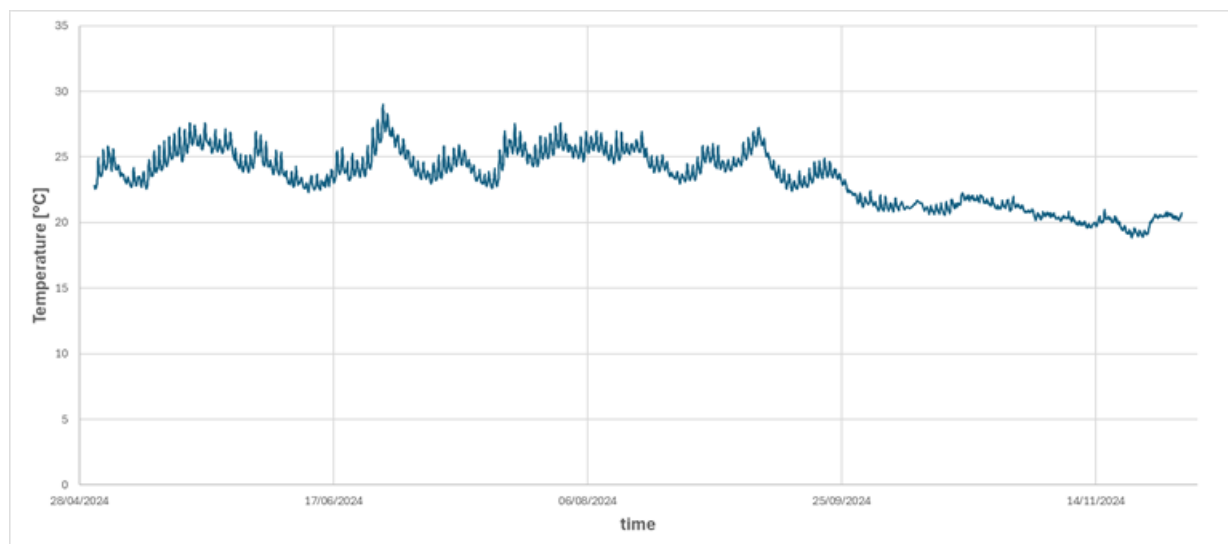


Figure 30: Trend of historical temperature data for room 24932 from 01/05/2024 to 30/11/2024

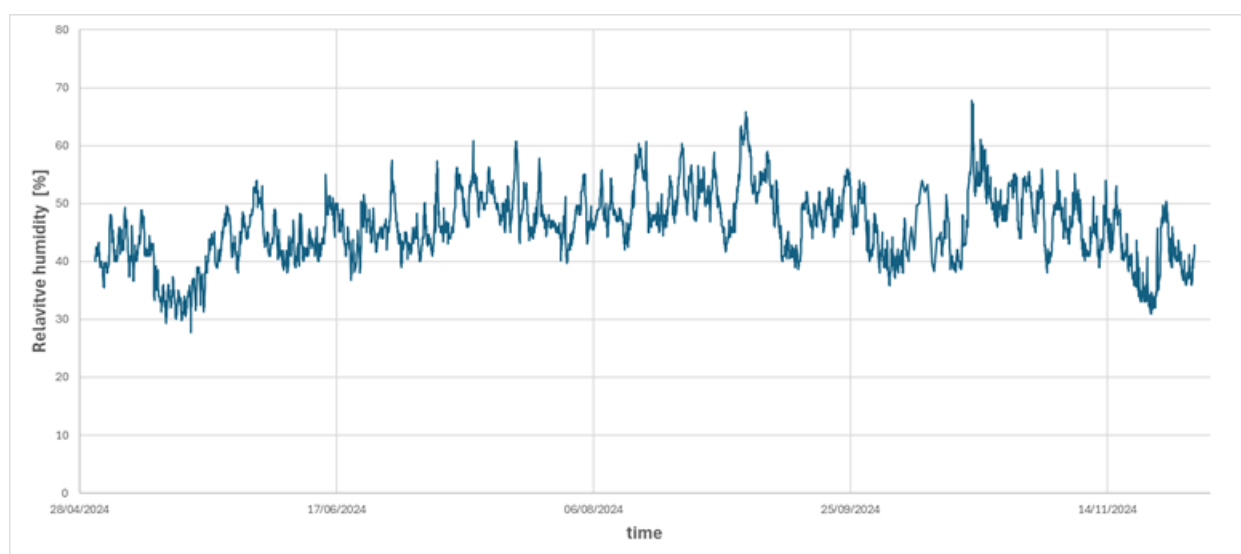


Figure 31: Trend of historical relative humidity data for room 24932 from 01/05/2024 to 30/11/2024.



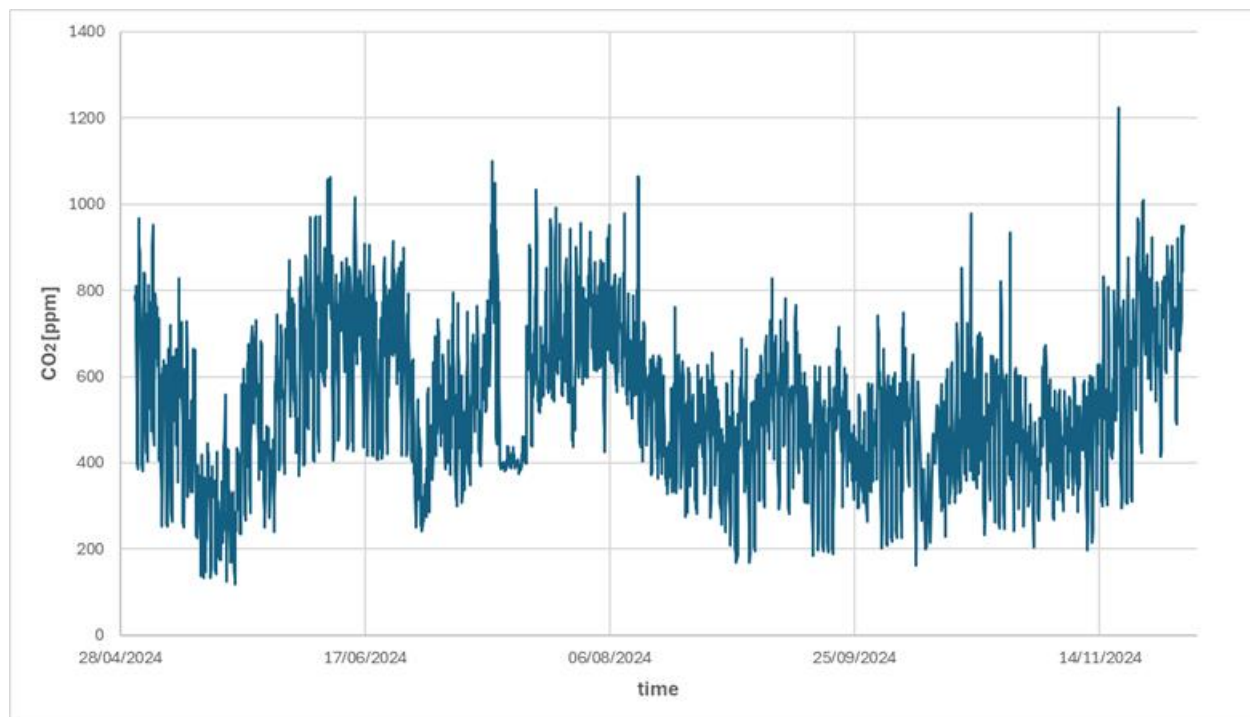


Figure 32: Trend of historical CO2 data for room 24932 from 01/05/2024 to 30/11/2024.

These data, after being pre-processed, were used to calculate the thermal comfort (sPMV), as shown in Figure 33. Additionally, CO2 data was also reported, as it will subsequently be used for the calculation of air quality. The predicted sPMV was calculated using ML models. Specifically, the LSTM model was employed to predict temperature (T) and RH, as shown in Figure 34 and Figure 35. The predicted T and RH values were subsequently used to calculate the predicted sPMV, illustrated in Figure 36.

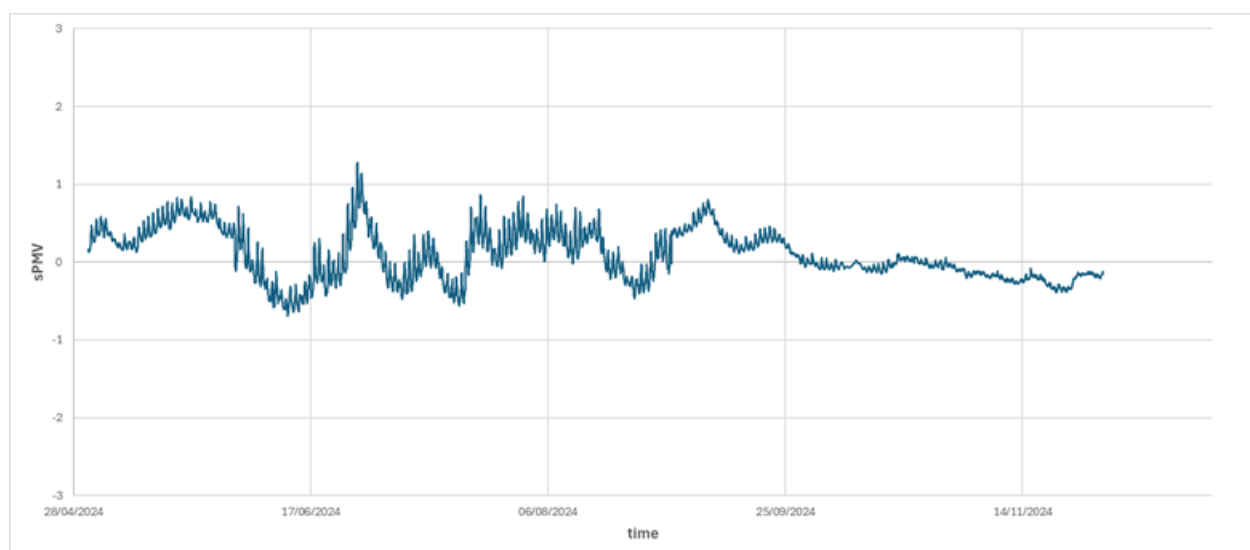


Figure 33: Thermal comfort (sPMV) calculated for the room 24932 from 01/05/2024 to 30/11/2024.



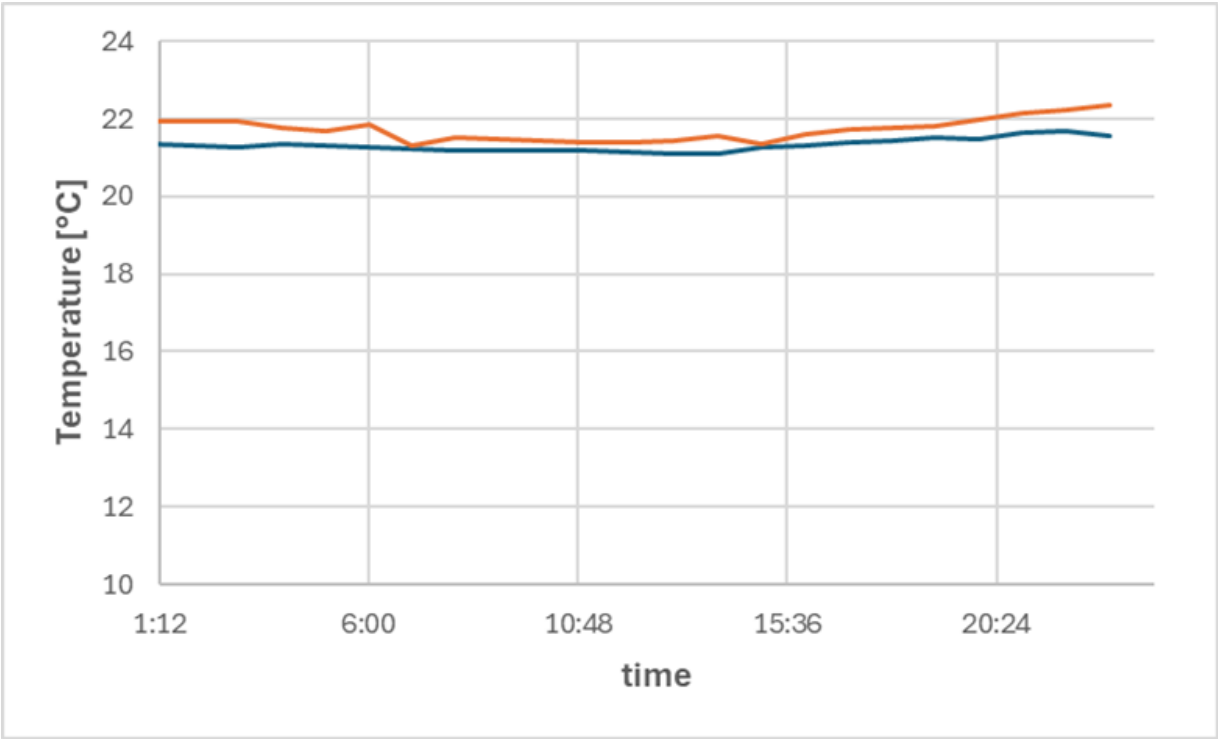


Figure 34: Prediction of temperature data for the day 30/11/2024 in orange, the real trend of the temperature for the day 30/11/2024, for the room 24932.

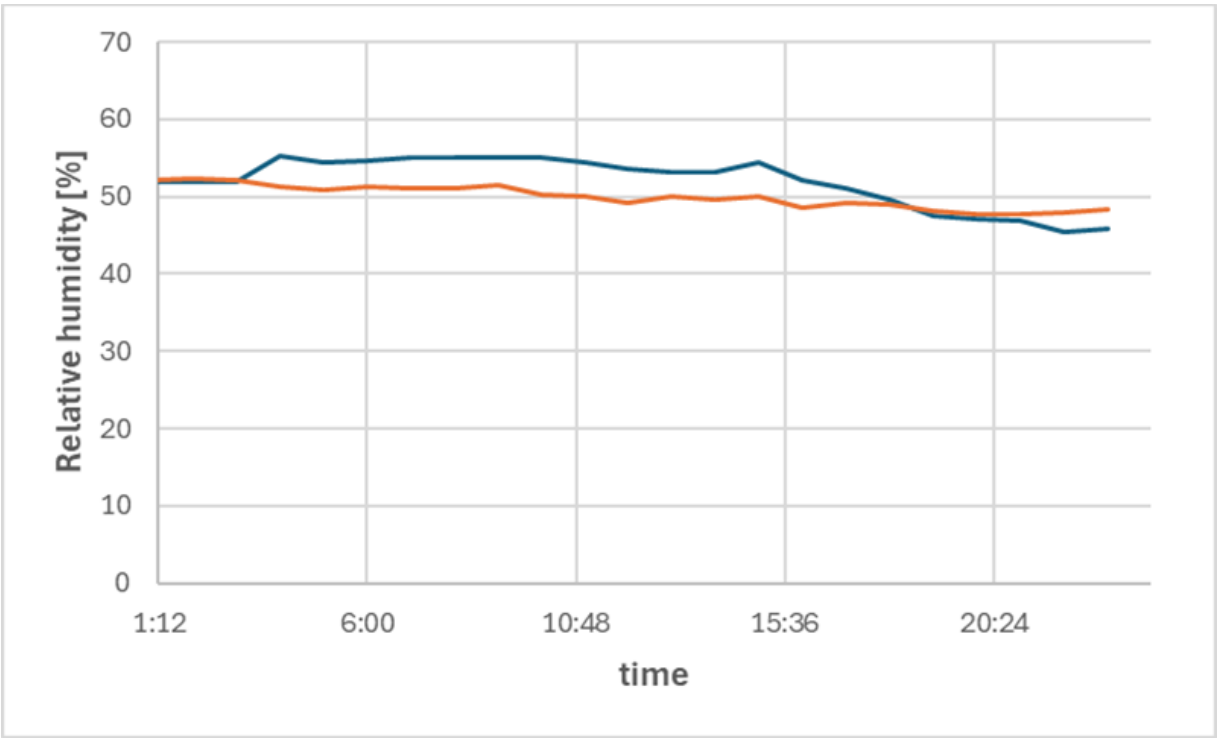


Figure 35: Prediction of relative humidity data for the day 30/11/2024 in orange, the real trend of the temperature for the day 30/11/2024, for the room 24932.



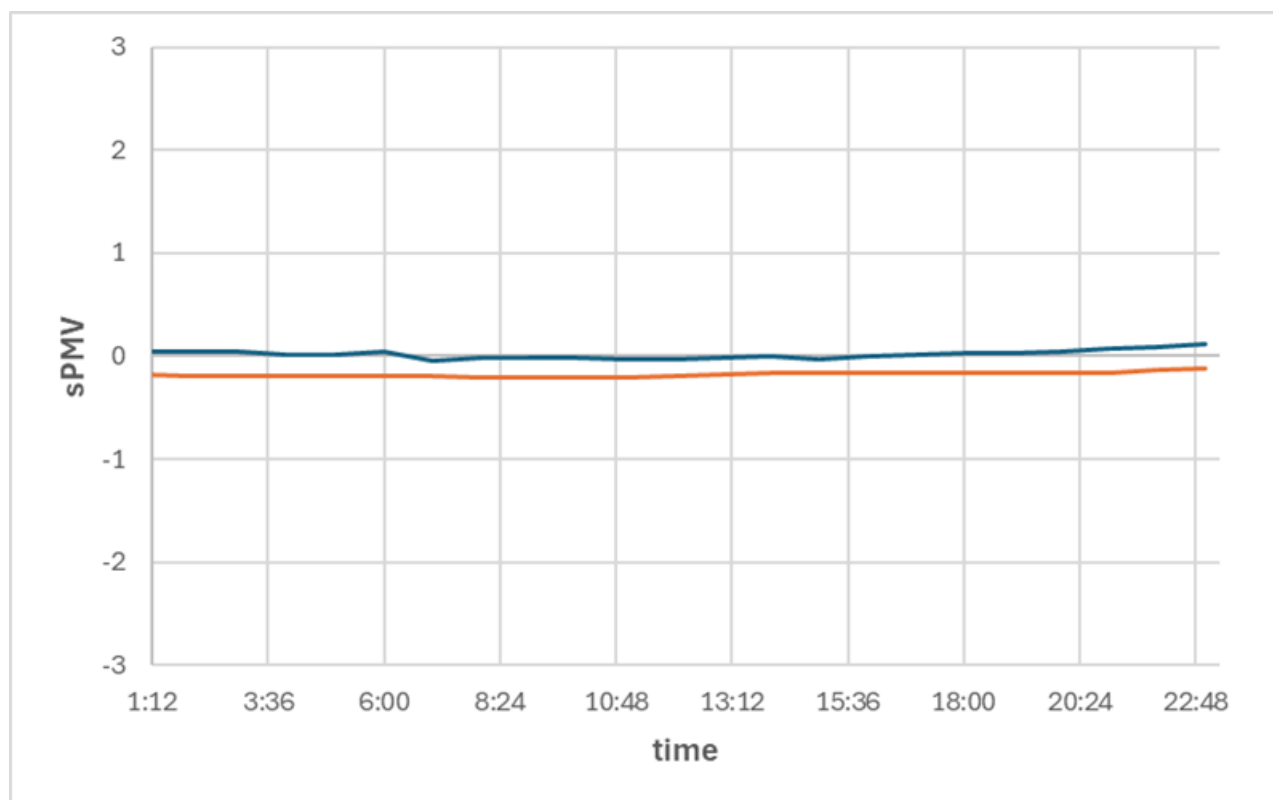


Figure 36: Prediction of thermal comfort (sPMV) data for the 30/11/2024 in orange, and the real trend of the thermal comfort for the 30/11/2024 in blue, for the room 24932.

These are the results that could be provided based on the data currently available from the individual pilots. As shown in the 3.2 the TSV and predicted TSV are required to fully utilize the comfort flexibility model. At this moment, a notification from the pilot leaders is awaited to begin the user feedback collection process. Once this phase is initiated, it will also be possible to obtain the TSV and predicted TSV, thereby completing all the parameters necessary for the proper functioning of the comfort flexibility model and enabling the demand and response feature for users. In collaboration with NTUA, strategies for demand and response are being further developed for each pilot, based on the specific characteristics of each pilot.

3.6.2. Pilot 4 – Italy

The well-being of aging individuals in their living environments is increasingly critical, particularly considering the COVID-19 pandemic, which has emphasized the need for older adults to remain at home to minimize health risks. As the global aging population rises, with projections showing an increase from 12.5% in 2021 to 20% by 2050, "aging in place" has become a key policy objective in many countries. However, the current housing stock is often inadequate to support the well-being of older individuals throughout their life course. IEQ, encompassing factors such as acoustic comfort, visual comfort, thermal comfort, and IAQ, is essential to ensuring the health and safety of aging individuals. Poor IEQ can contribute to a range of health issues, including cardiovascular disease, respiratory disorders, and cognitive decline. As such, optimizing IEQ is vital for improving the quality of life and promoting a safe, comfortable, and independent living



environment for older adults [11]. Based on this, special focus is given to the Italian pilot, where the residents are elderly individuals.

The data collected from the Italian pilot concern environmental variables from the two buildings: Casa Tito Garzoni and Casa Maddalena. Figure 37 and Figure 38 present the sensors installed in each apartment, the room IDs for each building, and the data availability status.

In each table, the types of sensors are reported as follows: T represents a temperature sensor, RH denotes a relative humidity sensor, Particulate Matter (PM) refers to a particulate matter sensor, LIGHTING indicates a lighting sensor, CO2 stands for a carbon dioxide sensor, and T outdoor represents the outdoor temperature, provided by a weather station, which in our case is sourced from an online service called 'WeatherBit'. The same types of sensors are also deployed in the other pilots.

The Figure 37 presents the data available for the Italian pilot, 'Casa Maddalena.' The green sections represent the data currently available, while the red sections indicate the data that is either unavailable or yet to be provided.

CASA MADDALENA						
5 APARTMENTS						
STARTING ACQUISITION		FROM 01/07/2024		TO 31-10-2024		
DATA AVAILABLE		ROOM ID				
SENSOR	VALUE	E144	E145	E146	E147	E148
T	°C					
RH	%					
PM4	µg/m3					
LIGHTING	lux					
PM10	µg/m3					
PM2,5	µg/m3					
PM1	µg/m3					
PM0,5	µg/m3					
CO2	ppm					
T OUTDOOR	°C					
TSV						

Figure 37: Pilot 4 - Data available for "Casa Maddalena"



Figure 38 Data available for the Italian pilot, 'Casa Tito Garzoni.' As outlined previously, the green sections represent the data currently available, while the red sections indicate data that is either unavailable or yet to be provided.

CASA TITO GARZONI												
STARTING ACQUISITION FROM 01/07/2024 TO 31-10-2024												11 APARTMENTS
DATA AVAILABLE		ROOM ID										
SENSOR	VALUE	E149	E150	E151	E152	E153	E154	E155	E156	E157	E158	E159
T	°C											
RH	%											
PM4	µg/m3											
LIGHTING	lux											
PM10	µg/m3											
PM2,5	µg/m3											
PM1	µg/m3											
PM0,5	µg/m3											
CO2	ppm											
T OUTDOOR	°C											
TSV												

Figure 38: Pilot 4 - Data available for “Casa Tito Garzoni”

The data were pre-processed by creating a data frame in which all available environmental parameters converge, required by the comfort flexibility model.

Figure 39, Figure 40, Figure 41 and Figure 42 show the environmental parameters of Temperature (T), RH, CO₂ concentration, and PM recorded in a sample room.

In the following plots there is a linear part due to missing data. The same procedure was applied to the rooms of the other buildings.

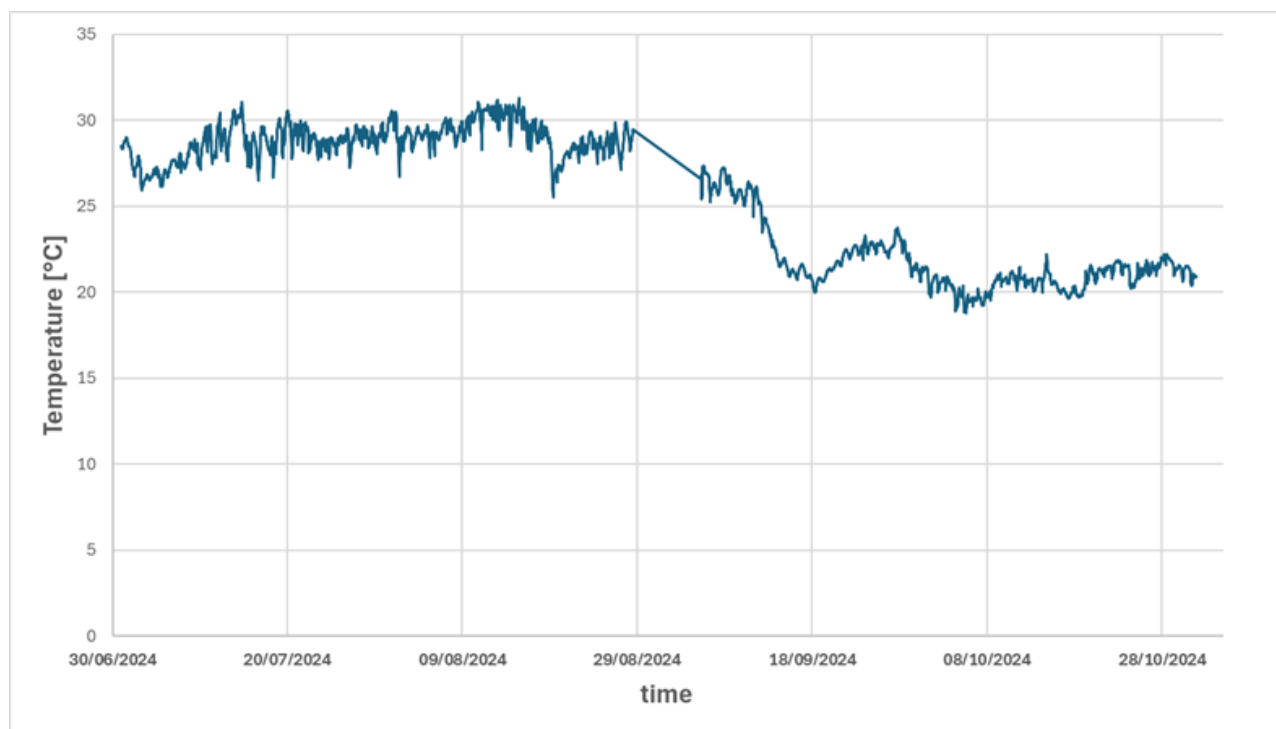


Figure 39: Trend of historical temperature data for room E145 from 01/07/2024 to 31/10/2024.



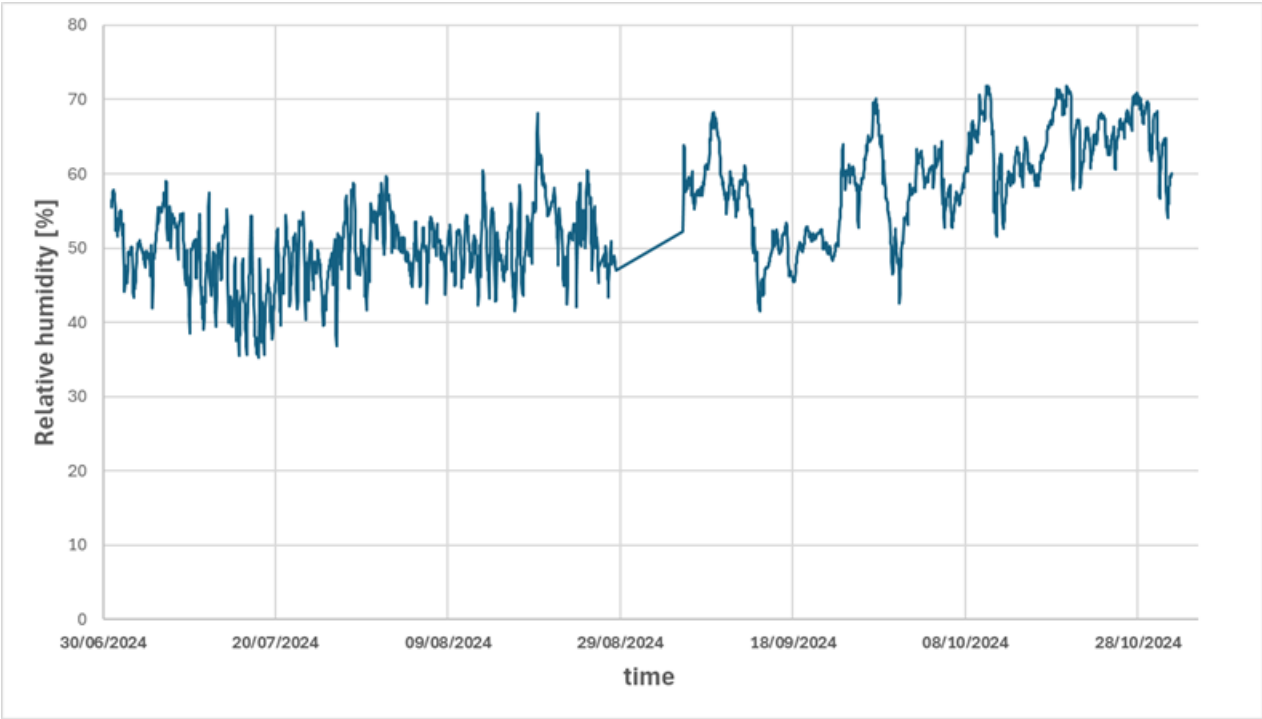


Figure 40: Trend of historical relative humidity data for room E145 from 01/07/2024 to 31/10/2024.

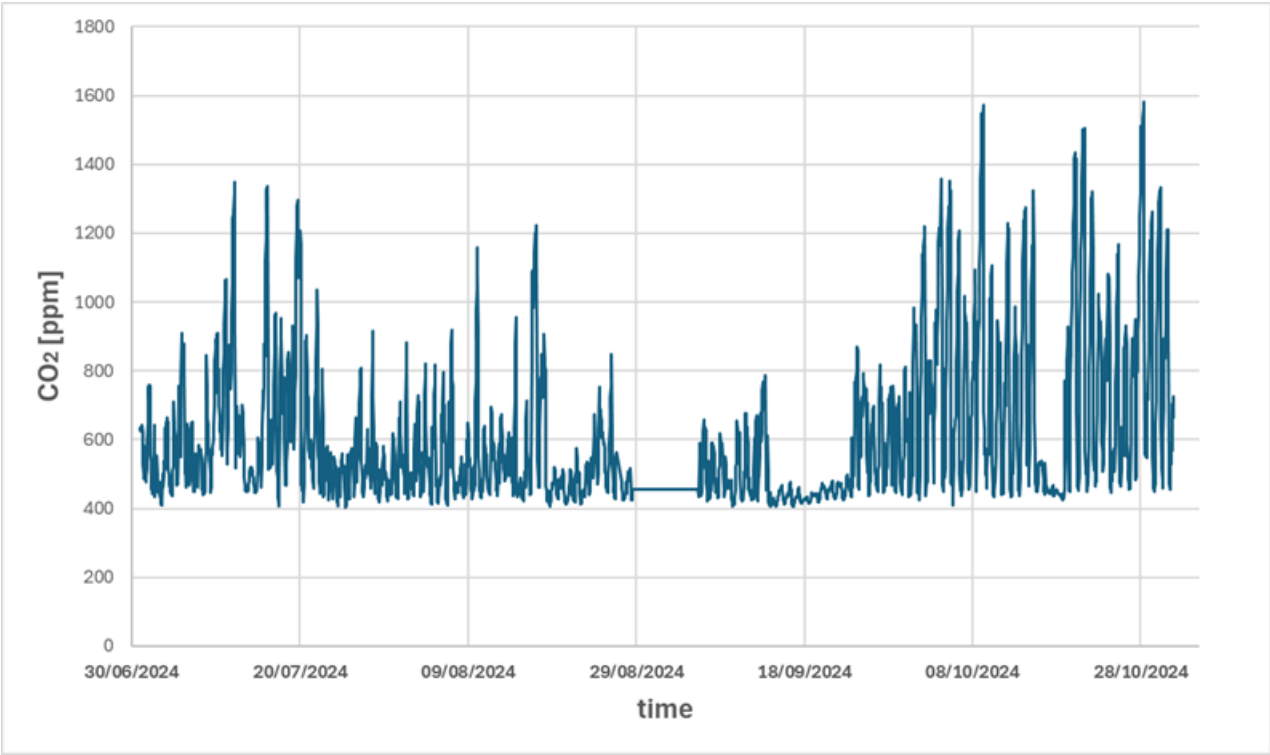


Figure 41: Trend of historical CO₂ data for room E145 from 01/07/2024 to 31/10/2024.



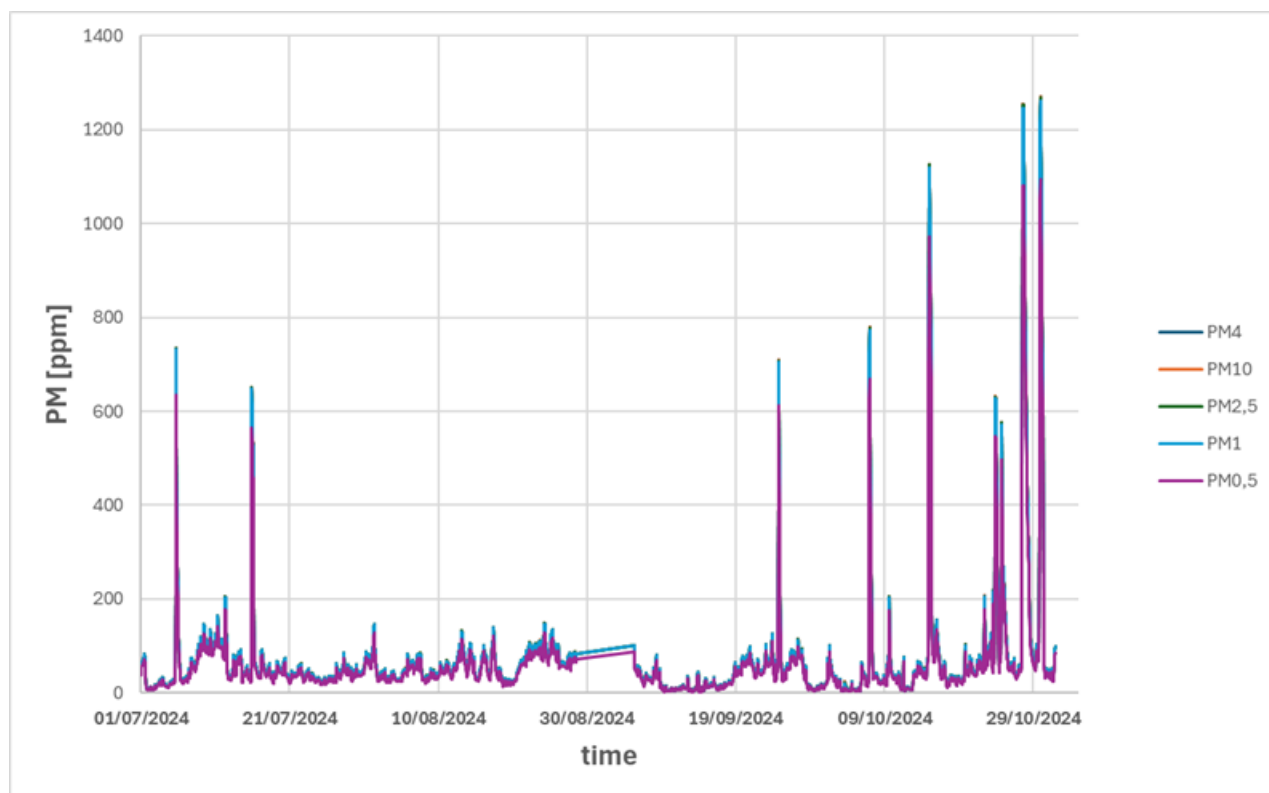


Figure 42: Trend of historical data of different matter (PM10, PM4, PM2.5, PM1, PM0.5) for the room E145 from 01/07/2024 to 31/10/2024.

These data, after being pre-processed, were used to calculate the thermal comfort (sPMV), as shown in Figure 39 with the methodology described in Deliverable 4.1 [5]. Additionally, air quality and lighting data were also reported, as they will subsequently be used for the calculation of visual comfort and air quality. The predicted sPMV was calculated using ML models. Specifically,



the LSTM model was employed to predict Temperature (T) and RH, as shown in Figure 43 and Figure 44.

The neural network used was configured with the following parameters:

- LSTM Layers: 2 layers with 64 and 64 units respectively.
- Dropout Layer: 2 layers with a dropout rate of 0.4.
- Dense Layer: 1 dense layer.
- Epochs: 10.
- Batch Size: 32.
- Train-Test Split:
 - Train: 80%
 - Test: 20%

The predicted T and RH values were subsequently used to calculate the predicted sPMV, illustrated in Figure 43.

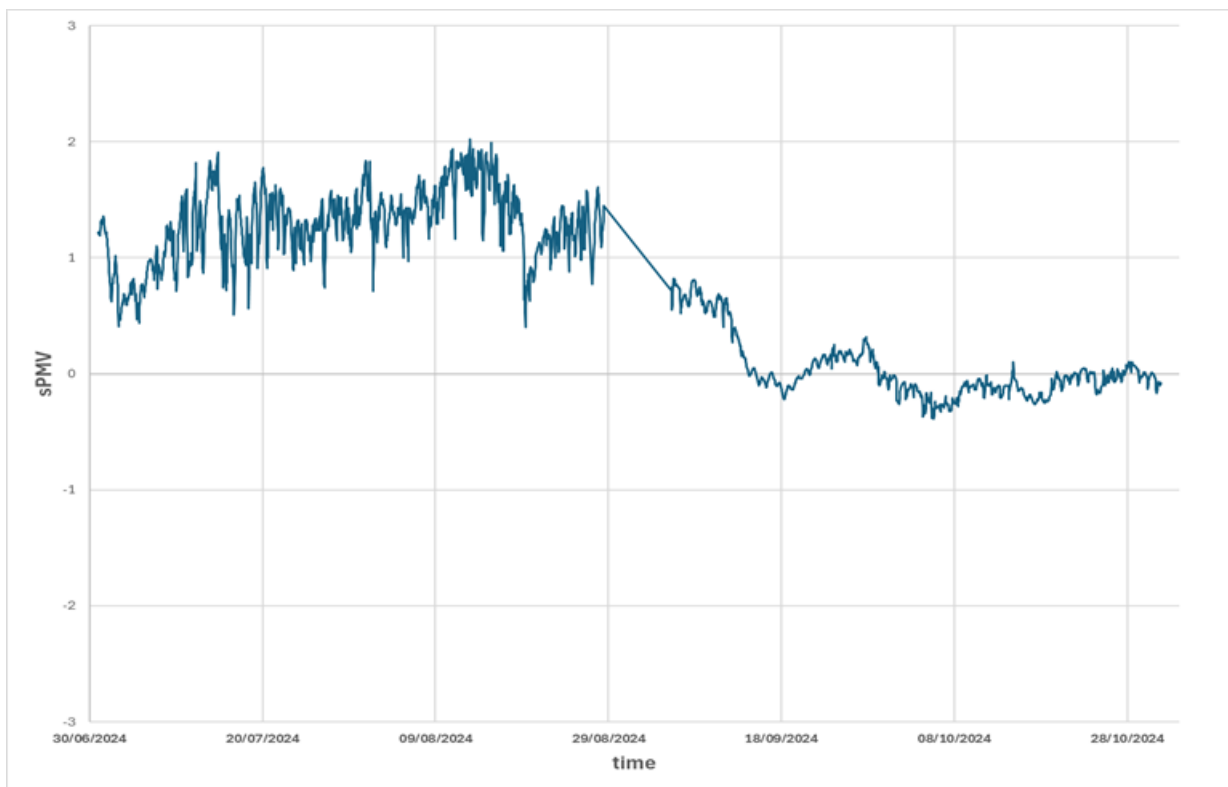


Figure 43: Thermal comfort calculated for the room E145 from 01/07/2024 to 31/10/2024.



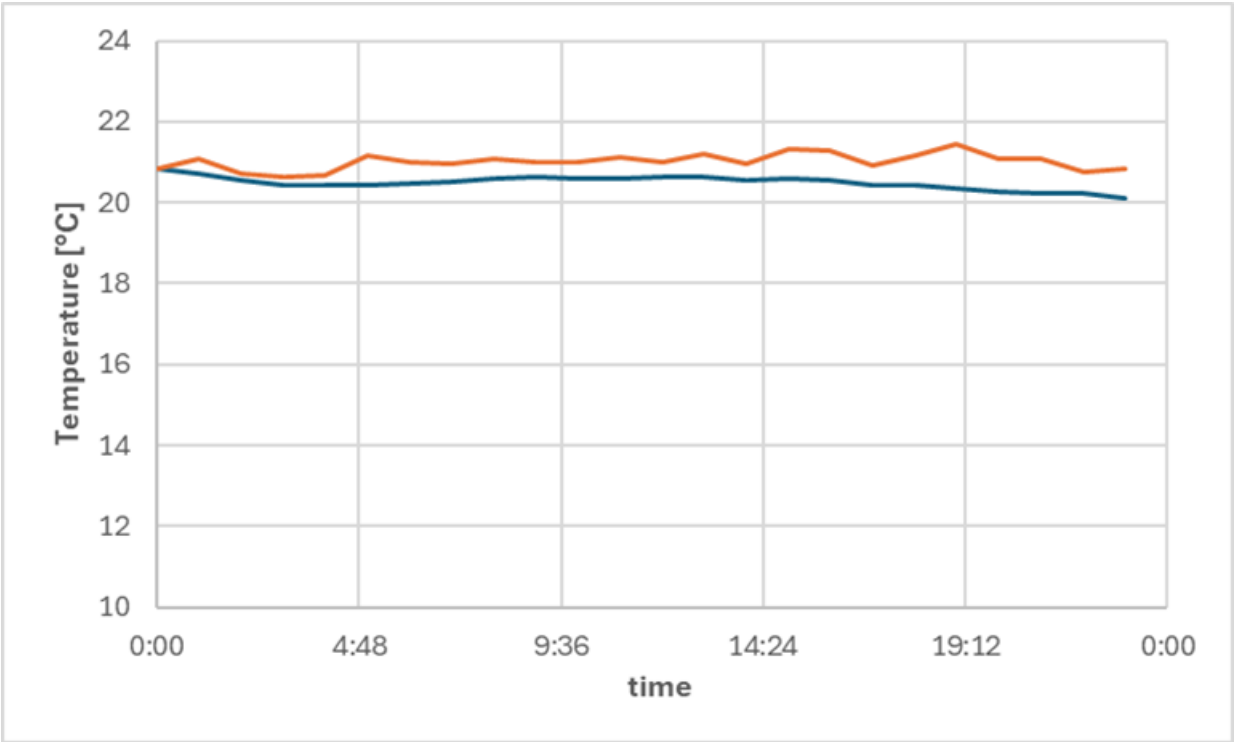


Figure 44: Prediction of temperature for the day 31/10/2024 in orange, the real trend of the temperature for the day 31/10/2024, for the room E145.

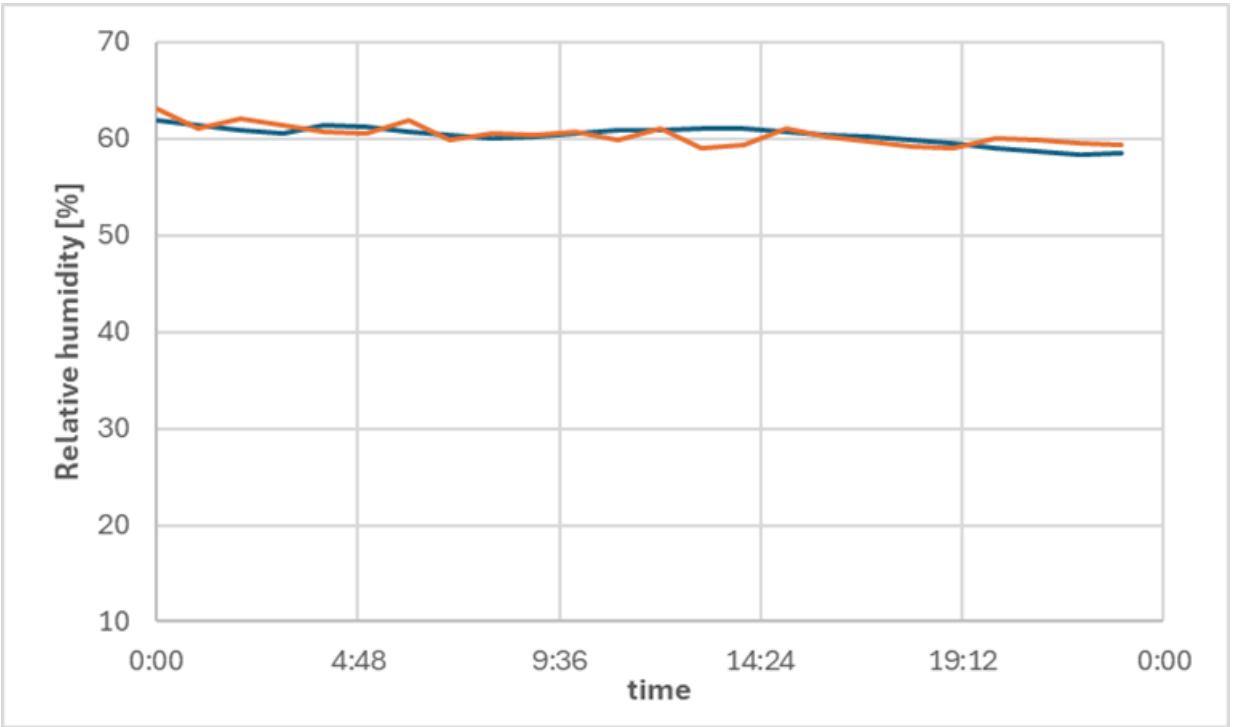


Figure 45: Prediction of relative humidity for the day 31/10/2024 in orange, the real trend of the relative humidity for the day 31/10/2024, for the room E145.



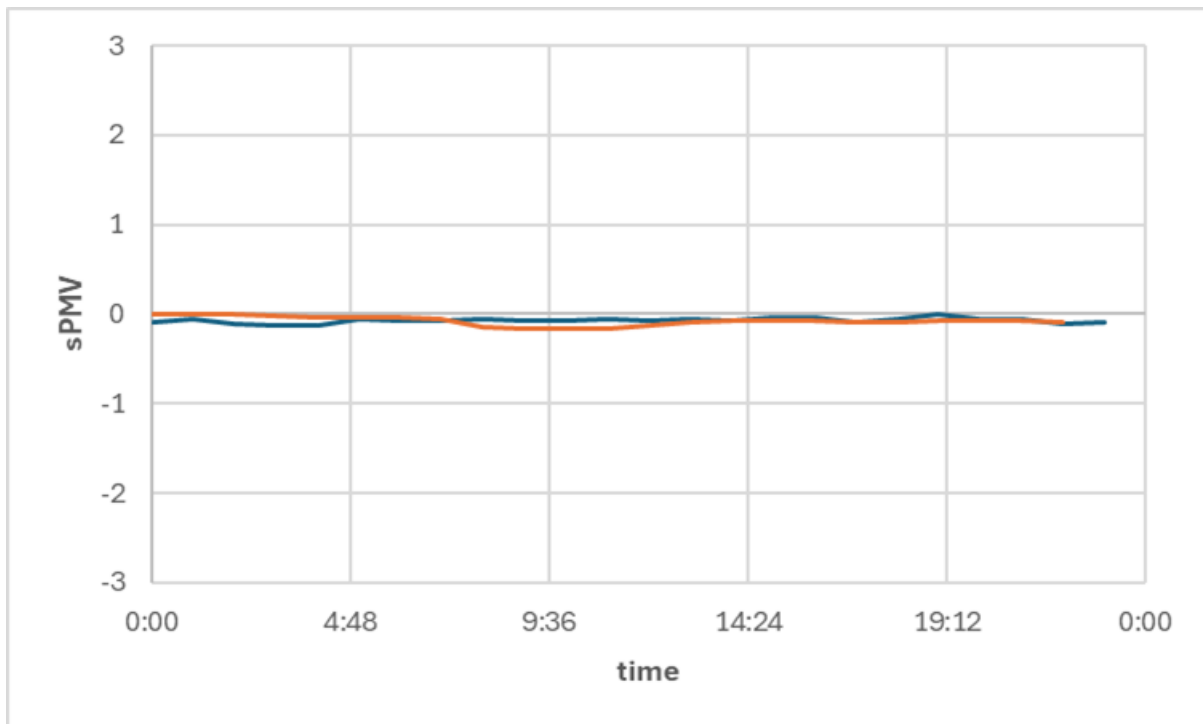


Figure 46: Prediction of thermal comfort for the day 31/10/2024 in orange, and the real trend of the thermal comfort for the day 31/10/2024 in blue, for the room E145

3.6.3. Optimization algorithm

The proposed approach suggests, based on external conditions, indoor environmental data and occupants' TSV, which internal condition (e.g., indoor temperature, relative humidity) should be achieved to optimize results in terms of consumption and comfort (i.e., TSV). This method does not directly indicate the control actions for the Heating, Ventilation, and Air Conditioning (HVAC) system, since it abstracts from its complex dynamics, but it provides suggestions and recommendations to the end-users who will directly act on the environment.

The method proposed relies on an historical series $S = \{s_1, \dots, s_T\}$ that describes the external environmental conditions, internal conditions of a specific building, and the corresponding



perceived comfort, is given. For simplicity, it is assumed that each observation is recorded at regular intervals. The i -th observation s_i is described by the tuple $(\tau_i, t_i, u_i, TSV_i, E_i, C_i)$ where:

- τ_i : External temperature (°C)
- t_i : Indoor temperature (°C)
- u_i : Indoor Relative humidity (%)
- TSV_i : thermal sensation vote
- E_i : Energy consumed in the interval between observations $i - 1$ and i
- C_i : CO₂ (ppm) emitted in the interval between observations $i - 1$ and i

Input: predicted external temperatures $\bar{\tau} = (\tau_{T+1}, \dots, \tau_{T+K})$ for the next K consecutive intervals

Output: suggested internal temperatures $\bar{t} = (t_{T+1}, \dots, t_{T+K})$ and internal humidity $\bar{u} = (u_{T+1}, \dots, u_{T+K})$ for the next K consecutive intervals.

Procedure:

1. Let $L = \{\mathbf{l}_1, \dots, \mathbf{l}_{T-K}\}$ be the list of $T - K$ tuples $\{(\mathbf{s}_1, \dots, \mathbf{s}_K), (\mathbf{s}_2, \dots, \mathbf{s}_{K+1}), \dots, (\mathbf{s}_{T-K}, \dots, \mathbf{s}_T)\}$, each composed of K consecutive observations from the historical series.
2. Determine the distance $d(\bar{\tau}, \mathbf{l}_i)$ between $\bar{\tau}$ and the temperature profile of each tuple $\mathbf{l}_i \in L$. One way to calculate this distance is using the norm:

$$d(\bar{\tau}, \mathbf{l}_i) = \sqrt{\sum_{k=0}^{K-1} (\tau_{T+k} - \tau_{i+k})^2} \quad \forall \mathbf{l}_i \in L$$

3. Identify the set B of tuples from L most similar to $\bar{\tau}$, for example, those with $d(\bar{\tau}, \mathbf{l}_i) \leq D$, where D is a given threshold.
4. For each tuple in B , calculate the performance indicators (consumption, emissions, and comfort):

$$C(\mathbf{l}_i) = \sum_{k=0}^{K-1} C_{i+k}$$

$$E(\mathbf{l}_i) = \sum_{k=0}^{K-1} E_{i+k} \quad \forall \mathbf{l}_i \in B$$

$$TSV(\mathbf{l}_i) = \sum_{k=0}^{K-1} |TSV_{i+k}|$$

5. Determine the Pareto-efficient frontier [12] F of B with respect to the objective functions (all minimization) related to consumption, emissions, and comfort. The efficient frontier consists of all non-dominated solutions concerning $C(\cdot)$, $E(\cdot)$, and $TSV(\cdot)$. A solution $\mathbf{l}_j$



dominates another solution $\mathbf{l}_k$ if $\mathbf{l}_j$ is strictly better than $\mathbf{l}_k$ in at least one objective function, i.e.:

$$C(\mathbf{l}_j) < C(\mathbf{l}_k)$$

$$E(\mathbf{l}_j) < E(\mathbf{l}_k)$$

$$TSV(\mathbf{l}_j) < TSV(\mathbf{l}_k)$$

6. The output consists of the internal temperature ($\bar{\mathbf{t}}_i$) and humidity profiles ($\bar{\mathbf{u}}_i$) of all the solutions on the Pareto-efficient frontier F .

$$\bar{\mathbf{t}}_i = (t_{T+1}, \dots, t_{T+K}), \quad i = 1, \dots, n$$

$$\bar{\mathbf{u}}_i = (u_{T+1}, \dots, u_{T+K}), \quad i = 1, \dots, n$$

3.7. Outlook for the future expected release

The following are the steps to be taken to proceed with the development of the service:

- Collection pilots' data to finalize the development of optimization algorithms.
- Implementation of Optimization DR strategy.
- Collection of TSV data from each pilot.
- Utilize ML and AI models to predict the TSV based on the feedback collected from the buildings' occupants.
- Start the integration of the service with Task 4.1 (Digital Twin interface), involving the implementation and development of the graphical user interface.



4. Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers

The aim of this chapter is to provide updated information about software expected to be developed as outcome of Task 4.3. This development corresponds to several services that aim to develop optimal DR strategies in the Greek and Austrian pilot sites and are linked to TE-10 Nudging application and TE-11 Multi-value DR service. TE-10 refers to the enhancement of the Greek and Austrian mobile applications to support nudging features to provide suggestions to the end-users for a more energy-efficient operation of their systems, while the TE-11 aims at providing DR capabilities by analysing and reporting the energy flexibility.

A quick overview can be found in Table 10, where the various services under development linked to DR strategies are listed.

Table 10: Task4.3 - Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
Design of optimal DR strategies for the gas pilot	TE10-FR1	UC15	TE-10	Pilot 3 - GR	In progress
Building performance and end-user profiling for the gas pilot	TE10-FR1	UC15	TE-1, TE10	Pilot 3 - GR	In progress
Energy disaggregation for the electricity pilot	NA	UC1, UC2, UC11, UC13	TE-4, TE-10	Pilot 3 - GR	In progress
Integration of tariff system for implicit and explicit electricity DR services	TE11-FR1	UC13, UC14	TE-11	Pilot 3 - GR	In progress
OurPower Nudging Application	TE10-FR2	UC22	TE-10	Pilot 1 - AT	In progress
Multi-value Demand Response for virtual clusters of building					
Integration of Implicit Demand Response service - Recommendation in the DT	TE8-FR5	UC11	TE-8	Pilot 3 - GR	In progress
Optimal DR control for heat-pump systems				Pilot 6 - IR	In progress



4.1. Service A - Design of optimal DR strategies for the gas pilot

4.1.1. Overview of the service

Table 11: Service A overview

Service A	Description
Service / Software title	Design of optimal DR strategies for the gas pilot
Responsible partner(s)	DOMX
Associated pilot site	Pilot 3, Greek – gas domain
Service brief overview	In this service we try to find out the gas pilot households with high energy savings potential in order to apply DR strategies to support flexibility. More specifically, each household that is equipped with the DOMX smart heating controller, supports a special parameter that defines how aggressively or how economically the heating system will operate when requested to heat up the place. Apparently, the same parameter results also in the energy consumption of the system. The end-users have the ability to parametrize this value according to their needs. The aim of this service is to calculate/simulate the energy consumption performance of each household under all possible configurations in order to identify the households that present potential of energy savings, without of course violating the comfort levels of the end users. To this aim DOMX is developing a new version of its simulator which will be trained using the datasets collected by the different households in order to calculate their energy under different configuration settings. By doing so it will be practical to identify the households with high energy savings potential and implement specialized nudges for these households. Moreover, we are developing a multivariate clustering algorithm based on deep learning techniques. The algorithm will be used to group the different households into subsets with similar characteristics in order to apply similar nudging techniques in the households that belong in the same cluster.
Associated TE	TE-10
Associated Use Case	HLUC7 / UC15
Functional Requirements (input by other activities)	TE10-FR1 Input has been received by the D2.3, from the definition of the parameters of the pilot site



Service A	Description
Data Input	Data provided by the DOMX smart heating controllers – this service is being developed with the aid of the data collected by the DOMX smart heating controllers that are installed in the gas boilers of the Greek pilot site. The smart heating controller collects all the parameters that are available from the gas boiler and is able to calculate the energy consumption for a given timeslot which is the main parameter used for the development of this service.
Service Output	Optimal DR strategies for the gas pilot
Repository Location	DOMX cloud system
License information	Proprietary
TRL	TRL 5-6
Current version number	V0.5
Expected date of finalization	May 2025, M25
Development Status	The second functional wave of the service has been released. Development for the final wave is in progress.

4.1.2. Architectural diagram

Figure 47 shows the architectural diagram of the DOMX service. We employ the data collected by the pilot households to feed the simulation algorithm that calculates the energy-consumption for every household under different settings. The households will be allocated in different groups with the aid of the multi-variate clustering algorithm depending on the heat growth each one presents to decide what kind of nudging activities will be applied.

The output of this service will be applied in the Greek pilot.



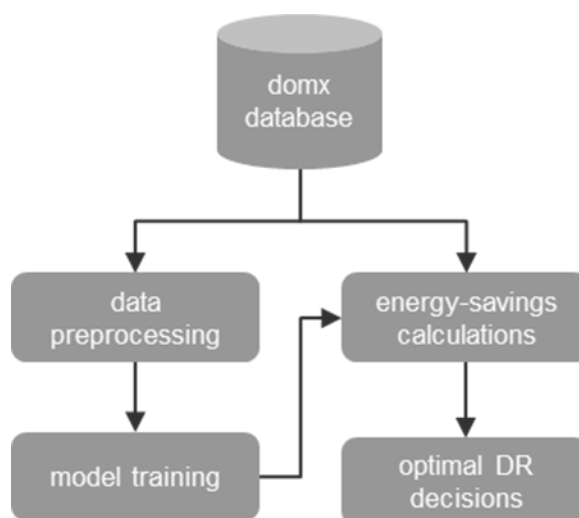


Figure 47: Service A - Architectural diagram

4.1.3. Main benefit and innovation

Benefit: The main benefit of this service is the development of a new simulator that will be able to precisely calculate the energy consumption of each household. This simulator can be used in various cases, such as the forecasting of the entire gas boilers portfolio under specific configurations, so that to derive to important decisions when required. Moreover, the development of the multi-variate clustering algorithm provides the ability to group the households according to their parameters, which is significant for providing similar suggestions and nudges to households that present similar behaviour.

Innovation: The simulator that is under development for this service takes into consideration the thermal dynamics of each household which provides an extra dimension in the prediction of the power consumption. This physics-based time series forecasting model is the latest version of AI in this field that increases the resulted accuracy.

Moreover, the clustering algorithm employs Multivariate Time-Series Clustering with Deep Learning techniques without the need of ground truth since it auto-trains itself to learn the provided data distribution.

4.1.4. Development activity

In this service it is being implemented a new version of the DOMX simulator to simulate the energy consumption performance of the various households to be able to calculate the energy consumption of an entire heating season under different configurations.

The existing DOMX simulator was presenting several limitations in the resulted performance, thus a new version that takes into consideration the thermodynamics of each household is being developed. In this version we consider an RC (Resistance-Capacitance) principle that models the



thermodynamics of the buildings in order to derive to more accurate results. In Figure 48 we present the model that we are using where the various temperatures are annotated with T , while the pace of their alteration is modelled with the Resistors in the scheme. Finally, the two capacitors emulate temperature storage in this principle. By using this scheme, we have developed a first version of the simulator, which is trained with data provided by the Greek pilot households.

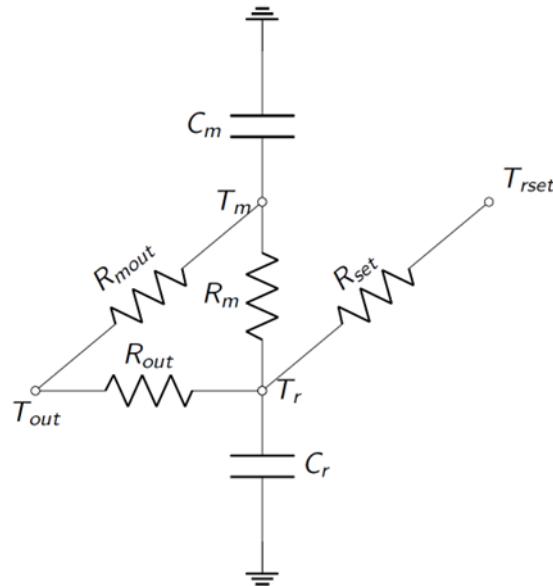


Figure 48: RC model used for the development of the new DOMX simulator

In Figure 49 we illustrate some indicative results from the development of the thermo-dynamics simulator. By using this service, we can predict the load of a household with minimum error, compared to the previous versions of simulators we had developed in DOMX. In this Figure 49 we illustrate the ground truth vs the predictive behaviour of a household for a duration of a few days for comparison reasons. In the first row we illustrate the performance of the household's boiler and the different power states that enters each time. Apparently, the predictive performance follows the actual one. We predict and use this parameter in order calculate the actual energy expenditure of the household. In the second row we illustrate the target temperature with the green line, while the blue indicates the real indoor temperature, and the red dashed line represents the simulated performance.





Figure 49: Indicative scenario of the performance of a household along with the results of the dynamic-load simulator that is used to predict the same parameters for comparison reasons

To identify households that belong in the same group, presenting similar characteristics we have developed a clustering methodology. We would like to make this allocation in groups to be able to select different nudges to be applied to different households. The clustering algorithm, we have developed, employs multivariate time-series and deep learning techniques. Thus, no ground truth is required since the algorithm uses auto-learn methodology for the provided data distribution. Moreover, we employ multi-variate clustering since the data by each household feature several different parameters that should be compared to derive to safe results.

In Figure 50 we illustrate an indicative snapshot from the first results we have from the clustering methodology performed in 20 households in total in data acquired by the Greek pilot. The main parameter used for the clustering of the different households is the heat growth they present. To do so, we have identified representative scenarios when requesting heat on each household and we have used this time-windows for the comparison among the different households. The algorithm returns a clustering allocation of all different snapshots (time-windows) of the involved households. Below, in Figure 50, we present just an indicative subset of the resulted clusters for illustration purposes. Apparently, the households that belong in the same cluster presents identical behaviour in the heat growth as seen with the blue line, while the orange line represents the target temperature. As said, we only use time-windows of specific time duration. In this clustering experiment we used time-windows of 3-hours for the comparisons. The time-windows have been selected by the data of each household depending on how often this behaviour is present. Another significant finding we have by the results of this methodology is the overshooting that the households present. Overshooting is the phenomenon that the indoor temperature substantially overpasses the given target temperature that leads to excess energy consumption and user discomfort. These findings (the households that present overshooting)



will be used to identify the households that will take part in the nudging interventions and to understand what kind of interventions should be applied on each cluster.

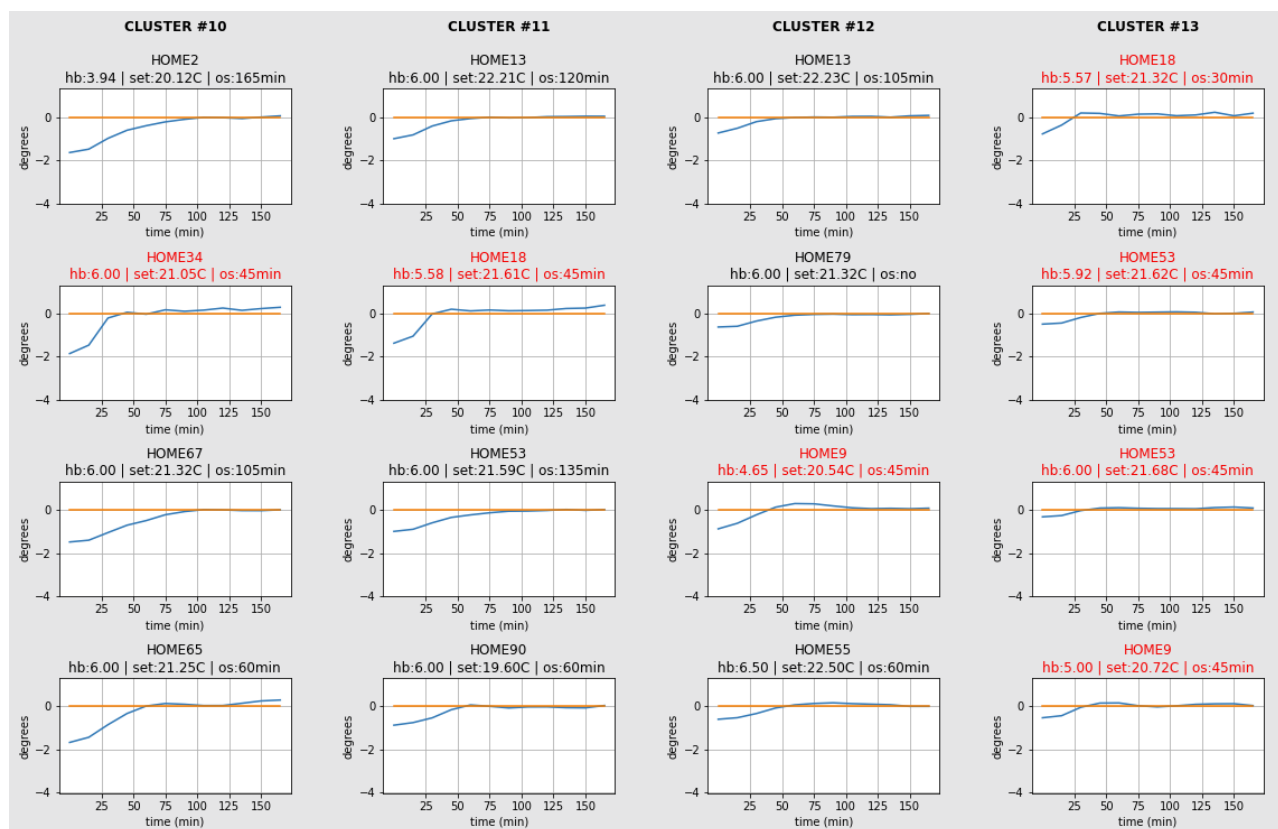


Figure 50: Indicative snapshot of a sub-set of the resulted clusters allocated using the heat-growth attribute employing the multi-variate clustering algorithm, using data by 20 households.

4.1.5. Outlook for the future expected release

The steps performed after the submission of the 1st wave of this deliverable is that we improved the performance of the developed simulator by testing more features and creating a more accurate version. To this aim the simulator was trained with data by the various households of the Greek pilot used it to estimate the energy consumption of the trained households under different configuration settings.

Moreover, a new clustering algorithm developed to group the different households into subsets with similar characteristics to apply similar nudging techniques in the households that belong in the same cluster. We endeavour to use this algorithm to derive to the final clusters depending on the scenario we would like to study.

As a next step, we identify the final definition of the households that should be involved in the nudging activity. Moreover, a nudging-enabled version of the domx mobile application is being prepared to provide suggestions to the end-users as well as to be able to measure the acceptance of each different approach proposed.



4.2. Service B - Building performance and end-user profiling for the gas pilot

4.2.1. Overview of the service

Table 12: Service B overview.

Service B	Description
Service / Software title	Building performance and end-user profiling for the gas pilot
Responsible partner(s)	NTUA (main contributor), DOMX (supporting)
Associated pilot site	Pilot 3, Greek – gas domain
Service brief overview	The objective is to unveil concealed intricacies within heating load profiles, affording a more distinctive comprehension of energy consumption dynamics and heating system operations focusing on identifying houses with the highest estimated energy savings, paving the way for sustainable DR programs
Associated TE	TE-1, TE-10
Associated Use Case	HLUC7 / UC15
Functional Requirements (input by other activities)	TE10-FR1
Data Input	Input Data are provided by DOMX and the gas part of the Greek pilot.
Service Output	Clusters of houses according to their estimated energy savings
Repository Location	DEDALUS GitHub
License information	Apache 2.0 open-source
TRL	TRL 7-8
Current version number	v0.3
Expected date of finalization	May 2025, M25
Development Status	2nd version ready and on-going development for the next version



4.2.2. Architectural diagram

The architectural diagram is as shown in Figure 51, and it spans from data collection to the creation of meaningful clusters according to each household energy-savings potential. The diagram is explained below in Section 4.2.4.

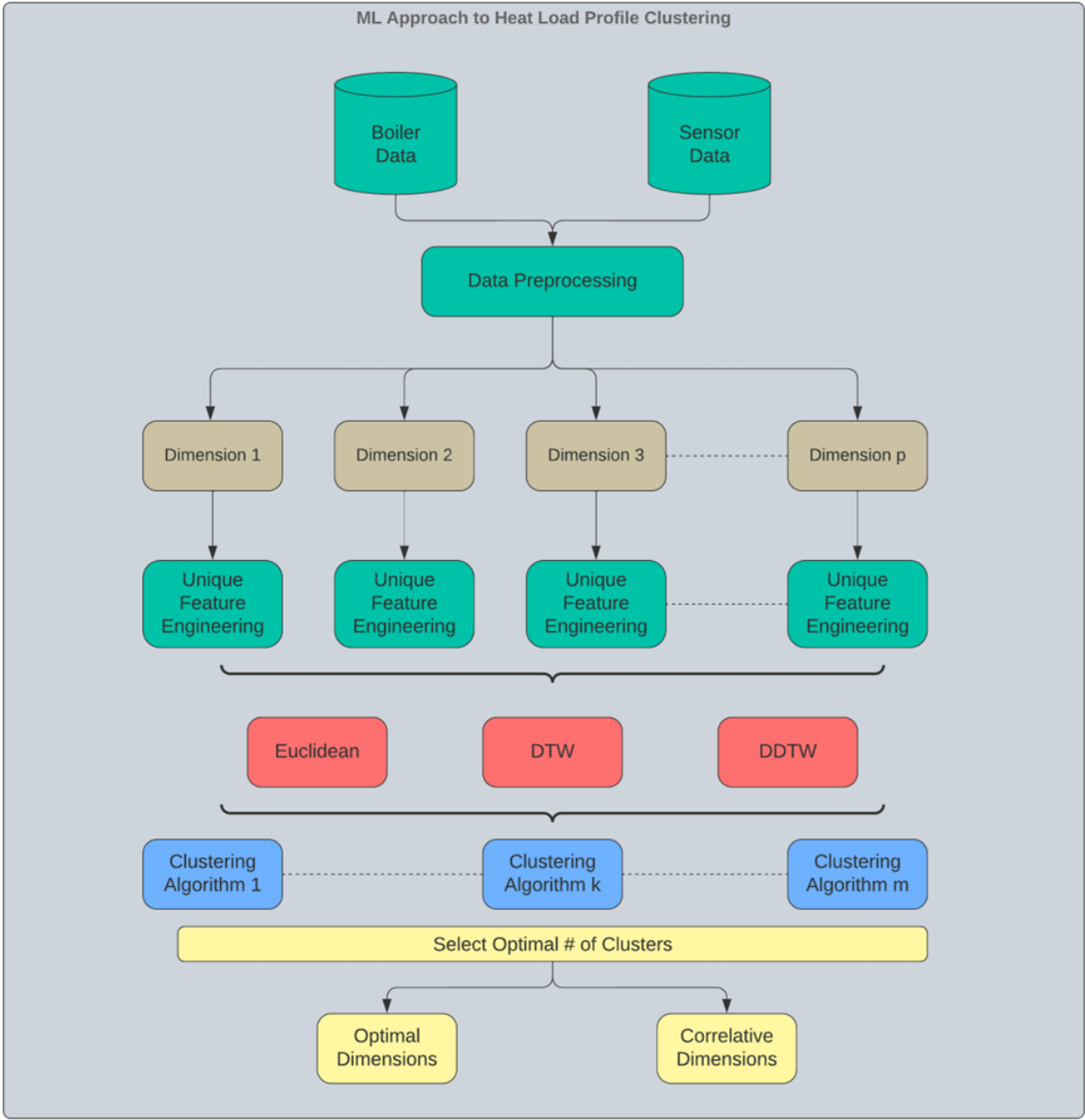


Figure 51: Service B - Proposed methodology



4.2.3. Main benefit and innovation

Main Benefit: The ability to deliver accurate demand predictions relies heavily on considering all the key aforementioned parameters, such as weather, building, user and heating system profiles. However, the significance amplifies when it comes to the optimal management of large portfolios of residential heating systems, for achieving energy savings or implementing DR actions towards environmental sustainability. As the population of participating households and the amount of available historical data continue to grow, the application of dedicated forecasting models at household level becomes challenging and practically impossible. This fact emphasizes the need for implementing innovative approaches for segmenting participating households into more homogeneous and manageable groups, which can reduce the variability and uncertainty of demand and ultimately simplify the demand profiling and forecasting procedures. Indicative application scenarios may include the ability to pinpoint residences with the highest potential for energy savings and/or for DR participation. As a result, heating load profiles grounded in diverse facets of the problem, which enable a better understanding of consumption patterns and behaviours, are the solution to the identified problem and at the same time the motivation for this work.

Innovation: The current work extends existing literature by incorporating a comprehensive set of parameters, including heat demand, temperature, building characteristics, boiler specifications, and user preferences. It is noteworthy to acknowledge that, to the best of the author's knowledge, no prior work has delved into the specific domain addressed by the current study. This pioneering exploration sets a significant benchmark for future research in the field, offering novel insights and avenues for further investigation.

4.2.4. Development activity

The methodology depicted in Figure 51 follows this sequence: Initially, data collection involves gathering boiler and sensor data from indoor and outdoor spaces in the building, which undergoes processing to identify outliers and anomalies. Next, key aspects of the problem are pinpointed, and a customized feature engineering process is conducted to match the characteristics of each aspect, specifically tailored for use in the Euclidean distance (ED) approach. Simultaneously, raw time-series data are employed for the Dynamic Time Warping (DTW) and Derivative Dynamic Time Warping (DDTW) methods, respectively. Clustering algorithms are then deployed on the prepared data across varying cluster numbers, ranging from two to nine. Evaluation is based on established metrics of similarity and dissimilarity to determine the optimal number of clusters for each aspect, distance measure, and the overall problem. Ultimately, the results from each scenario are evaluated to identify the most effective approach for each aspect and their interrelations within the problem domain.

The Heating Demand dimension of the problem has been addressed, and the methodology will stand as it seems for the rest of the project. As mentioned earlier, specific feature engineering strategies have been developed for each identified dimension to improve the outcomes derived from the ED approach, which will be comprehensively examined. Mean profiles for each dimension, for every minute of the day, have been extracted from the original data, forming the basis of the forthcoming analysis.



- Time-of-Maximum Analysis (ToM):
 - Identified peak periods of heat demand by determining the time of maximum load for each profile.
 - Categorized data into distinct time ranges, namely early morning, morning, noon, evening, night, and late night, and created binary columns indicating whether the time of maximum load fell within each specified range.
 - Analytically defined time intervals as follows:
 - Early Morning (5:00 - 8:00)
 - Morning (8:00 - 10:00)
 - Noon (10:00 - 17:00)
 - Evening (17:00 - 20:00)
 - Night (20:00 - 24:00)
 - Late Night (0:00 - 5:00)
- Statistical Analysis:
 - The statistical measures chosen for this analysis are as follows:
 - Mean
 - Standard Deviation
 - Maximum
 - Minimum
 - 25th percentile
 - 50th percentile
 - 75th percentile

Following feature engineering, the dataset underwent transformation to integrate the newly derived features. This involved expanding the original dataset to incorporate columns representing peak periods and statistical metrics computed for each heating load profile. The enhanced dataset formed the foundation for subsequent clustering analysis.

We experimented with two clustering algorithms, namely K-means Hierarchical Agglomerative clustering (HAC) and each algorithm demonstrated varying degrees of effectiveness in capturing the inherent structure of the data. Additionally, we evaluated the results based on three different distance metrics, ED, DTW and Derivative DTW, to quantify the similarity between heating load profiles. Furthermore, utilizing evaluation metrics is crucial in determining the ideal number of clusters for the dataset. Commonly employed metrics in scientific research, including Silhouette score (SIL), Davies-Bouldin Index (DBI) and Calinski-Harabasz Index (CHI), are applied to assess the clustering results' quality. These metrics aid in identifying the optimal number of clusters that accurately depict the underlying data patterns. Leveraging such evaluation metrics empowers researchers and practitioners to make well-informed decisions regarding clustering outcomes, enabling them to refine their analyses and improve the overall quality of load profiling processes. It is worth noting that the selection of the optimal number of clusters is based on choosing the clustering configuration that produces the highest SIL and CHI values, while aiming for a lower corresponding value of DBI.

Following the necessary data preprocessing a visual example of a typical household can be seen in Figure 52.



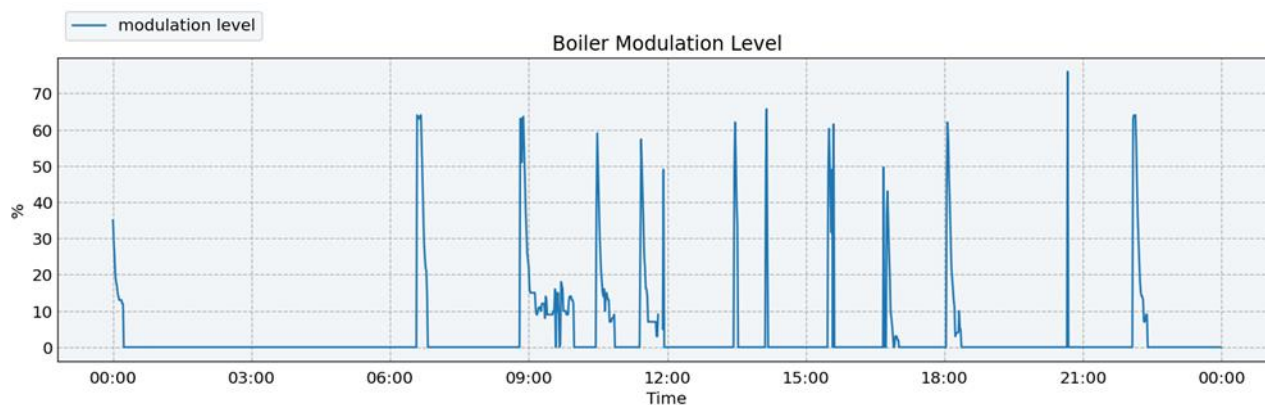


Figure 52: Visual example of a typical household

Furthermore, the mean daily modulation levels for all 29 houses, for the heat demand dimension are showcased in Figure 53. Those can be characterized as the modulation level profiles that will be used in this study. For the DTW and DDTW distances these timeseries are going to be used for clustering, as for ED the extra features described earlier will conclude the input to the clustering algorithms.

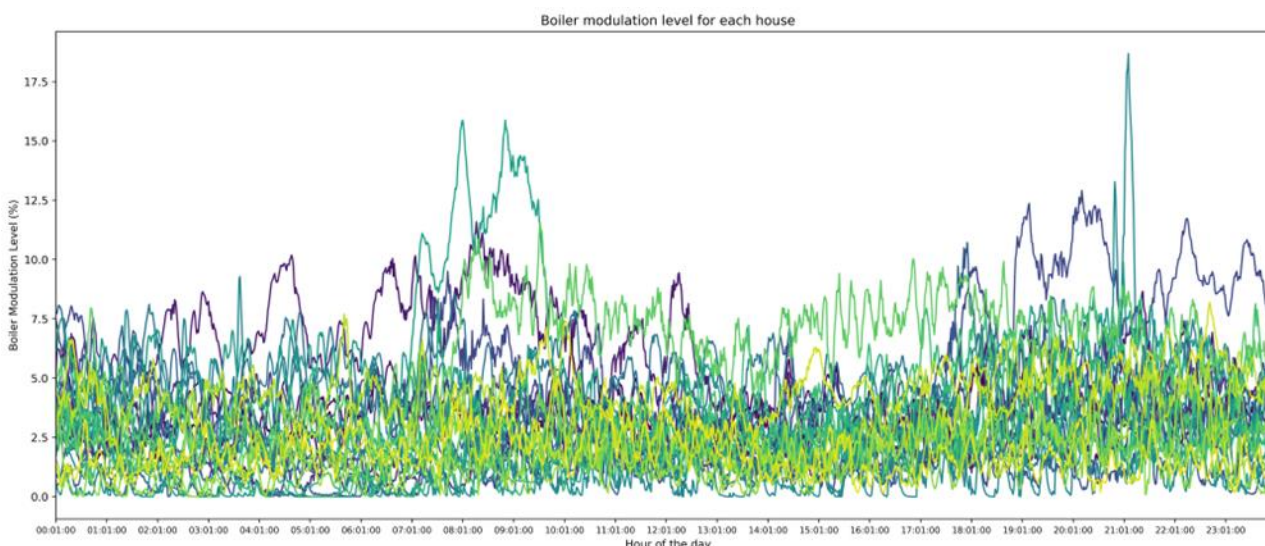


Figure 53: Heat demand dimension - mean daily modulation levels for all 29 houses

The best number of clusters in this case is 3. This result can be evaluated in Table 13 below, that results can be seen for each evaluation metric for the cluster number 3. As can be witnessed, the results for K-means with DTW are exceeding the rest and thus those results are going to be showcased later on.



Table 13: Cluster evaluation metrics for the heat demand dimension

Algorithm	Silhouette	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means with ED	0.174537	1.536718	9.738654
HAC with ED	0.216338	1.497139	9.249262
K-Means with DTW	0.428794	0.759205	34.586499
K-Means with DDTW	0.121640	1.693275	4.394497

In Figure 54, Figure 55 and Figure 56, the clusters that are the output of this clustering attempt are illustrated and the differences in the mean and median load profile of each cluster can be seen with different colours.

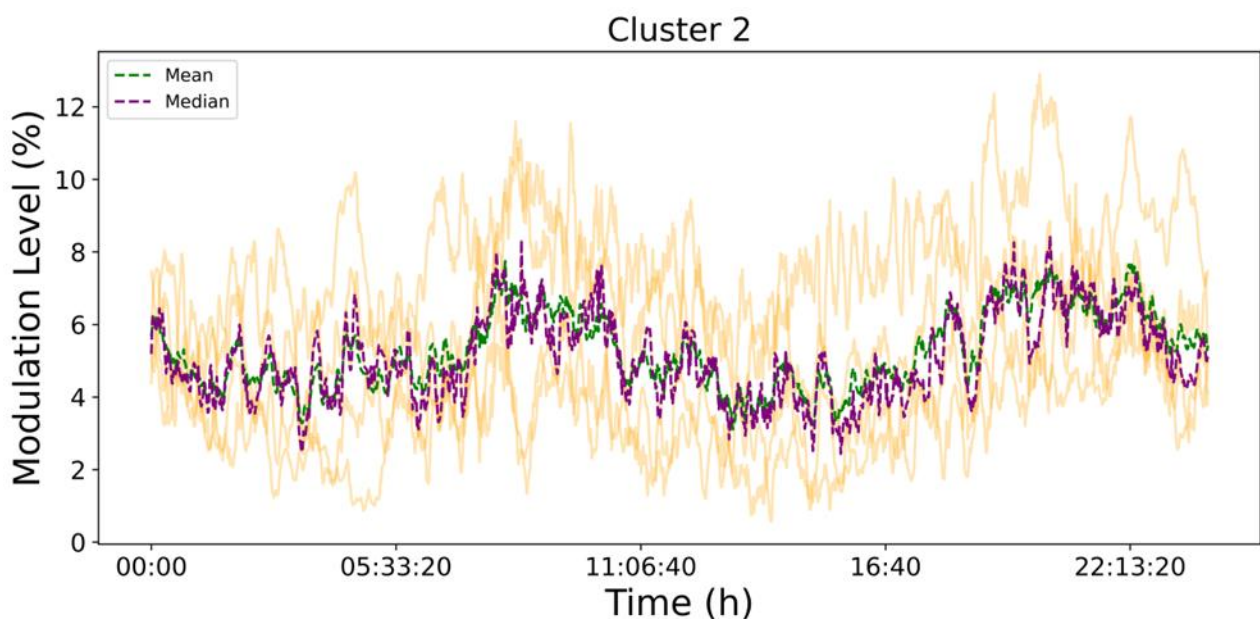


Figure 54: Cluster 2 for the heat demand dimension



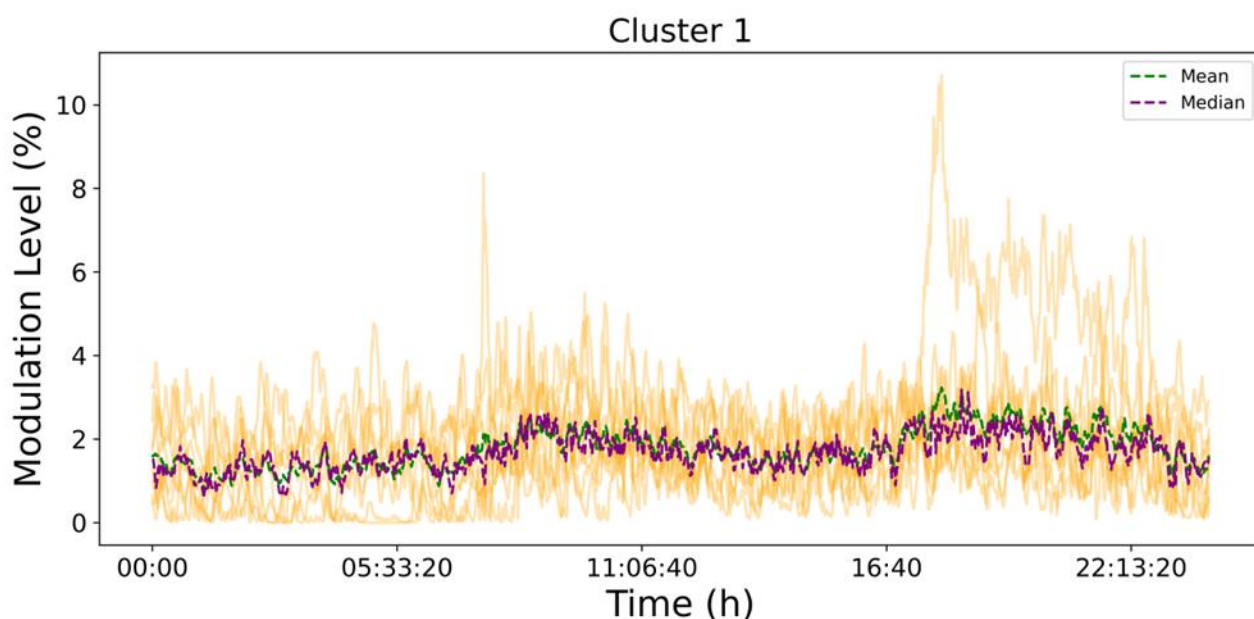


Figure 55: Cluster 1 for the heat demand dimension

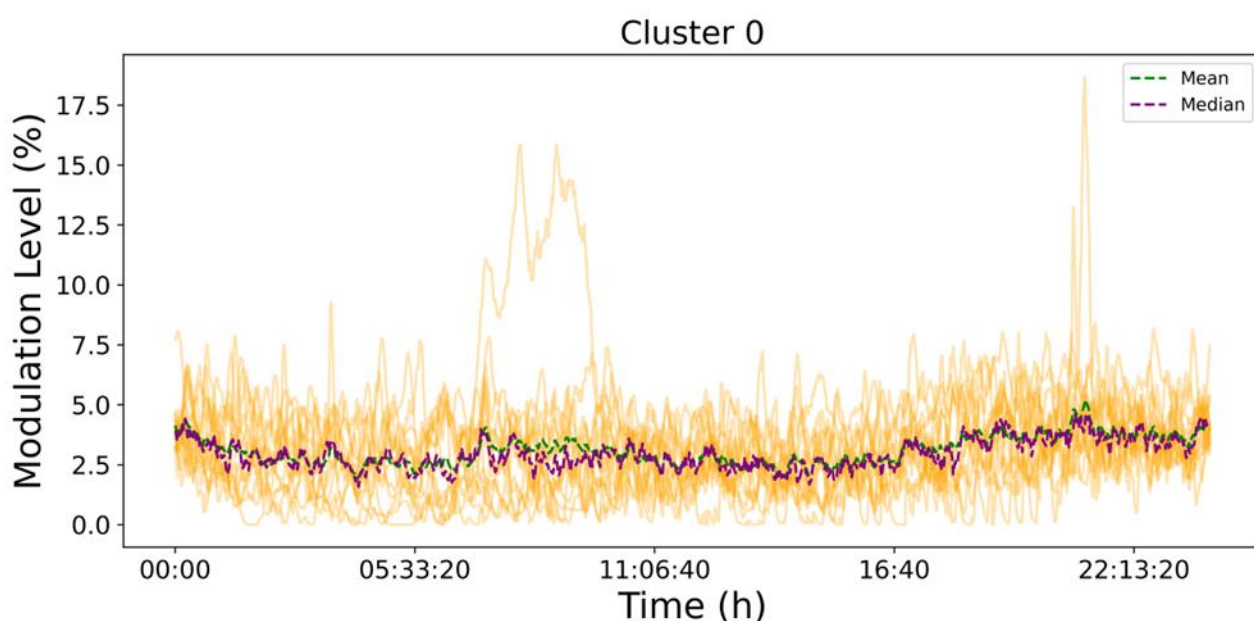


Figure 56: Cluster 0 heat demand dimension

Moving on, the next scenario involves the assessment of the variation in boiler and return water temperature differences for each house over the course of a day. Like the heat demand dimension the optimal number of clusters is 3. Table 14 displays the SIL, DBI, and CHI scores for each scenario, with the cluster number set to three since this is the optimal number of clusters identified by all metrics. It's important to highlight that the highest score is desirable for both SIL and CHI, indicating better clustering quality. However, for DBI, lower scores are



preferred, signifying better cluster separation. Notably, in the provided table, the values for each evaluation metric are optimal for DTW and they are the same for both algorithms, showcasing that in this case both algorithms produce the exact same labels.

Table 14: Cluster evaluation metrics for the boiler dimension

Algorithm	Silhouette	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means with ED	0.703640	0.284858	146.516525
HAC with ED	0.703640	0.284858	146.516525
K-Means with DTW	0.769452	0.178791	197.323287
HAC with DTW	0.769452	0.178791	197.323287
K-Means with DDTW	0.204362	1.565751	11.503813

The clustering algorithms produced identical labels for both ED and DTW metrics. This suggests that they achieved the same clustering results regardless of the algorithm used. However, the evaluation metric scores were notably higher for DTW. This indicates that the plain time-series data exhibited clear and distinct patterns, requiring no additional features for separation. While the results from the ED approach were promising with the inclusion of extra features, it's important to recognize that these outcomes might not be as accurate without those additional features. The effectiveness of the ED approach could vary depending on the complexity of the dataset and the nature of the underlying patterns.

In Figure 57, Figure 58 and Figure 59, a visualization that showcases all clusters individually, accompanied by vital statistical metrics such as mean, and median is depicted. This unified representation provides a holistic perspective of the clustering outcomes, facilitating a deeper comprehension of each cluster's characteristics. The mean serves as a measure of central tendency, representing the average behaviour of data points within a cluster, while the median provides insights into the central point of the distribution, serving as the basis for centre-based clustering algorithms.

Following our analysis, the fluctuation in outdoor temperatures recorded by sensors across various households over the course of the day are going to be analysed. In this case as well the resulting labels from K-means and HAC for both ED and DTW were the same, so no further analysis between algorithms is necessary at this point, while the optimal number of clusters again is three.



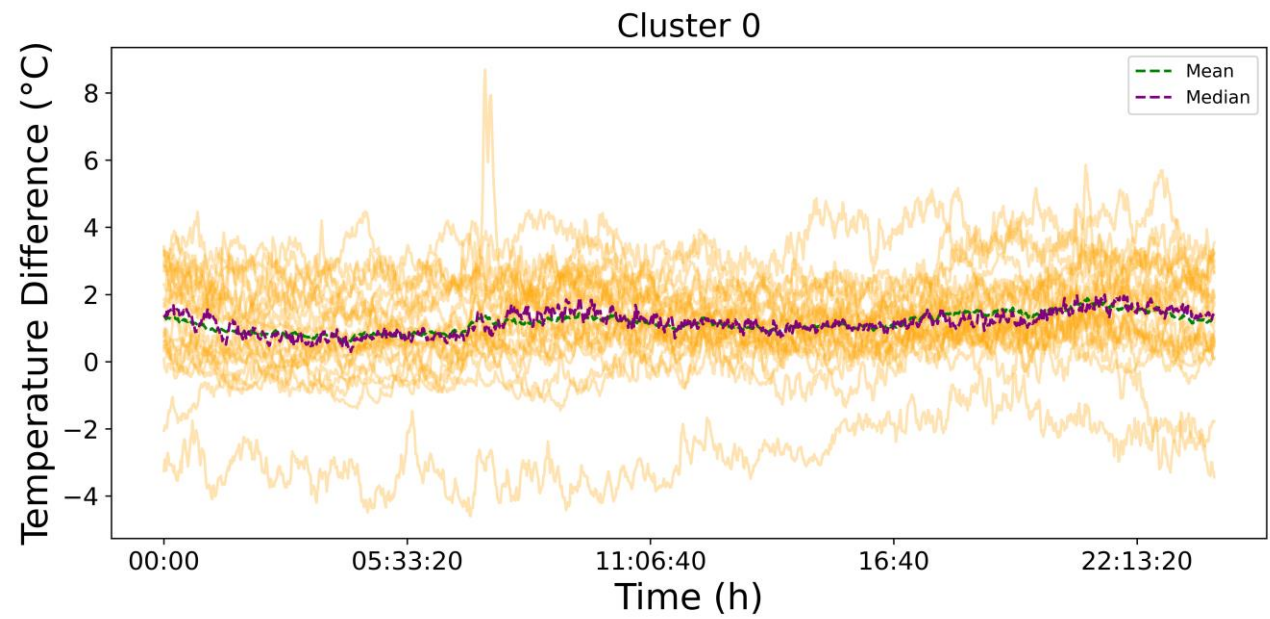


Figure 57: Cluster 0 for the boiler dimension.

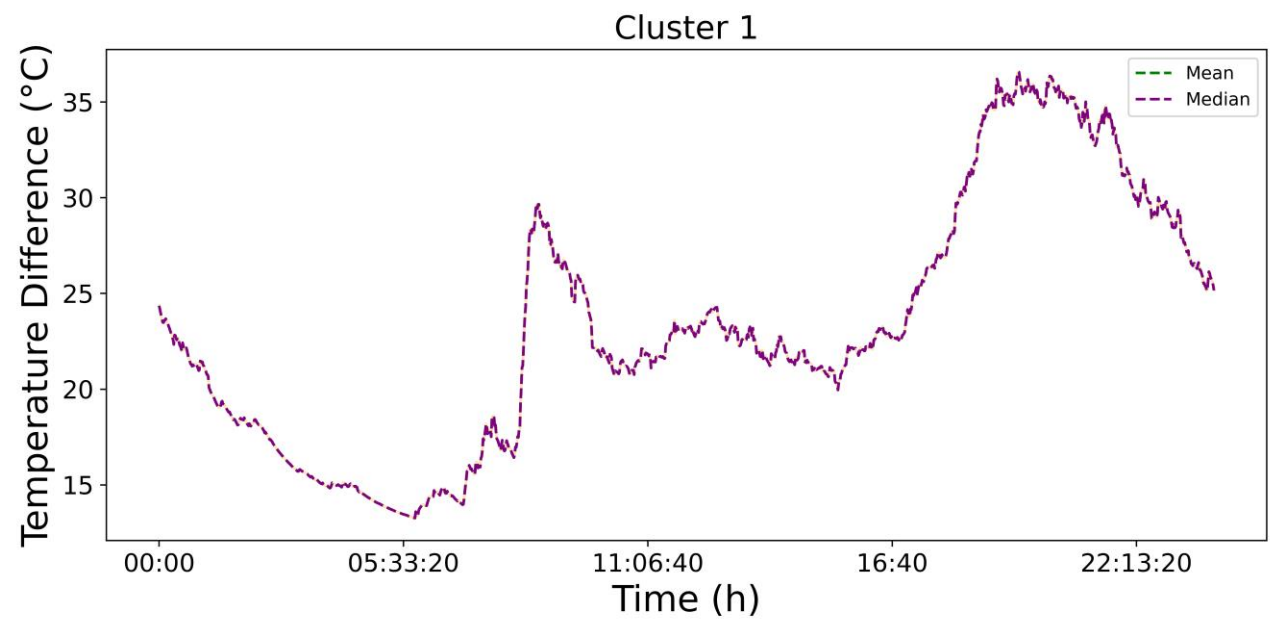


Figure 58: Cluster 1 for the boiler dimension.



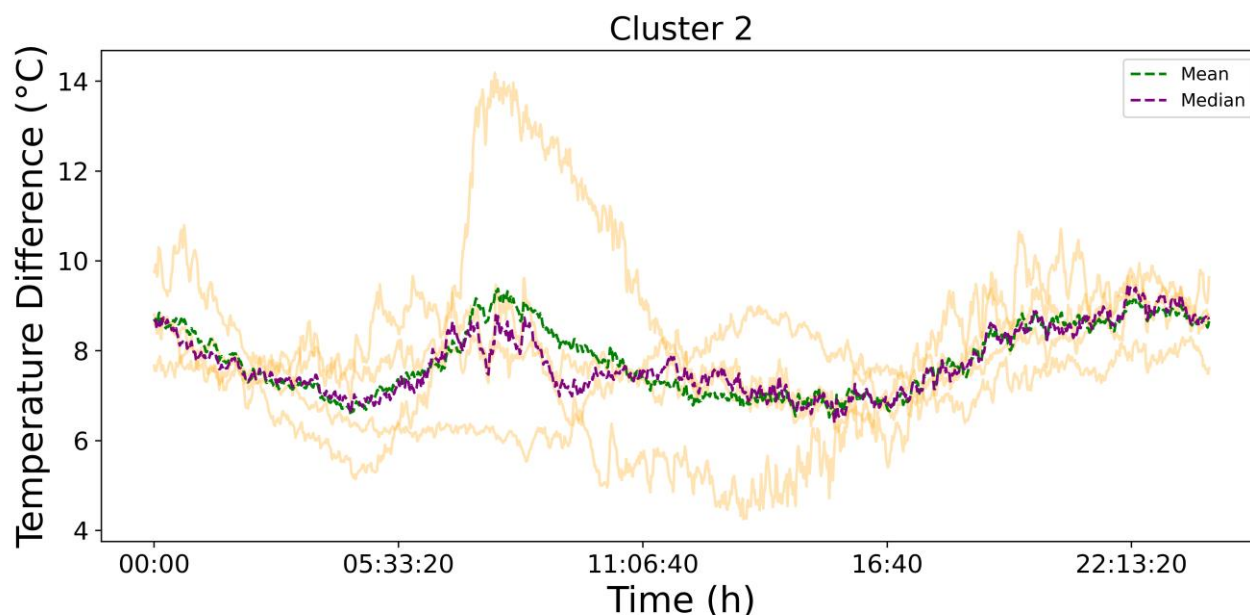


Figure 59: Cluster 2 for the boiler dimension.

In this case as well the resulting labels from K-means and HAC for both ED and DTW were the same, so no further analysis between algorithms is necessary at this point. Table 15 presents the evaluation metric scores for each scenario, with three clusters being the selected number. Interestingly, DTW consistently yields optimal values across all evaluation metrics for both K-means and HAC.

Table 15: Cluster evaluation metrics for the temperature dimension

Algorithm	Silhouette	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means with ED	0.358195	0.818539	25.022211
HAC with ED	0.358195	0.818539	25.022211
K-Means with DTW	0.674398	0.430472	33.784205
HAC with DTW	0.674398	0.430472	33.784205
K-Means with DDTW	0.325206	1,168875	20.332338

In summary, considering the results presented in Table 15 alongside these figures, the DTW approach emerges as the optimal solution for this problem. Visualizations of each cluster in Figure 60, Figure 61 and Figure 62, offer detailed insights into how data points are grouped based on their similarity in heat demand patterns, utilizing both K-means clustering and the DTW similarity measure. As can be witnessed again, the clusters are based on the amplitude of the values but also the kind of fluctuations that are present in each time series.



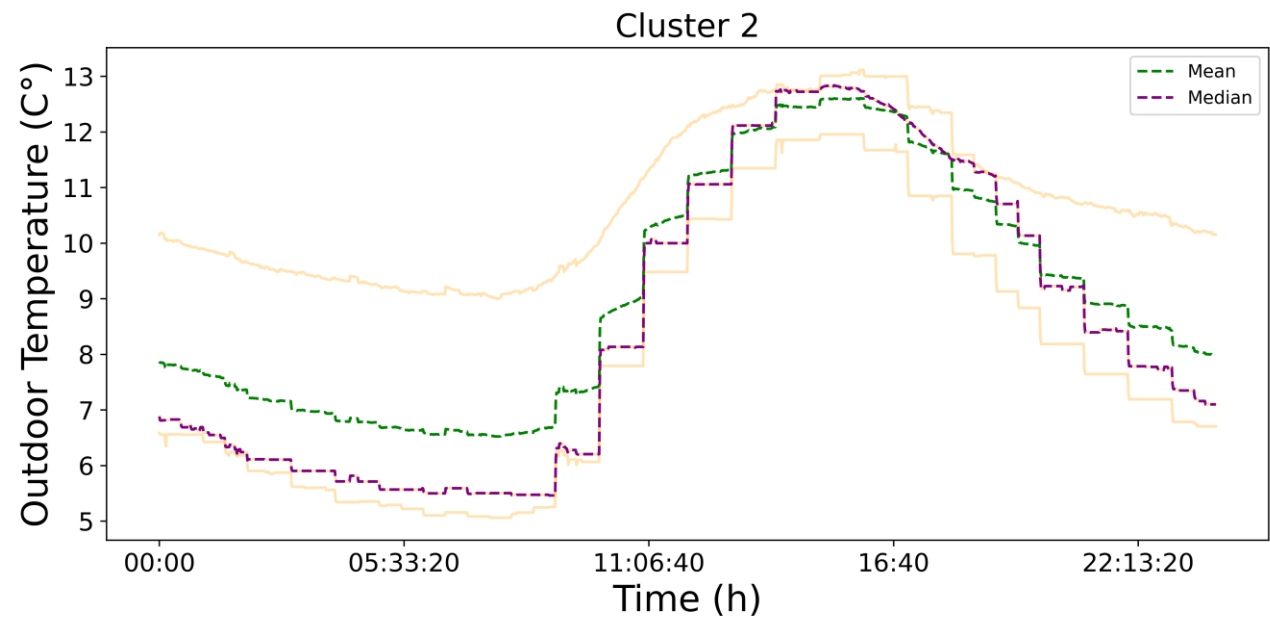


Figure 60: Cluster 2 of the temperature dimension

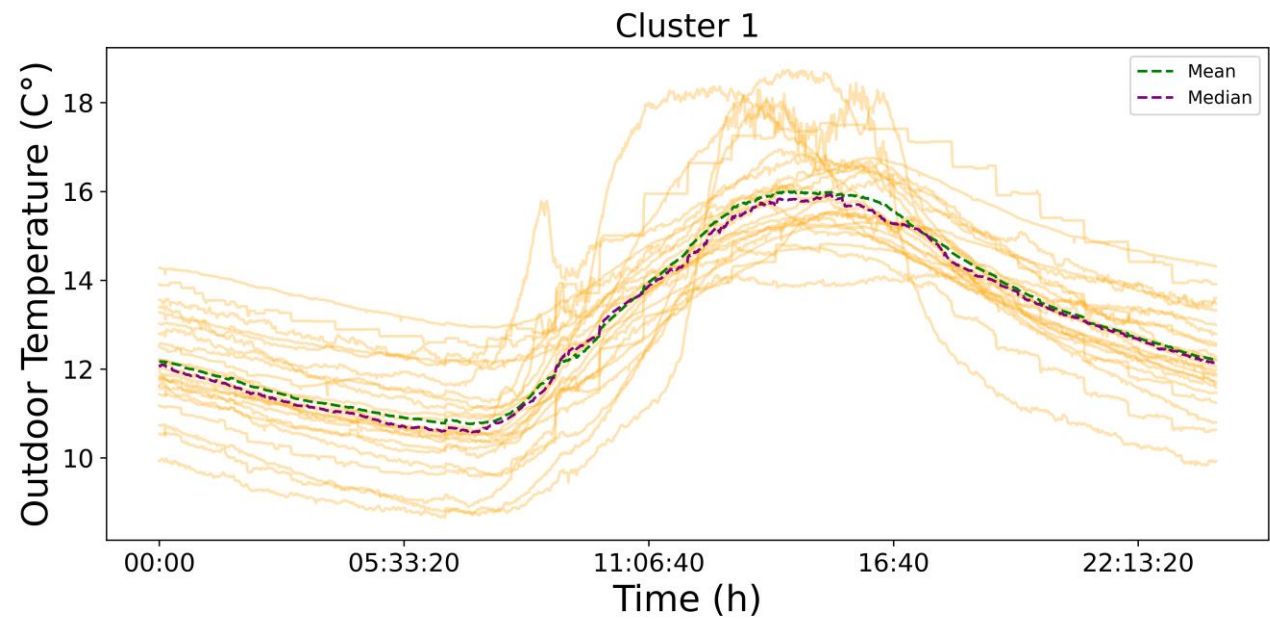


Figure 61: Cluster 1 of the temperature dimension



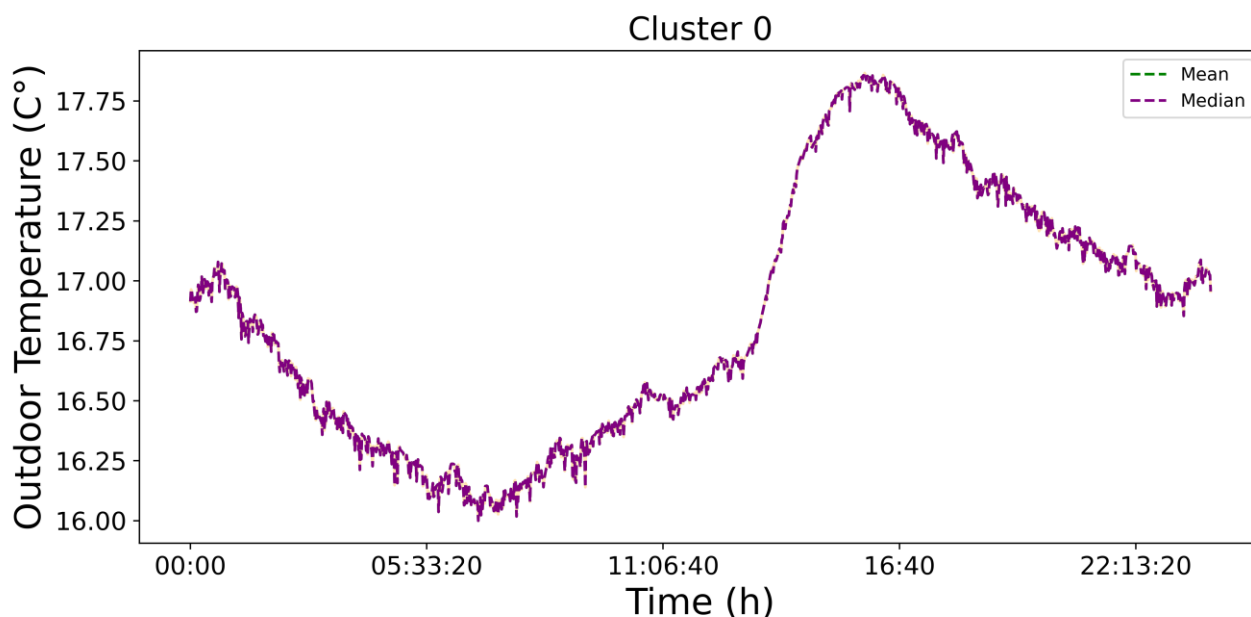


Figure 62: Cluster 0 for the temperature dimension

4.2.5. Outlook for the future expected release

Summing up, the current research endeavour highlights the critical importance of meticulous analysis in understanding heating load profiles and optimising heating system operations, particularly in the context of energy savings and DR. Through the evaluation of clustering algorithms and distance metrics across five dimensions, we have identified distinct patterns and correlations within the dataset, providing valuable insights for stakeholders.

In Figure 51 the methodology that we are going to follow into all releases is presented. As can be clearly seen from the diagram, we have worked into three dimensions (heat demand, boiler and temperature) of the problem and evaluated the results for all 3-distance metrics chosen for this task. Thus, the 2 dimensions that are going to be created, tested and evaluated are going to be the user and building dimension. Furthermore, the optimal dimensions of the problem will be showcased when the analysis has already taken places for all the facets, with the corresponding similarities between them. That way, the true intricacies of heating load profiles will be present for all the dimensions of the problem.



4.3. Service C - Energy disaggregation for the electricity pilot

4.3.1. Overview of the service

Table 16: Service C overview.

Service C	Description
Service / Software title	Energy disaggregation for the electricity pilot (HERON)
Responsible partner(s)	NTUA (main contributor), HERON (supporting)
Associated pilot site	Pilot 3, Greek– electricity domain
Service brief overview	This service utilizes Non-Intrusive Load Monitoring (NILM) techniques to disaggregate energy consumption data at the appliance level
Associated TE	TE-4, TE-11
Associated Use Case	HLUC1 / UC1&UC2, HLUC4 / UC11, HLUC5 / UC13
Functional Requirements (input by other activities)	n/a
Data Input	- Smart meter data: power - Smart plug data: power (in support and for the validation of NILM models)
Service Output	Disaggregated load of appliances as input in DR services and the Flexibility Modelling Tool
Repository Location	DEDALUS GitHub
License information	Proprietary
TRL	TRL 7-8
Current version number	v0.3
Expected date of finalization	May 2025, M25
Development Status	2nd version ready



4.3.2. Architectural diagram

Energy disaggregation, also known as NILM, is a process that aims to break down the total energy consumption of a household or building into the individual energy usage of each appliance. This technique allows users to gain insights into their energy consumption patterns without the need for intrusive monitoring systems. The proposed methodology is depicted in Figure 63. This methodology outlines the systematic approach taken to collect, preprocess, train, and evaluate the data, ultimately leading to the identification of individual appliance-level energy consumption. The process is divided into several key phases, each contributing to the overall goal of accurate and efficient energy disaggregation.

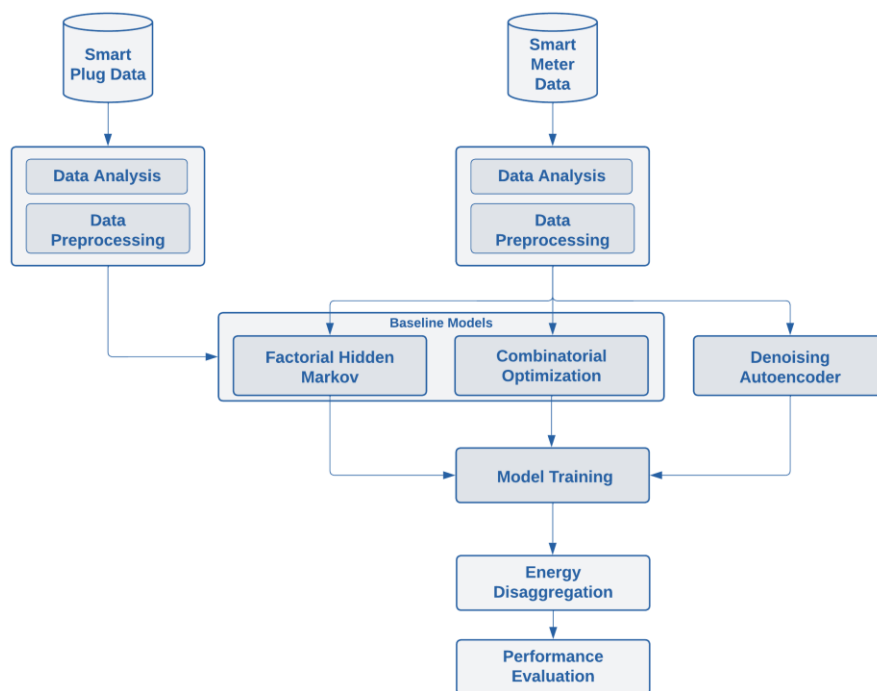


Figure 63: Energy disaggregation - Proposed Methodology

Here below the explain of each component:

- **Data Analysis and Data Preprocessing:** This phase involves analysing the collected data for patterns and anomalies. Data preprocessing includes cleaning, normalization, and feature extraction, ensuring the data is in optimal form for model training.
- **Model Selection and Training:**
 - **Denoising Autoencoder (DAE):** A neural network-based approach that learns a compressed representation of the input data while denoising it. This helps in extracting the features for energy disaggregation by reconstructing appliance-level energy consumption from noisy aggregate data.
 - **Baseline Models:** Advanced machine learning models such as FHMM and Combinatorial Optimization (CO) are used as baselines to compare the performance of the Denoising Autoencoder.



- **Energy Disaggregation:** The trained models disaggregate the total energy consumption into individual appliance-level data.
- **Performance Evaluation:** The disaggregated data is evaluated using metrics like accuracy, precision, recall, and F1-score to ensure the models' effectiveness.

The entire process is designed to be automated, with real-time data processing and analysis being conducted without the need for constant expert oversight.

4.3.3. Main benefit and innovation

Main Benefit: The primary benefit of this energy disaggregation approach is the ability to provide detailed, appliance-level energy consumption insights in real-time. This granular data enables users to understand their energy usage patterns, identify high-consumption appliances, and implement energy-saving measures more effectively. Additionally, utility companies can leverage this information to offer personalized energy-saving tips and incentives, ultimately contributing to overall energy efficiency and cost reduction.

Innovation: The integration of advanced machine learning models, including the DAE, FHMM, and Combinatorial Optimization (CO), represents a significant innovation in the field of energy disaggregation. The Denoising Autoencoder enhances the system's ability to handle noisy aggregate data and capture intricate energy consumption patterns with high precision. To implement the methodology, two open datasets, UKDAL⁹ and Reference Energy Disaggregation Data (REDD)¹⁰, which provide energy consumption data from real homes were used as well as the HERON platform's real residential dataset for training further enhancing the models' robustness and generalizability.

4.3.4. Development activity

The result of the development activity is summarised as follows:

1. Data Collection:

- **Smart Meter Data:** Collected overall power consumption data from smart meters installed at the pilot site, recording the total electricity usage of the entire household or building. Data from 28 houses were extracted.
- **Smart Plug Data:** Gathered data from smart plugs connected to individual appliances, measuring the power consumption of specific appliances to provide granular data for training and validation. The data includes 29 TVs, 23 A/C units, 10 dishwashers, 19 washing machines, 3 washer-dryer combinations, 3 dryers, 2 EVs, and 1 freezer as shown in Table 17.

⁹ <https://paperswithcode.com/dataset/uk-dale>

¹⁰ <https://tokhub.github.io/dbecd/links/redd.html>



Table 17: Appliances HERON’s dataset

Appliances									
	Houses	TV	A/C	Dish Washer	Washer Machine	Washer Machine / Dryer	Dryer	EV	Freezer
Number	28	29	23	10	19	3	3	2	1

2. Data Preprocessing:

- Data Cleaning: Removed any anomalies, missing values, or outliers in the data to ensure accuracy and consistency.
- Normalization: Scaled the data to a standard range, facilitating better model performance and faster convergence.
- Feature Extraction: Identified and extracted relevant features from the raw data, such as power spikes, usage patterns, and appliance-specific signatures

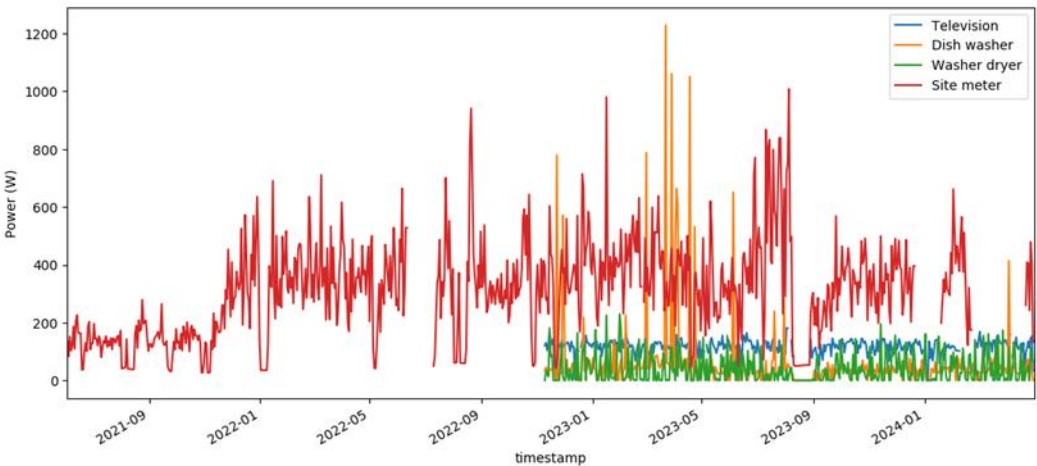


Figure 64: Overview of the consumption of house 2 (id39)

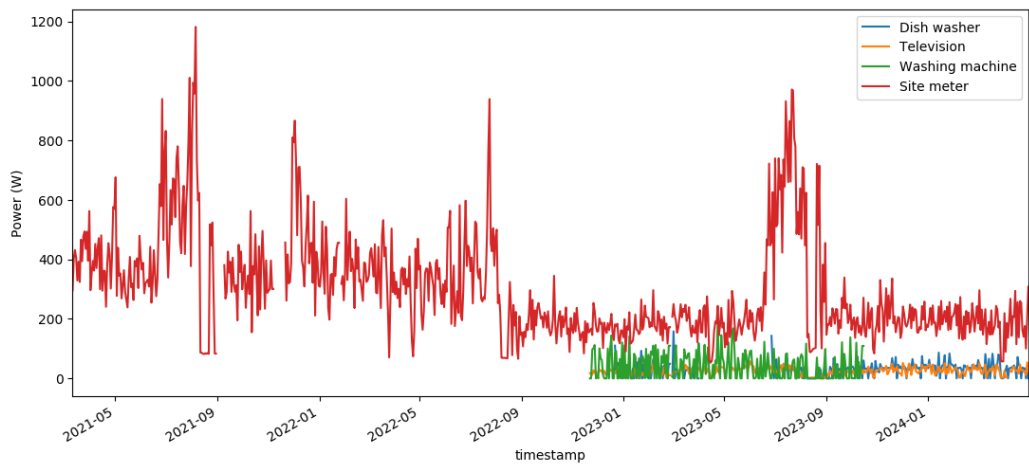


Figure 65: Overview of the consumption of house 3 (id81)



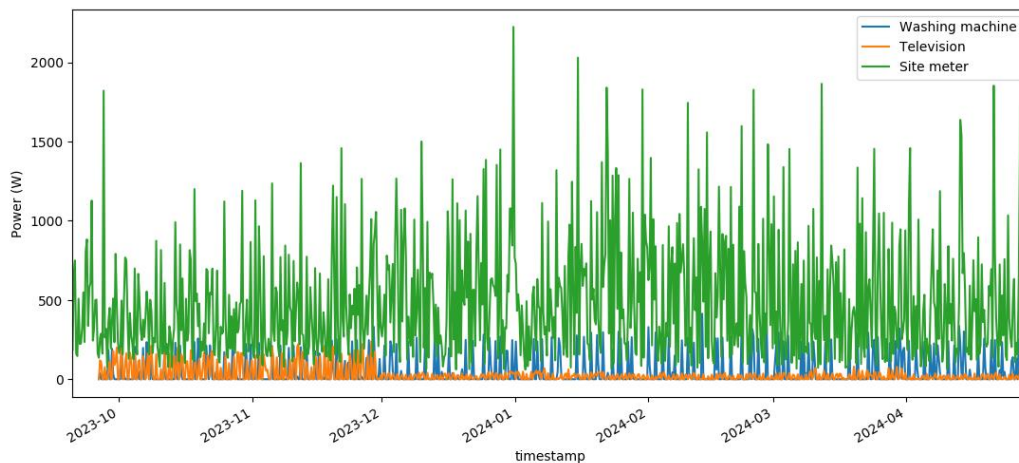


Figure 66: Overview of the consumption of house 27 (id281)

3. Training Phase:

- **Dataset Utilization:**
For the previous release publicly available datasets such as UK-DALE and REDD were employed for training purposes. The UK-DALE (UK Domestic Appliance-Level Electricity) dataset consists of appliance-level power consumption data recorded from several houses in the UK. It includes detailed, time-stamped power usage information, enabling fine-grained analysis and model training. The REDD (Reference Energy Disaggregation Data) dataset includes power consumption data from six houses in the US. It provides appliance-level and aggregate power measurements, serving as a benchmark for energy disaggregation research. In the current release, we utilize the HERON platform's real residential dataset, which includes comprehensive appliance-level energy consumption data from 28 houses, enabling robust model training and validation.
- **Model Selection:**
 - **FHMM:** A probabilistic model that represents multiple sequences of hidden states to capture the dynamics of power usage (Baseline Model).
 - **Combinatorial Optimisation (CO):** An algorithmic approach that solves optimisation problems by combining various parameters to achieve the best energy disaggregation results (Baseline Model).
 - **DAE:** The DAE is a neural network-based model specifically designed for energy disaggregation by separating the energy consumption of individual appliances from the noisy aggregate household consumption. Its architecture combines convolutional layers, dense layers, a flattening layer, and a reshaping layer to process the input and reconstruct the clean signal of a specific appliance. The input layer accepts sequences of energy data, while the convolutional layer extracts features such as transitions caused by appliances turning on or off. The flatten layer prepares the data for dense layers, where a bottleneck layer compresses the data to capture essential features, followed by layers that reconstruct the original structure of the appliance's energy consumption. The final convolutional layer reconstructs the clean appliance-level signal. The model minimizes the mean squared error (MSE) during training, penalizing large errors, and uses



the Adam optimizer to adjust learning rates efficiently. This architecture enables the DAE to denoise aggregate energy data, isolate individual appliance signals, and handle real-world noise effectively, making it a robust solution for NILM (Non-Intrusive Load Monitoring).

- **Training Algorithm:** Each model was trained on the dataset to recognize and predict energy consumption patterns of different appliances.

Table 18: DAE Model Architecture.

Model Layers	Layer Parameters
Input Layer	shape=(sequence_length, 1)
1D convolution layer	kernel_size=4, strides=1, filters=8, activation="linear"
Flatten layer	output_shape=(sequence_length * 8), activation='relu'
Dense layer	units=sequence_length*8, activation='relu'
Dense layer	units=128, activation='relu'
Dense layer	units=sequence_length*8, activation='relu'
Reshape layer	target_shape=(sequence_length, 8)
1D convolution layer	kernel_size = 4, strides=1, filters = 1, activation="linear"

4. Inference Phase:

- **Real-time Data Processing:** Fed pre-processed data from the pilot site into the trained models for inference.
- **Energy Disaggregation:** Each model disaggregated the total energy consumption into individual appliance-level consumption.

5. Evaluation and Comparison:

- **Performance Metrics:** Performance metrics such as accuracy, precision, recall, and F1-score were calculated for each model.
 - **Accuracy:** Measures the proportion of correctly identified appliances and their usage. In more detail, accuracy represents the overall correctness of the model by measuring the ratio of correct predictions (both true positives and true negatives) to all predictions made. It is calculated as the ratio of correctly classified predictions to the total number of predictions. Accuracy gives a general idea of how well the model performs, but it can sometimes be misleading, especially in cases where the dataset is imbalanced, resulting in one class significantly outweighing the other. For example, if an appliance is off for a long period, a model may have high accuracy simply by predicting off most of the time, even if it never captures when the appliances are on.



$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision: Calculates how many of the predictions where the appliance was considered to be operating were actually correct. It is especially important when the cost of false positives is high, meaning that if the model often misclassifies appliances as on when they are really off, it would lead to inaccurate disaggregation results. Precision is calculated as the ratio of true positives to the sum of true positives and false positives, and a high precision means that the model is reliable when predicting that an appliance is in use.

$$Precision = \frac{TP}{TP + FP}$$

- Recall: Measures how well the model can detect all actual instances of appliance activity. It is calculated as the ratio of true positives to the sum of true positives and false negatives. A high recall indicates that the model is effective at detecting appliances in use, even if it sometimes produces false positives. In scenarios where it is critical to capture every instance of an appliance being in use, even at the cost of some false positives, high recall is desirable. For example, in energy disaggregation, recall is important because failing to capture an appliance in use could lead to incomplete or incorrect energy usage data, distorting the energy consumption analysis.

$$Recall = \frac{TP}{TP + FN}$$

- F1-Score: It is particularly useful when we want to balance the trade-offs between recall and precision. It provides a single metric that considers both false positives and false negatives, making it particularly valuable when the cost of these two types of errors is equally important. A high F1 score indicates that the model performs well both in terms of recall and precision, meaning that it is effective in detecting when the appliance is being used and in avoiding misclassification when it is not being used. This balance is critical in NILM, where both overestimation and underestimation of appliance activity can lead to incorrect energy consumption predictions.

$$F1\ Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} = 2 \cdot \frac{TP}{2 \cdot TP + FP + FN}$$

Below, we present the experimental results for the REDD dataset. The disaggregation method DAE was trained on data from House 1 from April 18, 2011, to April 30, 2011, for the appliances: washing machine, and dishwasher. The effectiveness of it was then evaluated from April 30, 2011, to May 24, 2011, in House 1. We compared our model DAE against the built-in models of the



NILMTK¹¹, specifically Combinatorial Optimizatio¹² (CO) and the FHMM, to provide a benchmark for assessing performance.

Table 19: Performance Metrics REDD (Washing Machine)

	CO	FHMM	DAE
Recall	0.8762	0.7178	0.8152
Precision	0.2752	0.4785	0.6757
Accuracy	0.9207	0.9653	0.6107
F1 Score	0.4189	0.5742	0.7389

Table 20: Performance Metrics REDD (Dishwasher)

	CO	FHMM	DAE
Recall	0.588	0.6678	0.9796
Precision	0.0958	0.0501	0.7499
Accuracy	0.7349	0.4233	0.7398
F1 Score	0.1647	0.0933	0.8495

For the washing machine, CO achieves the highest recall (0.8762), effectively detecting most activations, followed by DAE (0.8152) and FHMM (0.7178). In precision, DAE leads with 0.6757, indicating fewer false positives compared to FHMM (0.4785) and CO (0.2752). FHMM stands out in accuracy (0.9653), surpassing CO (0.9207) and DAE (0.6107). The F1 Score highlights DAE (0.7389) as the most reliable model, followed by FHMM (0.5742) and CO (0.4189).

For the dishwasher, DAE achieves the highest recall (0.9796), significantly outperforming CO (0.588) and FHMM (0.6678). Precision further underscores DAE's superiority (0.7499), while CO (0.0958) and FHMM (0.0501) struggle with high false positives. In terms of accuracy, DAE (0.7398) and CO (0.7349) perform closely, while FHMM lags behind (0.4233). The F1 Score clearly favours DAE (0.8495) over CO (0.1647) and FHMM (0.0933).

Overall, DAE consistently outperforms the other models, particularly in recall, precision, and F1 Score, making it the most effective model for detecting appliance usage. While CO performs well in recall for the washing machine, its low precision impacts overall reliability. FHMM demonstrates high accuracy but lacks

¹¹ <https://nilmtk.github.io/>

¹² https://en.wikipedia.org/wiki/Combinatorial_optimization



in other metrics, particularly for the dishwasher. This positions DAE as the most balanced and reliable model for energy disaggregation.

We further tested the DAE using HERON's dataset in three houses (2, 3, and 27) for the appliances: TV, washer-dryer, dishwasher, and washing machine. These tests demonstrate the model's adaptability to diverse datasets and its effectiveness across multiple devices in real-world residential settings. The performance metrics for each appliance are summarized on the tables below:

Table 21: Performance Metrics HERON Dataset

Metric/device	TV	Washer Dryer	Dish Washer	Washing Machine
Recall	0.9984	0.0863	0.0449	0.0797
Precision	0.9798	0.9740	0.9814	0.9694
Accuracy	0.9783	0.1134	0.0741	0.1082
F1 Score	0.9890	0.1586	0.0860	0.1473

These results underscore the DAE's consistent strength in precision across all appliances, with exceptional performance for the TV, achieving an impressive F1 Score of 0.9891. While the model shows lower recall and accuracy for appliances such as the dishwasher and washer-dryer, it's important to note that in the context of DR, the primary goal is to reliably capture activation events. The DAE demonstrates a strong capacity to detect these activations with high precision, which is critical for enabling effective DR strategies. These findings highlight the model's adaptability and potential to provide valuable insights into appliance-level usage patterns across diverse houses and devices, with room for further refinement to enhance recall for specific applications.

6. Output Delivery:

Appliance Identification: Analysed disaggregated data to identify the operating appliances. The following plots illustrate the appliance-level predictions for each device:

TV:

Figure 67 displays the predicted (blue line) and actual (orange line) power consumption over time. The model captures the general trends and most activation events of the TV. However, there are noticeable discrepancies, such as underestimations during peak usage periods and occasional delays in detecting activation cycles. These issues suggest the model might be less sensitive to sustained usage levels or abrupt changes. Despite these challenges, the high precision (0.9798) and F1 Score (0.9890) indicate that the majority of the detected events are accurate, showcasing the model's overall strong performance in distinguishing TV usage from other appliances.



Disaggregation Results for TV

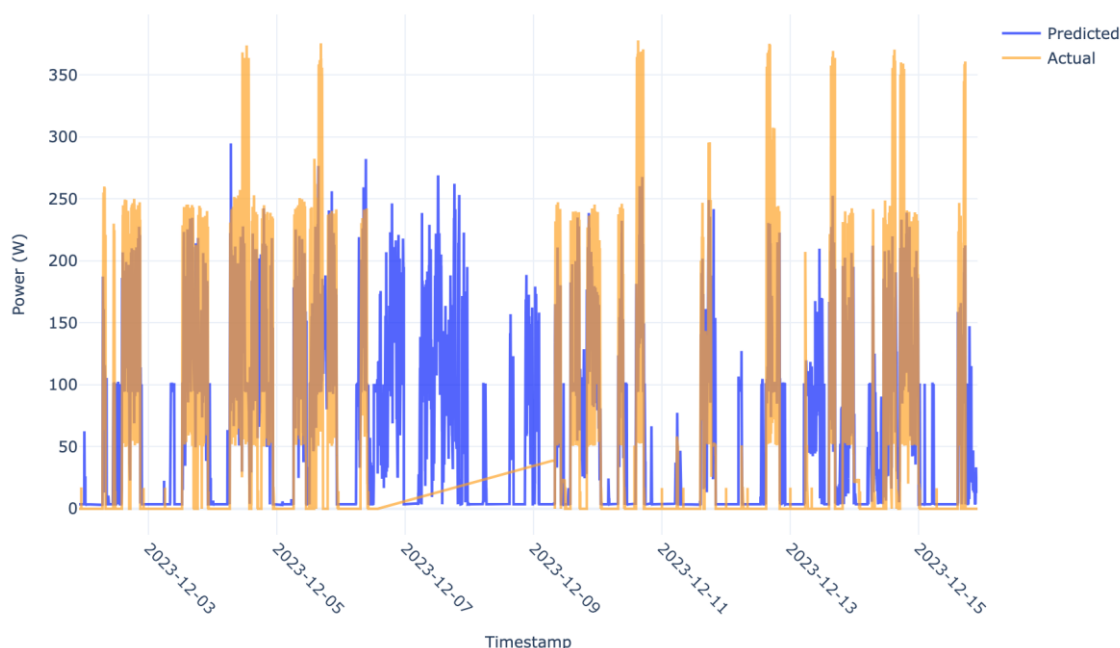


Figure 67: Disaggregation Results for TV

Dishwasher:

The dishwasher plot (Figure 68) demonstrates the predicted (blue line) and actual (orange line) energy consumption patterns. The model successfully identifies major activation events, demonstrated by the alignment of significant spikes. However, there are cases where the model overestimates power usage compared to actual consumption, indicating room for optimization in minimizing overpredictions. Additionally, small activation cycles are occasionally missed, impacting recall. This aligns with the performance metrics, which show high precision (0.9814) but lower recall (0.0449) and accuracy (0.0741), signifying a need for improvement in detecting all activation events.



Disaggregation Results for Dishwasher

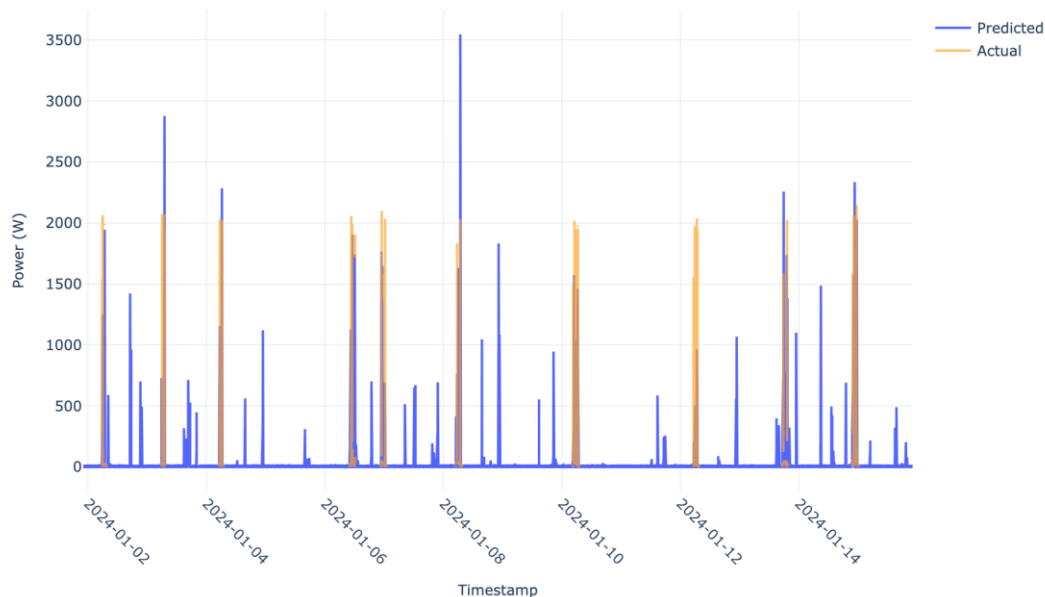


Figure 68: Disaggregation Results for Dishwasher

Washer-Dryer:

Figure 69 shows the predicted (blue line) and actual (orange line) power consumption over time. The model captures most major activation spikes accurately, with reasonable alignment during high-power events. However, there are noticeable instances of underprediction, where the predicted power does not fully match the actual usage, particularly during sustained or prolonged activation periods. Smaller activation cycles are frequently missed by the model, contributing to the low recall (0.0863). The high precision (0.9740) indicates that most of the detected activations are correct, but the low recall and accuracy (0.1134) highlight the need for improvement in detecting all activation events, especially those with lower power consumption or shorter durations.



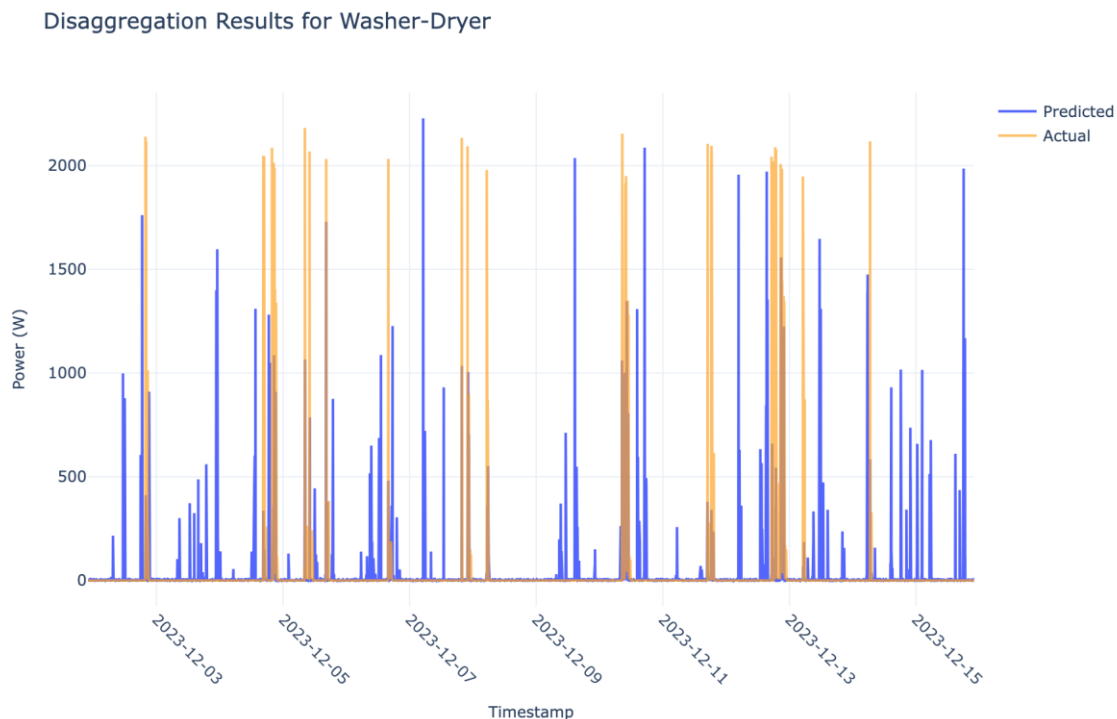


Figure 69: Disaggregation Results for Washer-Dryer

Washing Machine:

For the washing machine, Figure 70 illustrates the predicted (blue line) and actual (orange line) power consumption over time. The model identifies significant activation events accurately, with noticeable alignment during major usage spikes. However, discrepancies persist in cases where the model underpredicts actual power usage, particularly during sustained high-power periods. Smaller activation events are often missed, which contributes to the low recall (0.0797) and accuracy (0.1082). Despite these challenges, the high precision (0.9694) indicates that when an activation is predicted, it is generally correct, highlighting the model's strength in avoiding false positives but its limitations in detecting less prominent events.



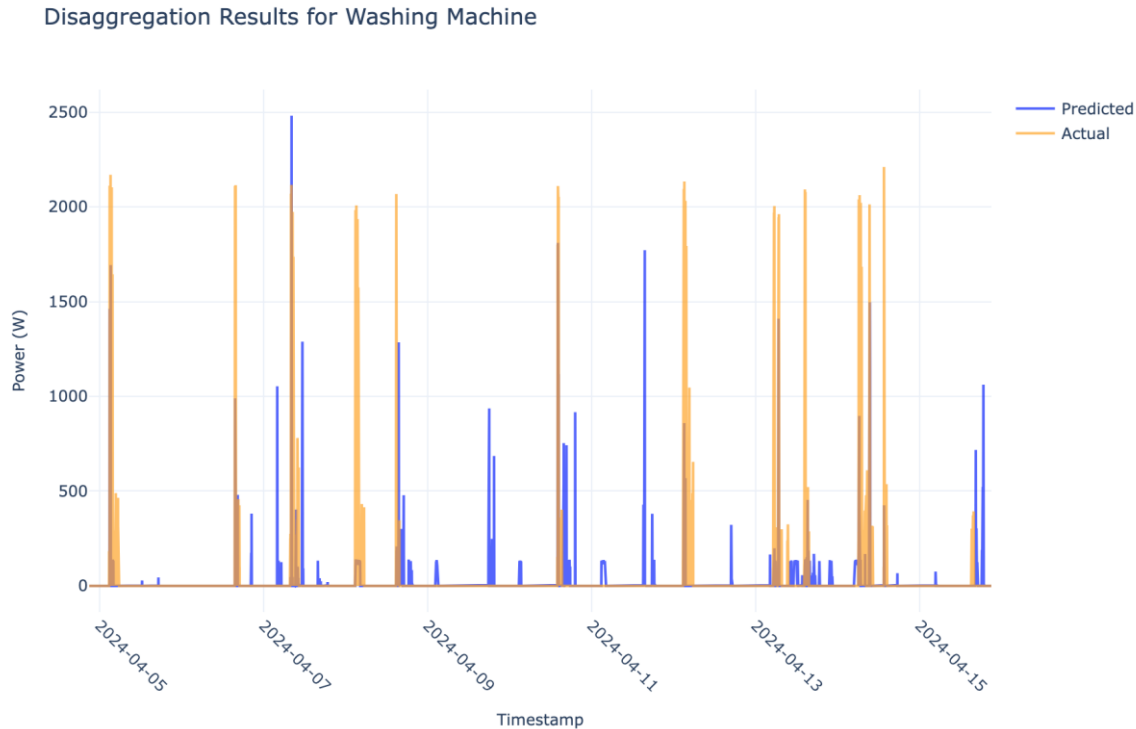


Figure 70: Disaggregation Results for Washing Machine

4.3.5. Outlook for the future expected release

In the upcoming release, we will further refine the current models and develop new ones to expand coverage across all remaining devices. Furthermore, the disaggregated loads generated by these models will provide critical inputs for DR services and the Flexibility Assessment Tool (TE-4), enabling more precise energy management and optimization.

4.4. Service D - Development of tariff system for implicit and explicit electricity DR services

4.4.1. Overview of the service

Table 22: Service D overview.

Service D	Description
Service / Software title	Development of tariff system for implicit and explicit electricity DR services
Responsible partner(s)	HERON
Associated pilot site	Pilot 3, Greek– electricity domain



This project has received funding from European Union's Horizon Europe Research and Innovation programme under the Grant Agreement No 101103998

Service D	Description
Service brief overview	This service aims at generating a tariff system based on multiple conditions and their combination. For example, the tariff can be based on environmental (i.e. shares of renewable) or market (day-ahead market) signals or their combination.
Associated TE	TE-11
Associated Use Case	HLUC5 / UC13, UC14
Functional Requirements (input by other activities)	TE11-FR1
Data Input	Day-ahead and Transmission System Operator (TSO) data. For GR pilot, such data is prepared via APIs from HERON.
Service Output	Files in JSON format with 24-hour tariff (electricity prices, RES shares or token points) as an incentive to engage in DR services
Repository Location	Business cloud system
License information	Proprietary for the hybrid version (combination of market and sustainability tariffs) and open-source versions for market data
TRL	TRL 7-8
Current version number	V2.0
Expected date of finalization	May 2025, M25
Development Status	Second version of the service utilising data from the Integrated Scheduling Programming (ISP) and the GR DAM is available.

4.4.2. Architectural diagram

Figure 71 shows the high-level architectural diagram of the service.



This project has received funding from European Union's Horizon Europe Research and Innovation programme under the Grant Agreement No 101103998

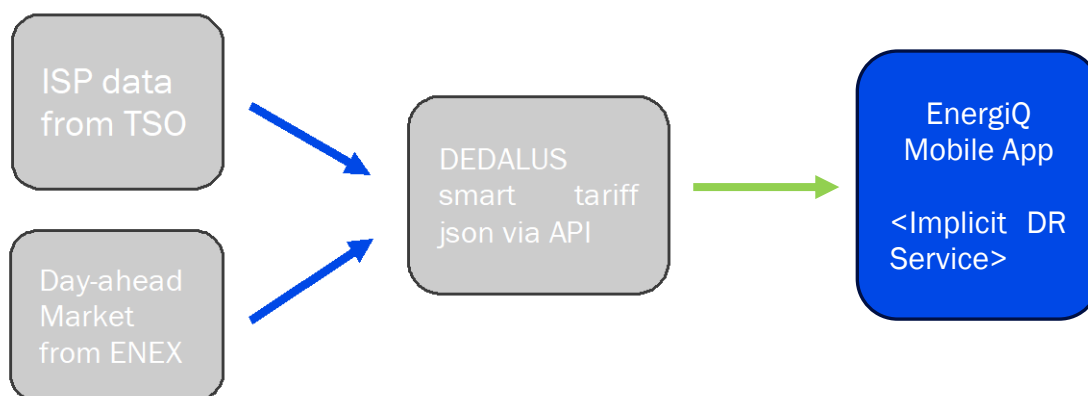


Figure 71: Service D - Architectural diagram v2.0

4.4.3. Main benefit and innovation

Benefit: The main benefit of this service is the development of a tariff that will be used for the DR service by utilizing publicly available data. It enables consumers to decrease their carbon footprint while they benefit from the wholesale market volatility and can be integrated seamlessly to existing DR services. Version 2.0 of this service is actively used in HERON's EnergiQ mobile app and is expected to be integrated in Implicit DR service (Service F).

Innovation: There is some innovation on the current version of the DEDALUS DR tariff as it sorts the Renewable Energy Sources percentage in the system and integrates multiple data sources to provide realistic 24 hour ahead information relying on public sources, without the uncertainty associated with forecasting methods. However, it should be noted that this greater innovation potential relies on the development of hybrid tariffs that combine market and system mix signals as indicated by the limited literature on dynamic market tariffs that take into consideration short-term estimation of Renewable Energy sources.

4.4.4. Development activity

HERON has developed the first version of the service calculating the percentage of system load satisfied by RES, drawn from Load and RES generation from the ISP provided by the TSO.

The ISP is solved as a co-optimisation problem considering the balancing energy and Balancing Capacity Offers of the BSEs as well as their respective constraints, transmission system constraints and the TSO's needs, in order to minimize the cost of balancing energy and balancing capacity procurement. Clearly, RES and System Load forecasts during the ISP process are as accurate as they can be, given the importance of ISP in the energy stability of the system.

For the Greek system, the ISP is managed by the Greek TSO (IPTO) and is executed at three scheduled times for each dispatch day D for half hour blocks as follows:



- ISP1 is executed at 16:15 EET on calendar day D-1 and concerns all dispatch periods (48 dispatch periods) of dispatch day D – Forecasts are published
- ISP2 is executed after ISP1, considering the updated input data. It is executed at 00:00 EET on calendar day D and concerns all dispatch periods (48 dispatch periods) of dispatch day D.
- ISP3 is executed at 12:00 EET on calendar day D, considering the updated input data. The time interval considered is from 13:00 EET till the end of dispatch day D (24 dispatch periods).

It may also be carried out on demand, each time the TSO deems that there are significant changes that need to be balanced. For the purposes of the Implicit DR solution under development we are collecting RES (HV assets) and load forecasts as they are generated from the TSO. The TSO's ISP forecasts are not issued based on the aforementioned ISP schedule.

According to the relevant code¹³, ISP forecasts are given as follows:

- ISP1 zonal load and RES forecasts and system requirements are published on IPTO's website until 09:30 EET on calendar day D-1 (v1 forecasts & requirements)
- ISP1 zonal load and RES forecasts and system requirements are updated and published on the website until 13:30 on calendar day D-1 (v2 forecasts & requirements)
 - ISP1 is executed at 16:15 EET, D-1
- ISP2 zonal load and RES forecasts and system requirements are published on IPTO's website until 21:00, D-1
 - ISP2 is executed at 00:00 EET, D
- ISP3 zonal load and RES forecasts and system requirements are published on IPTO's website until 09:00, D
 - ISP3 is executed at 12:00 EET, D

The data can be found from the Greek TSO's <website>¹⁴ under <Market / Market Statistics / Data> by downloading the "ISP Requirements" which includes RES and load forecasts and mandatory hydro declarations given that in Greece, Hydro, although a renewable energy source in principle, is allocated in the same category, in terms of system planning, with the dispatchable energy sources. Alternatively, the <File Download API> ¹⁵ can be used to automate file downloads.

Although this information is publicly available, handling the data requires some specialist knowledge due to TSO APIs and csv files containing Greek market specific data points (such as the list of pump storage hydro plants without specifying whether they are used as generators or as flexible loads). To alleviate such complexities, HERON automated the process through the development of an API which provides information to DEDALUS partners from further developing the DR and Digital Twins services, without technical knowledge of the Greek market.

¹³ IPTO Technical Decision: Integrated Scheduling Process (version 2.0, February 2021). Accessed from https://www.admie.gr/sites/default/files/users/dda/KAE/TECHNICAL%20DECISION%20ISP_eng_v2.pdf

¹⁴ <https://www.admie.gr/en>

¹⁵ <https://www.admie.gr/en/market/market-statistics/detail-data>



Tests identified issues with the final outcome due to characteristics of ISP schedule which combine market schedules, bids and observed generation and consumption from market entities. As it can be seen in Figure 72, RES percentages above 100% occurred. The reason for this is that the formula calculating v1.0 did not include exports which act as load.

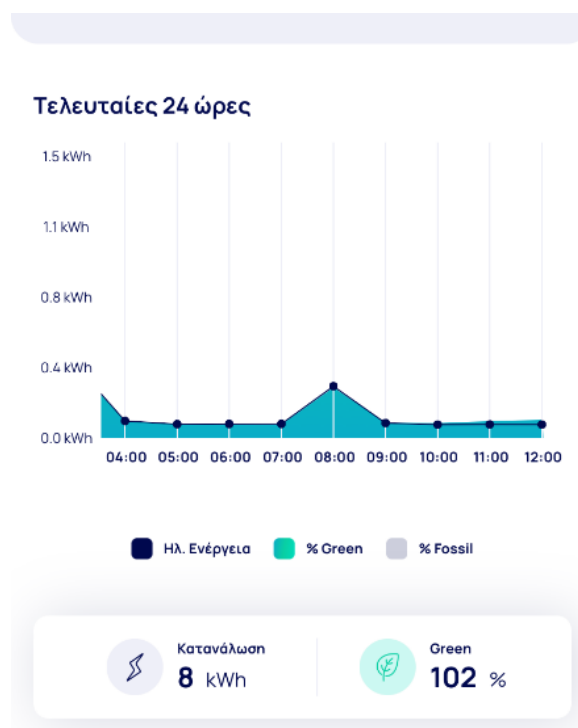


Figure 72: Green tariff v1.0 implementation

Including export decreased the percentage but required additional data to be integrated in the formula. The sum of exports over all interconnections known up to ISP1 results was calculated from data from DAM and CRIDA1 (GR intraday market solved before 20:00). The flow of the interconnection was taken into consideration, distinguishing from exporting load and importing generation based on the notation (i.e. XX-GR where XX is the interconnected country from which electricity flows to GR and GR-XX where XX is the interconnected country to which electricity flows from GR). The methodology was validated for specific past instances in which percentages above 100% in the first version of the service occurred as seen in Table 23.



Table 23: Demonstration of RES percentages for Service v1.0 and v2.0

Datetime	DR	Load	Losses	RES	Hydro	PERCENTAGE v1.0	PERCENTAGE v2.0
2024-03-11 07:00:00	0	5105	84.5	2690	511	61.68%	54.40%
2024-03-11 08:00:00	0	5500	90	3585	503.5	73.14%	63.50%
2024-03-11 09:00:00	0	5595	91.5	4550	361	86.36%	72.94%
2024-03-11 10:00:00	0	5640	102.5	5290	158	94.87%	77.90%
2024-03-11 11:00:00	0	5625	115	5775	195.5	104.02%	84.54%
2024-03-11 12:00:00	0	5660	124	5975	190.5	106.60%	85.51%
2024-03-11 13:00:00	0	5635	133.5	5950	190.5	106.45%	84.13%
2024-03-11 14:00:00	0	5535	134.5	5695	183	103.68%	82.05%
2024-03-11 15:00:00	0	5515	132	5155	185	94.56%	76.70%
2024-03-11 16:00:00	0	5385	126.5	4410	232.5	84.23%	68.51%
2024-03-11 17:00:00	0	5330	127	3700	423	75.55%	62.68%
2024-03-11 18:00:00	0	5640	149.5	3345	900.5	73.33%	60.05%

4.4.5. Outlook for the future expected release

- Development of a hybrid tariff combining market and environmental elements i.e. identify the intervals with lowest prices and relative high shares of renewables (+50%).
- Integration of the hybrid tariff as an API to be used in DR related services for frontrunner and multiplier.
- Integration of data sources for the multiplier pilots (IT, ES)
- Development of environmental, market and hybrid tariffs for multiplier pilots (IT, ES)



This project has received funding from European Union's Horizon Europe Research and Innovation programme under the Grant Agreement No 101103998

4.5. Service E - OurPower Nudging Application

4.5.1. Overview of the service

Table 24: Service E overview.

Service E	Description
Service / Software title	OurPower Nudging Application
Responsible partner(s)	OurPower, BluePrint
Associated pilot site	Pilot 1, Austria
Service brief overview	The OurPower Nudging Application enhances user engagement and optimizes energy consumption within communities. By providing personalized advice and enabling user responses to energy-saving opportunities, the service aims to foster proactive behaviour changes and enhance community-wide energy efficiency. Furthermore, the nudging application serves as a key information channel, offering data on energy production and consumption.
Associated TE	TE-10
Associated Use Case	UC22
Functional Requirements (input by other activities)	TE10-FR2 The system will deliver personalized energy utilization (saving) recommendations to users.
Data Input	• Energy Production Data: Information about the historic and forecasted energy generation from Photovoltaic (PV) systems. • Energy Consumption Data: Data on historic and forecasted energy consumption by the community and individual users. • User Preferences and Historical Responses: Information on user behaviour, preferences, and previous responses to nudging requests to tailor advice more effectively. • Community Event Information: Details about upcoming community dialogs and online events to be communicated to users.
Service Output	• Personalized Energy Consumption Advice: Specific recommendations for users to adjust their energy usage based on historic and forecasted data. • Improved Energy Management: Contribution to overall community energy optimization by encouraging and facilitating individual actions in response to dynamic energy situations.



Service E	Description
	• User Interaction Data: Records of how users respond to the advice and interact with the app, which can be used to further refine the nudging strategy and improve the app. • Enhanced Community Engagement: Increased user participation in community through direct communication and feedback opportunities within the app.
Repository Location	OurPower private Github repository
License information	Proprietary
TRL	6
Current version number	v0.2
Expected date of finalization	April 2025, M26
Development Status	The second version integrating new design and nudging mechanisms has been developed, development of all features is still ongoing.

4.5.2. Architectural diagram

The architectural diagram specification of the OurPower Nudging Application and its connected components/services is shown in Figure 73. At the core lies the Pocketbase¹⁶-based OurPower Nudging Application backend, serving as the foundational element for data management and processing. This backend integrates seamlessly with various data sources, including energy monitoring services and the OurPower Community Manager, which provides metering data retrieved from the EDA portal. The metering data is not stored directly in the nudging application backend but is accessed from the OurPower Community Manager through established integrations. Additionally, the backend will interface with forecast services to acquire energy and weather forecasts crucial for generating accurate nudging recommendations. On the frontend, the application utilizes Ionic¹⁷, a versatile SDK, to create an intuitive interface for end-users across multiple platforms, including Android, iOS, and web. A landing page will serve as the initial entry point for participants to onboard onto the OurPower Nudging Application.

¹⁶ Pocketbase. <https://pocketbase.io/>

¹⁷ Ionic Framework. <https://ionicframework.com/>



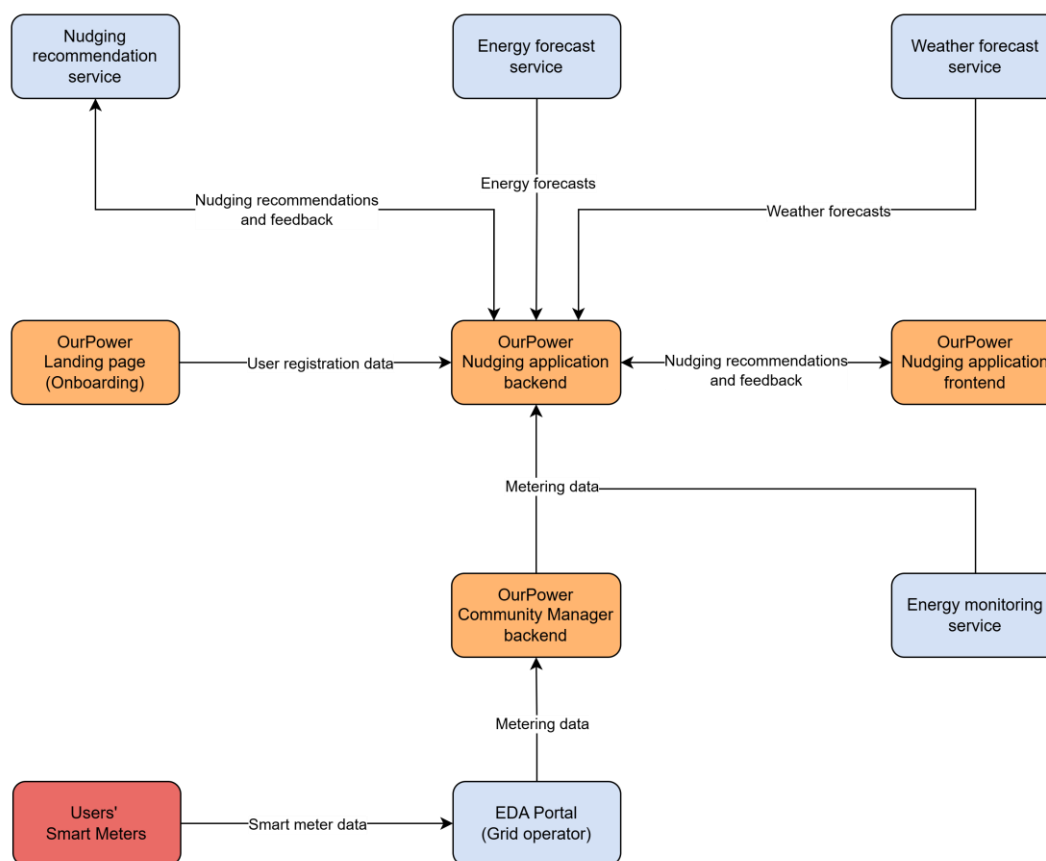


Figure 73: Architectural diagram of the OurPower Nudging Application and connected components/services

4.5.3. Main benefit and innovation

The OurPower Nudging Application presents a transformative solution for optimising energy consumption and fostering proactive behaviour changes within communities. By leveraging personalized advice and user responses to energy-saving opportunities, the service enhances user engagement and community-wide energy efficiency. Its innovative approach serves as a key information channel, providing valuable insights into energy production and consumption. The main benefit lies in its ability to deliver personalized energy utilization recommendations, empowering users to make informed decisions based on historic and forecasted data. Moreover, the application contributes to improved energy management at the community level by encouraging individual actions in response to dynamic energy situations. Through user interaction data and enhanced community engagement, the OurPower Nudging Application not only drives efficiency but also cultivates a more responsive and sustainable energy ecosystem.

4.5.4. Development activity

The nudging application utilizes Pocketbase for backend services and Ionic for the frontend, ensuring a robust, scalable, and user-friendly experience. Pocketbase is an open-source backend solution that combines database and server functionalities into a single package. The backend is hosted on servers managed by OurPower, leveraging the organization's existing



infrastructure. Ionic is a powerful HTML5 SDK that facilitates the development of mobile applications using web technologies. It is particularly well-suited for cross-platform applications like the nudging application, which will be available to end-users as Android, iOS, and web applications.

In the past six months, development efforts have focused on finalizing the user interface, integrating the Austrian Energy Data Exchange (EDA) via the OurPower Community Manager backend and setting up the backend of the nudging application. EDA data is processed and stored within the OurPower Community Manager, which serves as the intermediary for retrieving this data. The nudging application backend then accesses this data through the established integration, ensuring seamless availability for analysis and recommendations. The backend has been configured to efficiently support these processes, leveraging OurPower's existing infrastructure for reliability and scalability.

Following the validation of earlier mock-ups during the co-creation workshop in WP2 in May 2024, additional feedback sessions with selected users were conducted. These sessions provided valuable insights into the application's navigation, user interface, and key features. These inputs informed refinements to align the application's functionality with user needs and expectations. Mock-ups of the latest version of the application are shown below in Figure 74.



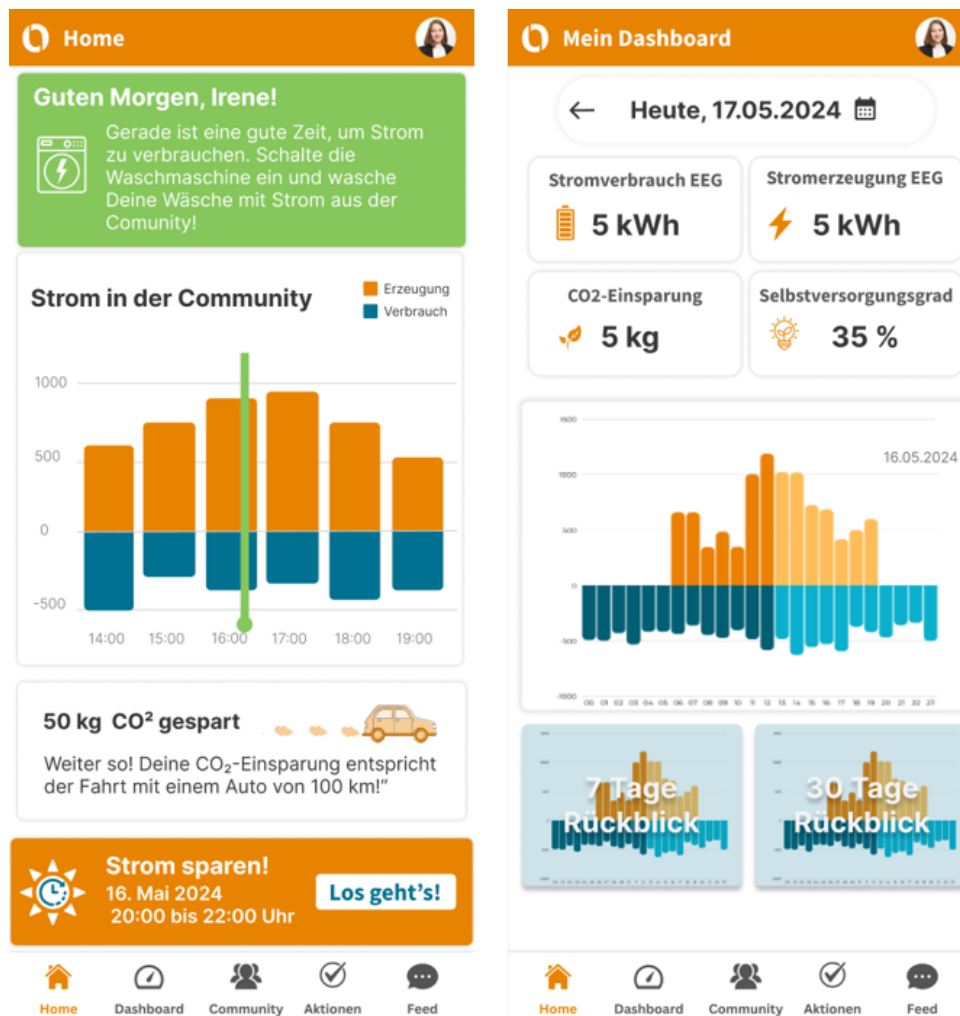


Figure 74: Mock-ups of the OurPower Nudging Application

Based on the user feedback and research into additional nudging mechanisms, the feature scope of the application has been expanded to include:

- **Individual View:** Displays data and statistics on energy consumption and production, including forecasts and impact evaluations.
- **Community View:** Provides aggregated statistics on energy consumption and production for the energy community, shared forecasts, impact evaluations, and access to essential documents and invoices.
- **Daily Nudges:** Offers general, actionable suggestions, integrates a feedback mechanism, and includes calendar functionality to support energy-saving actions.
- **Community Feed:** Facilitates interactions between managers and members of the energy community, serving as an engaging nudging tool.
- **Education and Quiz Features:** Delivers interactive content to promote energy awareness and reinforce energy-saving behaviours.
- **Personalization Options:** Allows users to customize profiles, notifications, and app design, enhancing engagement and the effectiveness of nudges.



These features are under active development, with the focus on seamless integration and an optimal user experience. To streamline development and simplify implementation, the nudging recommendation algorithm is being revised to deliver personal nudges based on aggregate data and predefined rules. This approach provides meaningful and actionable suggestions while maintaining simplicity and reducing computational complexity.

4.5.5. Outlook for the future expected release

The planned developments for future expected releases of this service include:

- Finalize feature development.
- Complete refinement of the nudging algorithm.
- Develop and deploy landing page for the nudging application.
- Integration with other components, e.g., pre-aggregation framework.
- User testing and validation.

4.6. Service F - Multi-value Demand Response for virtual clusters of building

4.6.1. Overview of the service

Table 25: Service F overview.

Service F	Description
Service / Software title	Multi-value Demand Response for virtual clusters of building
Responsible partner(s)	HERON (main contributor), NTUA (supporting)
Associated pilot site	Pilot 3, Greek – Electricity domain vector
Service brief overview	The Multi-Value DR service for virtual clusters of buildings is designed to optimise energy consumption by utilising optimisation strategies to synchronise it as much as possible with renewable energy availability and minimising carbon footprint. Firstly, the household assets are categorised into three types: Run-to-Completion assets (e.g., washing machines, dishwashers), Start-Stop assets (e.g., boilers, air conditioners, EV chargers), and Always-On assets (e.g., any other asset that runs continuously). This classification enables tailored recommendations for efficient energy usage and ensures that energy demand could be shifted to align with periods of high renewable energy availability. By leveraging forecasted RES share through the integration of green dynamic tariffs and household-specific constraints, the service aims to promote and shift long-term environmental sustainability by



Service F	Description
	reducing reliance on carbon-intensive energy sources through informed and flexible energy management strategies.
Associated TE	TE-11
Associated Use Case	UC 13
Functional Requirements (input by other activities)	HLUC5
Data Input	24 hour ahead scheduled RES based on the_ green dynamic tariff system (Service D) - Forecasted Total Energy Consumption - Energy Consumption per Asset (Service C) - User-defined preferences for specific appliances (preferred start times, maximum delay allowances) Device-Specific Constraints: - Run-to-Completion Assets: Operation time (duration of cycles). User-defined flexibility for start times (e.g., within a specific time interval). - Start-Stop Assets: User-defined flexibility for start times (e.g., within a specific time interval). - Always-On Assets: Continuous power consumption (fixed load profile).
Service Output	Json via an API
Repository Location	Private HERON GitHub
License information	MIT
TRL	TRL 5
Current version number	V0.2
Expected date of finalization	September 2025, (M29)
Development Status	The service has made significant progress in its development, focusing on the following key milestones: - Definition of service objectives, balancing RES utilization and carbon footprint minimization. - Classification of household assets into operational categories: Run-to-Completion, Start-Stop, and Always-On.



Service F	Description
	• Completion of architecture for data collection, processing, and storage. • Validation of the optimization framework, including data flows, inputs, outputs, and decision-making logic. • Simplistic RES share forecasting integrated into the service. • Preparatory integration for retrieving energy consumption forecasts and appliance-level disaggregation data. • Finalization of the HERON survey to gather household constraints and flexibility preferences. • Preliminary activities to define UX specifications for recommendation formats

4.6.2. Architectural diagram

The Demand Response service has been developed to optimise household energy consumption by aligning it with RES availability and carbon footprint reduction goals. The service architecture is designed to integrate data collection, processing, and optimisation into a seamless workflow while incorporating user preferences and constraints. Below, in the high-level architecture, the main components and their interactions are described in detail. The graph aims to outline the core components and data flow, focusing on modular design, data sources and overall system interoperability.

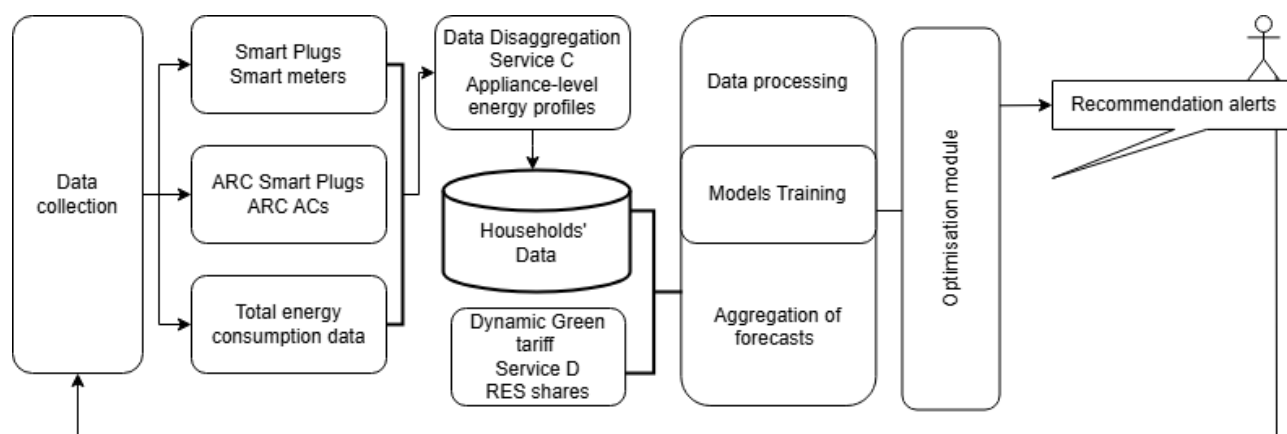


Figure 75: Demand Response high level architecture

The architectural framework begins with data collection, gathering information from various sources to create a comprehensive energy profile for households. Smart plugs and smart meters provide granular device-level energy consumption data, while ARC devices (smart plugs and ACs) contribute additional device-specific information. These data streams are stored in HERON's database and made accessible through an API, which enables seamless communication and data exchange with the other modules.



The disaggregation of device-level energy consumption data is performed by Service C. This service processes the data to generate detailed appliance-level disaggregation and sends the results the DR service via APIs. This ensures that the disaggregated data is accurate and readily available for optimisation and decision-making. In addition to the household data, the architecture integrates grid-level information by integrating the daily (24 hour ahead) expected RES shares calculated by importing data from the market operator (enex) and the TSO (IPTO) databases. Data from the abovementioned sources is accessed via APIs and stored in centralised databases, enabling smooth communication between system components.

All the components are under the interoperability layer, to facilitate seamless data exchange between all system components. This layer ensures that the disaggregation service, demand response module, and other interconnected systems work together cohesively, supporting a streamlined workflow. The integration of ontology-based semantic modelling standardises data formats and enhances interoperability. This approach facilitates seamless communication between the service and external modules, ensuring compatibility across the DEDALUS framework.

The Figure 76 shows the detailed architecture of the demand response where the optimisation engine takes actions once the appliances used at each time have been classified.

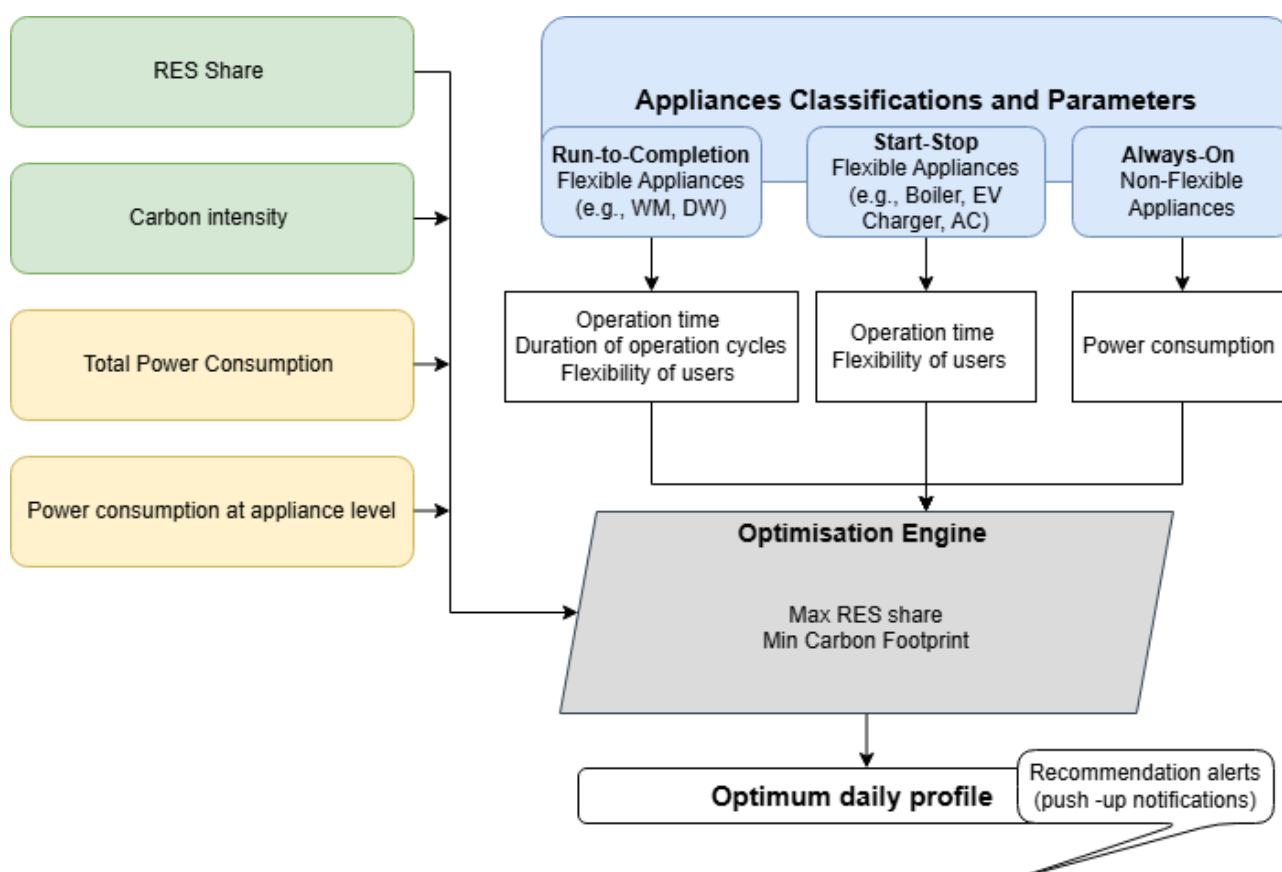


Figure 76: Demand response optimisation engine architecture



Appliances are classified into three distinct categories based on their operational characteristics:

- **Run-to-Completion Flexible Appliances:** These include appliances such as washing machines and dishwashers, which require their operation cycles to be completed once started. Flexibility for these devices is determined by the user's willingness to delay the start time.
- **Start-Stop Flexible Appliances:** This category comprises devices such as electric boilers, EV chargers, and ACs, which can pause and resume operation without compromising functionality. For these appliances, flexibility is defined by the users' willingness to interrupt and defer operation at a different time intervals.
- **Always-On Non-Flexible Appliances:** These are appliances that operate continuously or have fixed power consumption patterns and therefore offer no flexibility for scheduling.

For each appliance category, the system calculates operational parameters, including operation times, cycle durations, and user-defined flexibility time intervals. These parameters are combined with forecasted data, such as the expected RES share, carbon intensity, total household energy consumption, and appliance-specific consumption levels. The resulting dataset is fed into the optimisation engine to develop schedules that align with carbon-aware demand response objectives while respecting user-defined constraints.

The user preferences and constraints gathered through a dedicated survey designed by HERON. This survey collects key information such as the user's willingness to adjust energy usage, preferred flexibility levels, and any specific operational constraints for their appliances. These preferences are stored and utilised in the optimisation process, ensuring the service is user-centric and tailored to individual needs.

The optimization process is the core of the demand response service, as illustrated in Figure 77.



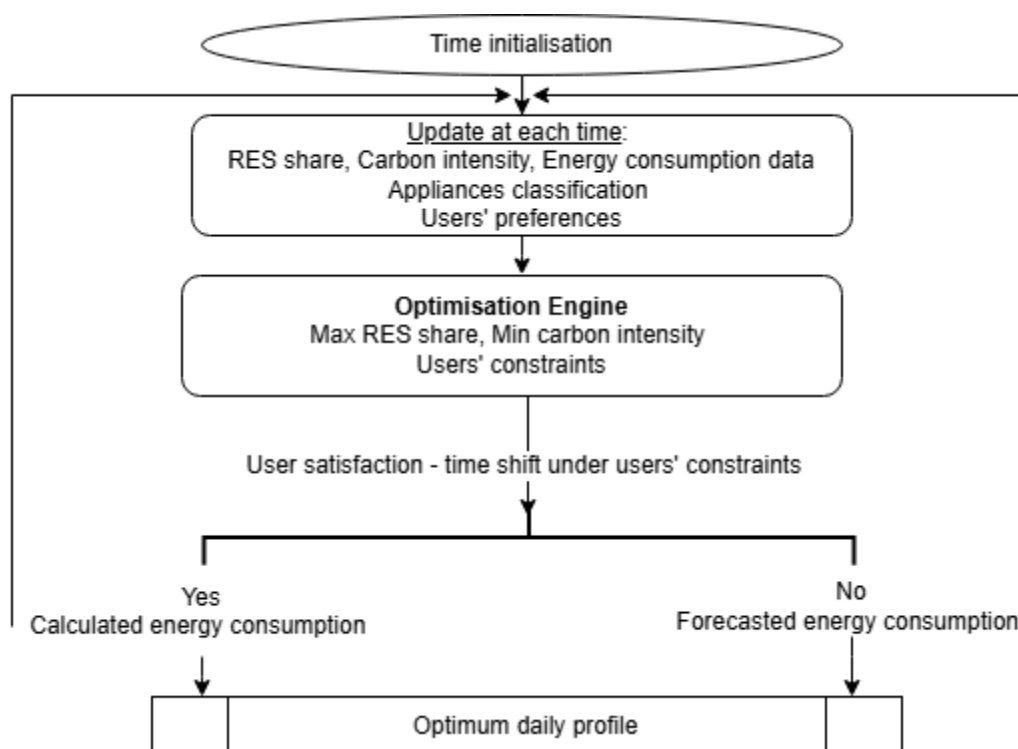


Figure 77: Demand response optimisation workflow.

The process is executed with a time horizon of 24 hours at hourly time steps. At each time step, the system updates critical inputs, including forecasted RES share, carbon intensity, household energy consumption, appliance classifications, and user preferences. This ensures that the optimization engine operates on the most up-to-date data, enabling it to adapt dynamically. The optimization engine processes these inputs to generate schedules that maximize RES utilization and minimize carbon emissions. The engine is programmed to balance multiple objectives, including RES availability, user satisfaction, and operational constraints. For instance, it prioritizes shifting energy-intensive tasks to periods with high-RES availability while adhering to user-specified constraints, such as preferred appliance operation times.

The optimization workflow incorporates a feedback mechanism to verify whether the calculated energy consumption aligns with forecasted energy needs and users' needs. If discrepancies are detected, the schedule is recalibrated, ensuring that energy consumption forecasts are met without compromising the user's constraints. This iterative process continues until the optimized daily energy profile is achieved, which balances utilization of RES potential with individual household needs.

Once the optimum daily profile is finalized, the system delivers the proposed schedules to users in the form of recommendation at a fixed hour with the day (i.e. 20:00 at the previous to recommendations day). These recommendations are sent via push notifications in HERON's EnergiQ mobile app, providing users with actionable insights and enabling them to align their energy usage with the recommendations. The alerts, tailored to each household, clearly communicate the timing and duration of recommended appliance operations, helping users make informed decisions.



4.6.3. Main benefit and innovation

Main Benefit

A key benefit of this service lies in the ability to align household energy consumption with periods of high-RES availability. By leveraging advanced forecasting models and real-time optimization techniques, the service enables households to shift energy-intensive activities to periods when RES potential is highest. This approach reduces reliance on carbon-intensive energy sources and also empowers users to understand their energy behaviour and actively contribute to it towards carbon neutral profiles. By avoiding periods of high carbon intensity, the service helps users minimize their overall carbon footprint.

Another major benefit is the user-centric design of the service. From the early stages of development, user engagement has been a priority, with a dedicated survey designed by HERON to gather detailed information on user preferences and constraints. This survey collects critical data, such as flexibility levels and operational requirements for specific appliances, enabling the system to tailor energy schedules to meet individual needs while adhering to optimization objectives. The push notification mechanism ensures that users receive clear and actionable recommendations, making it simple for them to participate in the demand response process. This personalized approach enhances user satisfaction fostering their active engagement.

Push notifications provide users with detailed information about their energy consumption patterns, RES utilization, and carbon savings. By offering a clear view of their contributions to sustainability, the service fosters behavioural changes and strengthens the connection between users and their energy management goals.

Innovation

The service's innovation lies in the combination of machine learning forecasting and dynamic multi-objective optimization. The optimization problem is formulated as a multi-criteria Mixed Integer Program (MIP) balancing RES utilization, carbon footprint reduction, and user satisfaction. These algorithms enable efficient exploration of complex solutions, identifying optimal trade-offs among competing objectives.

Machine learning models, including regression, neural networks, and ensemble techniques, predict RES share, carbon intensity, and household energy consumption with high accuracy. Furthermore, the appliance-specific classification and scheduling mechanism adds another layer of precision to the optimisation process. By categorising appliances into Run-to-Completion, Start-Stop, and Always-On groups, the service respects the unique operational characteristics of each appliance and use different algorithmic techniques according to their needs.

User engagement is further enhanced by the transparency and insights provided through the service. The push notifications inform users about energy consumption patterns, RES utilisation, and carbon savings, offering them a clear view of their contributions to sustainability. This level of engagement encourages behavioural changes and fosters a stronger connection between users and their energy management practices.



4.6.4. Development activity

The development activities so far have focused on refining core functionalities, enhancing user interaction, and preparing the system for deployment and testing. All source code and development artifacts have been committed and hosted to a centralized cloud-based repository managed by HERON and accessible via HERON's GitHub platform. This repository serves as the main development hub, enabling collaboration among project contributors while ensuring version control and transparency. Access to the repository has been restricted to authorised personnel to maintain security and integrity during the development phase. Contributors (NTUA, ENG) have requested access and integration activities will start in a later stage of the development phase.

The development activities so far have included:

- Integration of RES shares and forecasting models.
- Definition and performance testing of multi-criteria optimization (MIP).
- Appliance classification and parameterisation.
- Design and deployment of the HERON survey to gather user constraints and preferences.
- Incorporation of user preferences into the optimisation process.
- Development of enhanced APIs for interoperability.
- Preparatory activities for ontology adaptation under the DEDALUS framework.

4.6.5. Outlook for the future expected release

The foreseen development activities include:

- Improving the accuracy and granularity of forecast models and validating appliance-level predictive models.
- Advanced behavioural analytics to refine recommendations based on user profiles.
- Enhancements to multi-objective optimization techniques, enabling real-time data processing.
- Development of intuitive visualization tools for user recommendations and energy insights.
- Integration of the DEDALUS components.
- Development of MVP for integration testing with the digital twin, and the front runner pilot and multipliers.
- Integration with the digital twin.
- Integration with multipliers (IT, ES).
- Final validation.

4.7. Service G - Integration of Implicit Demand Response service - Recommendation in the DT

This service aims to integrate the DR recommendations generated as output of Service F with the Digital Twin, providing an intuitive interface for facility managers. The following sections outline the main objectives, key benefits and development activities related to this service.



4.7.1. Overview of the service

Table 26: Service G overview

Service G	Description
Service / Software title	Integration of Implicit Demand Response service - Recommendation in the DT
Responsible partner(s)	ENG
Associated pilot site	Pilot 3, Greek – Electricity domain
Service brief overview	This service aims at integrating the outcome of Service F (Implicit DR recommendations) within the context of the Digital Twin TE-8 in Task 4.1.
Associated TE	TE-8
Associated Use Case	UC 11
Functional Requirements (input by other activities)	TE8-FR5
Data Input	Service G will take as input the outcome of service F.
Service Output	Visualization of DR strategies in the Digital Twin.
Repository Location	DEDALUS GitHub
License information	MIT
TRL	TRL 5
Current version number	V0.1
Expected date of finalization	September 2025, (M29)
Development Status	First version available

4.7.2. Architectural diagram

The middleware API is one component of the Digital Twin developed in the context of Task4.1 and the related architectural diagram is shown in Figure 2, Section 2. There isn't any specific architectural diagram associated with this service.



4.7.3. Main benefit and innovation

The main innovation brought by this tool resides in the core of the seamless integration of DR services with the Digital Twin. In a nutshell, the facility manager will be provided with an easy and intuitive way to understand potential DR strategies in the DT UI. The integration will also be provided with clear specifications, and easiness of communications, facilitating the onboarding of both sides of the chain (service side and DT's side).

4.7.4. Development activity

The integration of DR programs in the Digital Twin will follow the implementation of the respective TE. Particularly, a middleware service will be developed to send the incoming communications towards the DT interface. The current version of this component provides the possibility to perform some validity checks on the input received, that for now is structured in JSON. Given this data, the API will validate them (e.g. the middleware will check if the home_id is valid and present in the accepted IDs), providing errors in case of non-valid data or data that are not expected. The development is available in the GitHub repository¹⁸.

Since this implementation is still early stages, additions to the functionality of the middleware may be reported in the next development cycle.

Meanwhile, an example of how the output of service F will be displayed within the digital twin is in DR recommendations section of Figure 8.

4.7.5. Outlook for the future expected release

For the following development cycle, the development plan includes:

- Refinement of the first version of the middleware.
- Integration with DR recommendation services.
- Integration with the Digital Twin.

¹⁸ https://github.com/dedalus-technology-release/T4.3_Pilot3_Middleware/tree/main



4.8. Service H - Optimal DR control for heat-pump systems

4.8.1. Overview of the service

Table 27: Service H overview

Service H	Description
Service / Software title	Optimal DR control for heat-pump systems
Responsible partner(s)	CENTRICA
Associated pilot site	Pilot 6, Irish
Service brief overview	As a part of this service different optimal control policies will be implemented to control individual heat pumps. The main goal of this optimal control will be to minimize the total energy cost for end-users while taking into account their comfort level. The dynamic of the system will be provided to the controller from digital twins model.
Associated TE	TE-13
Associated Use Case	UC 1-5
Functional Requirements (input by other activities)	TE8
Data Input	Real data collected from the pilot (sub-metering data), weather data, market data
Service Output	Optimal control strategy for managing heat pumps to minimize household energy cost while respecting the end-users comfort level
Repository Location	TBD
License information	TBD
TRL	TRL 7
Current version number	On hold
Expected date of finalization	TBD
Development Status	On hold

Due to the issue with Irish pilot, no actual development has been done as a part of this service yet. This service will be updated after finalizing the next amendment in the project.



This project has received funding from European Union's Horizon Europe Research and Innovation programme under the Grant Agreement No 101103998

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5. Local flexibility management for energy communities and self-consumption/cost optimization

The aim of this section is to provide information about the local flexibility management for energy communities and self-consumption/cost optimization service as outcome of Task 4.4.

5.1. Overview of the service

This service focuses on optimising and managing local energy flexibility within communities and districts, targeting economic, social, and environmental benefits. Leveraging OurPower's P2P digital platform (TE-12) and CARTIF's district level DR optimisation (TE-13), the service enhances DR management through a sophisticated decentralized management framework and enables trading flexibilities on the flexibility marketplace. A critical component is the development of a pre-aggregation framework that interfaces with local assets, enabling the identification and utilization of flexibility opportunities. These flexibility opportunities are either showcased to a nudging app (TE-10 developed in Task 4.3 as explained in section 4.5) or executed by automated DR, e.g., by controlling EV charging stations and battery storage. As part of the service the Universal Secure Gateway (TE-6) and Secure and Privacy/Confidentiality Preserving Data Transactions (TE-7), both developed by COMS in Task 3.6, will be utilized for resource virtualization and real-time management, which automates the DR management processes, thus enhancing participation and supporting local sustainability goals. As the service will be tested in two demos, not all the components will be deployed in both demos. In this sense, the Austrian demo focuses on TE-12 while the Spanish demo focuses on TE-13, sharing the same approach detailed in this section.

Table 28 contains the functional specifications of the service and the mapping with the associated technology enablers and use cases.

Table 28: Task4.4 - Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
The system enables flexibility management / trading for pre-aggregated building flexibility.	TE-12-FR1	UC18, UC20, UC22	TE-12	Pilot 1 (AT)	In progress



Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
The system provides optimal operation profiles for energy assets to maximize self-consumption or minimize operation costs at district level.	TE-13-FR1	UC18, UC19	TE-13	Pilot 1 (AT), Pilot 5 (SP)	In progress

5.2. Architectural diagram

Figure 78 and Figure 79 show the high-level service architecture designed to enhance local flexibility management within energy communities, with a focus on optimising self-consumption and cost-efficiency. This service integrates advanced monitoring, control mechanisms, and trading functionalities to ensure optimal energy usage and enhance community engagement.

Figure 78 focuses on the architectural diagram for the deployment of the service in the Austrian demo, while Figure 79 focuses on the architectural diagram of the Spanish demo. The differences between both architectures are mainly related to the way the market is integrated in the service and to the monitoring platforms used in each demo. In the development of this work in the Spanish pilot, it has been evidenced that it is necessary to count as well on external data-sources, accessible through the integration of API, to take relevant information for the optimization process, such as the energy prices throughout the day. Therefore, new data inputs have been included in the data model.



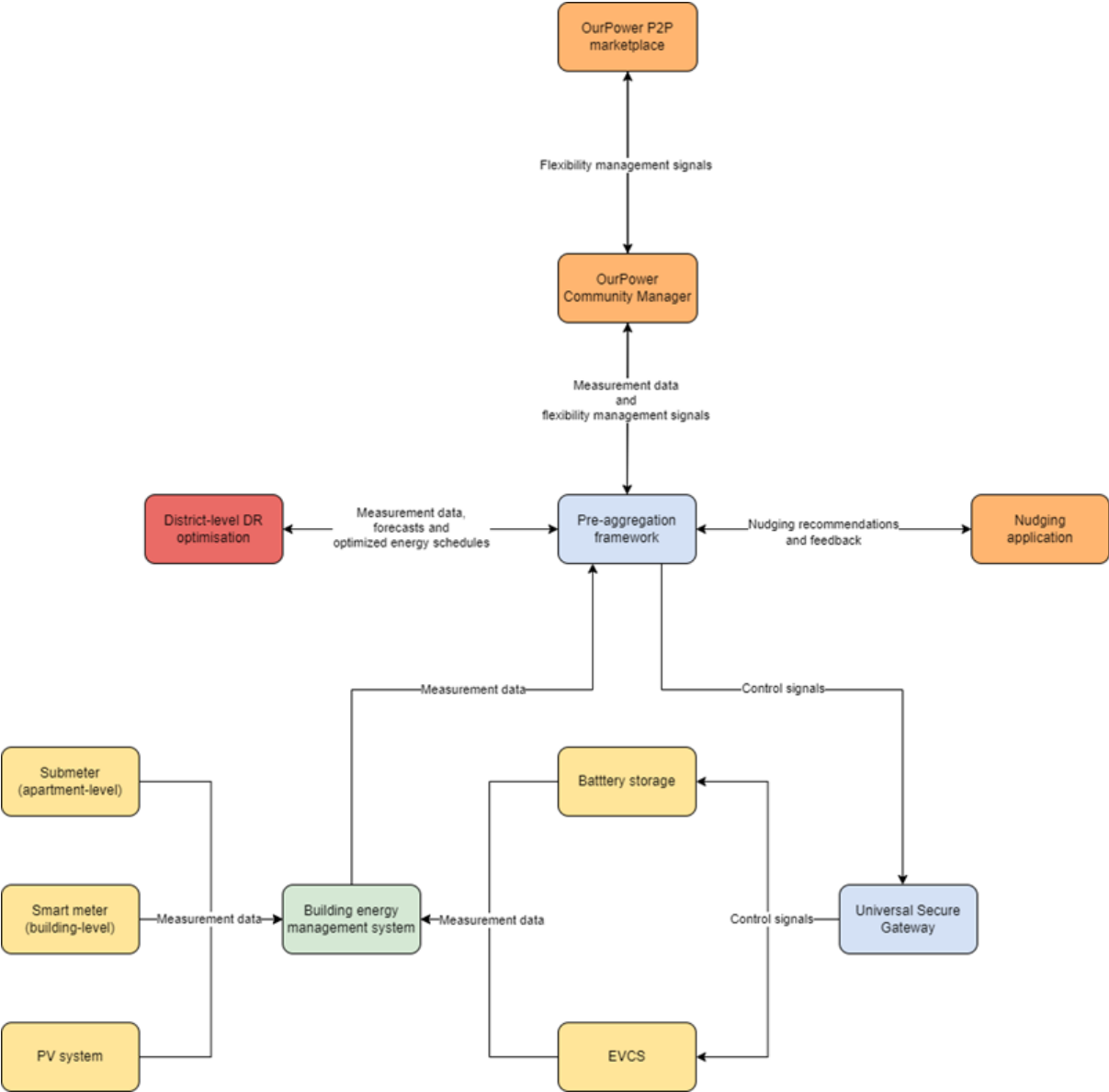


Figure 78: Architectural diagram for the deployment of the service in the Austrian demo.



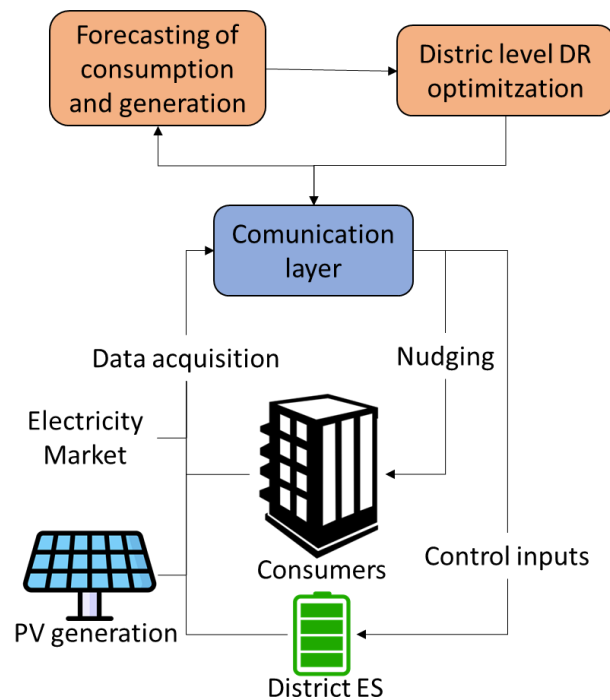


Figure 79: Architectural diagram for the deployment of the service in the Spanish demo

In the case of the Austrian pilot, the architecture of the service is centred around a few components which are seamlessly integrated to manage the energy dynamics of the community effectively. These include the Pre-Aggregation Framework, Universal Secure Gateway, District-level DR optimisation, OurPower Community Manager, and the OurPower P2P Flexibility Marketplace. Additionally, the Nudging App, developed under Task 4.3, interacts with this service, promoting manual DR activities.

At the heart of the service is the Pre-Aggregation Framework, which is tasked with the continuous monitoring of energy metrics across the community. This framework gathers extensive data from various sources, such as energy consumption data from apartment and building meters, energy production data from photovoltaic systems, usage data from electric vehicle charging stations, and the operational state of battery storage systems. This rich dataset provides a detailed real-time view of energy usage and production, forming the basis for all subsequent optimisation processes.

The district-level DR optimisation complements the Pre-Aggregation Framework by specifically addressing the unique energy management needs at a broader district level. This system leverages the comprehensive data collected – from residential energy consumption patterns to the operational status of communal energy assets – to execute sophisticated DR strategies that are tailored to the district's scale and complexity. By optimising DR at the district level, the system can manage peak load reductions more effectively, distribute energy resources efficiently across a larger network, and enhance the reliability and stability of the power supply.

Control within the service is twofold – automated and manual. Automated demand response is facilitated through the Universal Secure Gateway, which controls the operation of EV charging



stations and battery storage based on predictive analytics and real-time data. This ensures that energy storage and consumption are optimised according to current network conditions and storage capabilities. Concurrently, manual demand response is supported by the nudging app and personalized nudging recommendations. These recommendations encourage users to adjust their energy consumption behaviours, enhancing the community's overall energy efficiency and participation in demand response programs.

The OurPower Community Manager plays a pivotal role in integrating community energy management with the flexibility marketplace in the Austrian demo. It processes and analyses data received from both, the Pre-Aggregation Framework and the DSO, crucial for effective community billing and accounting. It also interfaces with the OurPower P2P Marketplace, managing the offer and sale of aggregated community flexibility, responding to market demands, and forwarding flexibility signals back to the Pre-Aggregation Framework for execution.

Also, in the Austrian demo, the OurPower P2P Marketplace, which is the basis for the OurPower virtual balancing group, is responsible for flexibility management and handles the trading of flexibility through signals such as requests, offers, and responses. It efficiently matches these elements to optimise the utilization of energy resources within the community, fostering a responsive and economically beneficial energy ecosystem.

In the case of the Spanish pilot, architecture is currently under development. The communication layer module is about to be finished. In the Spanish case the consumption and generation forecasting, is under development. The main block of district level DR optimization service is in a test stage with historical data. In the Figure 81, an example of a daily linear programming optimization, performed with historical data, for the Spanish pilot is presented with the different element that take place in the Spanish pilot.

5.3. Data for service execution

As it was already introduced in previous reports, for the execution of this service and further the effective operation of the local energy flexibility management service the acquisition and integration of following types of data inputs are required:

- **Energy consumption data:** This data is meticulously collected via advanced smart meters and sub-metering systems. These meters are crucial for monitoring real-time energy usage within the community, enabling detailed analysis and management of energy consumption at a very granular level. The precision of this data allows for the optimisation of energy distribution and load balancing, which are vital for effective demand response execution.
- **Energy production data:** Specifically focusing on PV systems, this data tracks the amount of energy generated by solar panels within the community. Accurate and timely production data is essential for optimising the integration of renewable energy sources into the grid and for facilitating effective flexibility trading within the P2P marketplace.
- **Asset information:** The service requires detailed and up-to-date information on the capabilities and availability of local energy assets, such as solar panels, batteries, and EV charging stations. This data is instrumental in optimising the utilization of these assets, maximizing their efficiency, and integrating them seamlessly into the overall energy management strategy.



- **Weather data:** The inclusion of weather data is pivotal for the comprehensive operation of the local energy flexibility management service. This data provides critical insights into environmental conditions that significantly affect energy production, particularly from PV systems. By integrating real-time and forecasted weather information such as solar irradiance, temperature, and cloud cover, the service can more accurately predict energy generation patterns. This predictive capability is essential for optimising the use of PV systems within the community. Additionally, weather data enhances the service's ability to perform effective load forecasting and DR operations, as weather conditions directly influence energy consumption patterns and needs.
- **Market data:** This encompasses vital information about energy prices, demand, and supply dynamics. Access to market data ensures that the service can make informed decisions, enabling users to engage in flexibility trading and DR activities that are economically advantageous.

5.4. Service output

As it was defined in the previous deliverable D4.1 [5], the service outputs are designed to provide actionable information and tools for users, community managers, and energy providers, focusing on optimising energy use, enhancing sustainability, and supporting community engagement.

These outputs are:

- **Optimised Energy Schedules:** the service provides detailed schedules that recommend optimal times for energy consumption and production, leveraging near real-time and forecasted energy consumption and production data. These schedules help users and building/community managers to maximize the utilization of renewable energy sources, aligning energy consumption with periods of high renewable energy production.
- **Flexibility Utilization Schedules:** Based on flexibility utilization schedules, users receive near real-time alerts and adjustments related to their energy usage and local availability of renewable energy. These alerts inform users about the best times to consume or conserve energy based on current renewable availability and demand dynamics, facilitating effective participation in demand response programs.
- **Community Energy Reports:** Comprehensive reports are generated detailing the community's overall improvements in energy efficiency, increases in self-consumption of produced energy, and reductions in carbon emissions. These reports provide insights into the effectiveness of energy management strategies and the community's progress toward sustainability goals.
- **Flexibility Management Reports:** Reports on flexibility management within the P2P marketplace are provided. These summaries help users track their flexibility activities and assess the impact of their decisions on promoting a greener energy grid. The service's outputs aim to empower users with comprehensive insights and practical tools needed to actively participate in and benefit from a sustainable energy ecosystem. This not only leads to enhanced energy efficiency and environmental benefits but also fosters a culture of active participation and awareness regarding energy sustainability within the community.



5.5. Main benefits and innovation

The service delivers significant benefits across economic, social, and environmental dimensions by optimising local energy flexibility management. By harnessing OurPower's P2P digital platform and CARTIF's district level DR optimisation (TE-13), it innovatively integrates DR with a decentralized management approach, enhancing energy system efficiency and reducing costs. This service innovatively uses a pre-aggregation framework that interfaces with local assets to identify flexibility opportunities, making it a pioneer in enhancing residential customers' ability to participate in flexibility markets. Socially, the service fosters a more engaged and responsive community by utilizing non-financial incentives and a nudging app, which encourages participation in DR programs and promotes sustainable energy practices. Environmentally, the service contributes to reduced carbon footprints through optimised energy distribution and consumption, aligning with broader sustainability goals.

5.6. Development activity

As the previous initial phase focused primarily on conceptualizing the service and planning the integration of its various components, this phase focused on further developing concepts for specific parts of the service and starting the implementation and integration of necessary components.

OurPower has begun developing and implementing a concept for managing a virtual balancing group based on the OurPower P2P marketplace, which will be used as part of the flexibility management. Therefore, adjustments on the OurPower P2P marketplace had to be done to integrate dynamization of the existing load profile system. A database leveraging DuckDB¹⁹ was established to store and process virtual balancing group data, including flexibility requests, adjustments, and forecast-based scheduling. Build upon this database a system is being developed to request flexibility from the energy community based on the virtual balancing group's situational needs. The energy community shall respond and coordinate flexibility measures internally. A remuneration and reward concept for flexibility management are currently under development, focusing on costs associated with the virtual balancing group. Initial analyses have been conducted, and further work is ongoing. In collaboration with ENG, forecasting capabilities for energy production from photovoltaic and wind power plants are under development. These forecasts will support virtual balancing group management, optimizing energy distribution and consumption within the energy community.

The development of the OurPower Community Manager focused on integrating critical data sources, enhancing functionality for flexibility management, and improving user interfaces. Key integrations included the EDA system, enabling the processing of 15-minute interval metering data, which is received as XML file daily by mail, and the EPEX market data, which provides daily prices fetched through automated scripts. Additionally, the integration of data from the building energy management system at the Austrian pilot (EUDT), storage systems (Sonnen), and EV charging platforms (EcoFactor) has been ongoing to facilitate monitoring and control.

¹⁹ <https://duckdb.org/>



Enhancements to the community dashboard introduced visualization of energy consumption, production metrics, and insights, enabling community managers to oversee the energy data of the energy community effectively.

Universal Secure Gateway (USG) is a part of deployment in the Austrian pilot providing interfaces to the local flexible and unflexible assets with ability to control the selected flexible assets according to the schedules and optimization targets defined in the pre-aggregation platform. USG is deployed in a local network of the Austrian pilot (GreenPoint building) in a way to be able to access the data interfaces of local building energy management system (BEMS) and asset control and data acquisition interfaces. It is securely connected to the pre-aggregation platform to enable secure and privacy-preserving data collection and asset control which prevents any unauthorised person or application from accessing data or take control of the assets.

The main updates of the USG were focused on extending the core functionality with the custom interfaces to the assets deployed inside the Austrian pilot. The first and the most important interface to the local assets is the BEMS interface which collects the data from the building energy management system. BEMS is connected to all important smart meters and submeters in the GreenPoint building and thus provides the measurement information important for the billing and estimation of the energy costs related to the building (energy community) consumption and production profiles. USG connects to the BEMS data sources over a secure web API.

USG connects to the battery storage units by connecting to the individual web APIs accessible in the local network inside the GreenPoint building. Each battery storage unit provides the raw live stream of energy flow data which includes the current PV production, battery charging/discharging status, grid-side energy flows and local consumption information. The interface is used to manage and control the operation of the battery storage unit according to the schedules and set points created during an optimization phase in the pre-aggregation platform.

Electric vehicle charging stations (EVCS) are integrated using a dedicated secure web API provided by the Ecofactor (EVCS provider). The integration provides mainly the data collection for the status of charging station and acquiring the charging session data. Currently, the control actions are not yet supported but the possibilities to control the charging station parameters will be further investigated and implemented (e.g., charging schedules, charging power limits, pricing signals, etc.).

The functionality of pre-aggregation platform was extended by integration of secure and privacy-preserving data transactions component (TE-5) with API gateway component which provides identity and access management to the data collected inside the pre-aggregation platform. Each pre-aggregation platform user which is also a data owner controls who has access to his/her data which can be personal or sensitive. The fine-grained access control and identity management gives community manager control to manage which external services have access to the data API inside the energy community and each individual platform user must give a consent for each individual data source before that service can access his/her data. Each consent can be revoked by the data owner at any time. All data access events are also logged inside the secure and privacy-preserving data transactions component which gives data owner an overview who and when accessed individual data sources they are exposing to other users



and services. Figure 80 presents the view of the secure and privacy-preserving data transaction component management user interface inside the pre-aggregation platform giving user a control over access to the data and the view on the log of all accesses to their data.



Resources

Name	Path
Home	/home
Users	/users

New Resource

Access Requests

User	Client	Resource Path	Scope
external_user_1	external_client_1	/assets	Read

Authorisation Log

Resource	User	Client	Status	Time
/users	external_user_1	external_client_1	Allow	2024 12 10 12:32:24
/users	external_user_2	external_client_1	Denied	2024 12 10 11:32:13

Figure 80: Secure and privacy-preserving data transactions component tab inside the pre-aggregation platform user interface providing access management functionality to the user.

The next important development was the implementation and deployment of digital twins for the flexible assets that are deployed in the Austrian pilot. The process of development and integration of digital twins is yet not finalized and will continue until the beginning of the full piloting phase just short after delivery of this report. The digital twin development activities were done for the battery storage systems, EV charging stations, PV plants and building section or whole building consumption.

Digital twin for the battery storage system is designed around the installed battery storage systems provided by Sonnen²⁰. Local battery storage system API is connected to the USG for the data collection and asset control. Battery storage systems in Austrian pilot are connected to four electrical interfaces: consumption port, production port, battery storage port and the grid port. Consumption port is the connection of the consumption part of the building to the battery storage system. From the consumption port the energy from the grid and the battery storage flows to the

²⁰ <https://sonnengroup.com/>



building. Production port connects PV plants to the battery storage system where the power generated by PV plants can be used for storage, for consumption, for sending the produced energy to the grid or for the combination of all three options. Battery storage port is the interface to the battery of the battery storage system and provides the functionality to charge or discharge the battery. The grid connection point connects the grid with battery storage system and connected PV plants and building. The live data provided by the battery storage system API is used for visualization of energy flows and providing historical data and statistics to the building manager. Battery storage system is modelled to provide functionality of emulating the battery storage systems to support forecasting and optimization of local energy self-consumption and reduction of energy expenses.

Digital twin for EV charging stations is presenting the live data and statistics available by the connection to the Ecofactor (EVCS provider) interface. The information provided by the EVCS digital twin is the status of the charging station, statistics of the charging station and forecasts for the future charging events which can help make better plans for local energy consumption to increase self-consumption and reduce energy expenses.

Digital twin for the PV plants provides the information about the past PV production and provides the insight into the forecasted PV production. The output of forecasts together with the forecasted information about the building section consumption are a crucial information supply for the digital twins of battery storage systems and EV charging stations to be able to provide forecasts of complete energy flows and emulated results that are used during the self-consumption optimization algorithm strategy evaluation before deployment of the control schedules to the assets.

In the case of the Spanish pilot, the implementation of the architecture is currently ongoing. Communication Layer module is about to be finished. In the Spanish case the consumption and generation forecasting, is under development. The main block of District-level DR optimization service is in a test stage with historical data. In the Figure 81, an example of a daily linear programming optimization, performed with historical data, for the Spanish pilot is presented with the different elements that take place in the Spanish pilot.

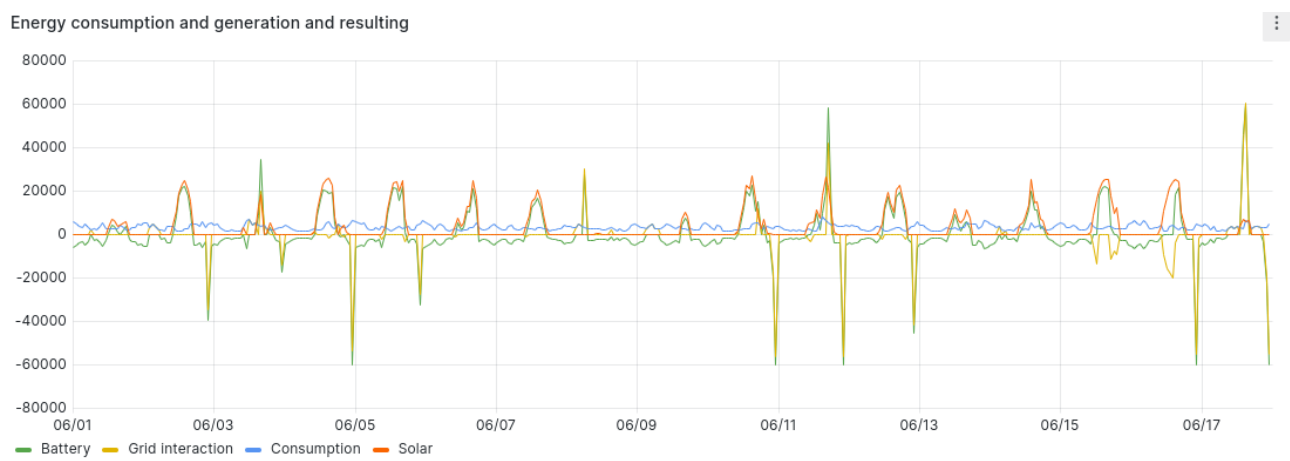


Figure 81: Daily linear programming optimization for the Spanish pilot



The data inputs for energy consumption and production data, asset information and weather data have been easily integrated, meanwhile the market data is the one that offers higher difficulties, as it implies sensible data that should be requested to energy commercializing companies. Nevertheless, it is expected to be integrated with no remarkable issues throughout the execution of the project. In addition, it is worth to mention that the requirements change depending on the country, hence at the different pilots the boundary conditions are different.

A first version of the software supporting the service TE-13 for Pilot 1 has been successfully completed and is intended for deployment during the demonstration phase. The algorithm focuses on strategies for managing the flexibility of shared renewable generation, storage and shifting loads and is highly adaptable to various community configurations. It incorporates the operational characteristics of the assets and maximizes economic benefits by minimizing energy costs. The following pseudocode outlines the Python-based implementation of the algorithm:

Service execution starts.

1. Read the information about the community members and their associated assets.
2. Read the required data input: renewable energy production and non-flexible demand forecasts, energy pricing, flexible energy demand, operational interactions agreed with the grid.
3. Call the optimal management module.
4. Create the required decision variables and parameters for the OCP.
5. Create the constraints.
6. Create the objective function.
7. Call the solver
8. Retrieve and output the optimal commands for the assets.

Service execution ends.

The initial steps for the service execution involve reading the necessary data for the control algorithm. Dedicated modules were designed to pass a dictionary that maps each community member to their corresponding assets, including renewable generators as well as non-flexible and flexible demand devices. This stage ensures that the configuration of the community is accurately incorporated in the problem formulation.

The call to the optimal management module is next in the execution and it represents the more important part of the service. This module generates the expected optimal operational profiles by solving a constrained non-linear optimisation problem with continuous decision variables. Therefore, the optimal management module firstly creates all the required variables involved in the modelling of the asset's dynamics.

The primary objective of the optimization is to minimize the cost of the operation, as reflected by the objective function (39), where P_{CEbuy} and P_{CEsell} are the powers purchased from and injected into the grid, respectively, while p_{CEbuy} and p_{CEsell} are the energy prices.

$$\sum_T (P_{CEbuy} \cdot p_{CEbuy} - P_{CEsell} \cdot p_{CEsell})h \quad (39)$$



The constraints of the problem can be classified into three groups based on their intended purpose. The first category gathers the variable bounds, which in most of the cases are directly related to the operational limits of the quantities they represent. These constraints are in the form of (40), where b_l indicates the lower bound of X and b_u denotes its upper bound.

$$b_l \leq X \leq b_u \quad (40)$$

The second group of constraints pertains to the dynamic modelling of the electric storage assets. The energy in the batteries is modelled following (41), which computes the state of charge at time k by adding or subtracting the energy received or delivered to the energy from the previous time step. This model assumes that the battery cannot simultaneously charge, and discharge as defined by (42), with $P_{ch,k}$ and $P_{disch,k}$ as the charging and discharging powers at time k , respectively. Under this modelling framework, the energy increases during charging and decreases as power is supplied back to the microgrid.

$$SOC_k = SOC_{k-1} + (P_{ch,k} - P_{disch,k})h \quad (41)$$

$$P_{ch,k} \cdot P_{disch,k} = 0 \quad (42)$$

As it is possible for the community to have charging points for electrical vehicles, the additional constraints (43), (44) are imposed, so the EV batteries are charged within a predefined charging window $[k_i, k_f]$. In this sense, the EV are treated as flexible loads, allowing the optimization to generate optimal charging profiles that meet user requirements while adhering to the specified constraints.

$$P_{chEV,k} = 0, \quad k < k_i \vee k > k_f \quad (43)$$

$$SOC_{EV,k} = SOC_{EVreq}, \quad k = k_f \quad (44)$$

The third group of constraints addresses the energy balances that must be maintained according to the community configuration. In general terms, these constraints take the form of (45), (46) and (47), which relate the powers demanded from and supplied to the grid with the total power consumed and generated by the users. It is important to note that the power offered by a member originates from renewable energy sources and the stored energy in their batteries. Similarly, the total power demanded by a user is the combination of the flexible and nonflexible loads along with the energy allocated for storage.

$$P_{CEbuy} - P_{CEsell} = \sum_{u \in U} (P_{ubuy} - P_{usell}) \quad (45)$$

$$P_{usell} = P_{PV} + P_{disch} \quad (46)$$

$$P_{ubuy} = P_L + P_{ch} + P_{chEV} \quad (47)$$

The above approach to construct the optimal optimisation problem was implemented using the Pyomo modelling language and the interior point solver Ipopt. Preliminary stand-alone tests using historical data were carried out with the Austrian demo as a target community. The Austrian



demo is a residential building with two floors, each equipped with its own photovoltaic generation system. The aggregated load demand for each floor is actively monitored. Additionally, a set of batteries is treated as a shared resource, allowing the stored energy to be utilized to meet the overall demands of the community.

The results from the preliminary test, shown in Figure 82, confirmed that the algorithm performs as expected, with shared solar production utilized to meet the building's demand, and any surplus energy stored as shown in figure.

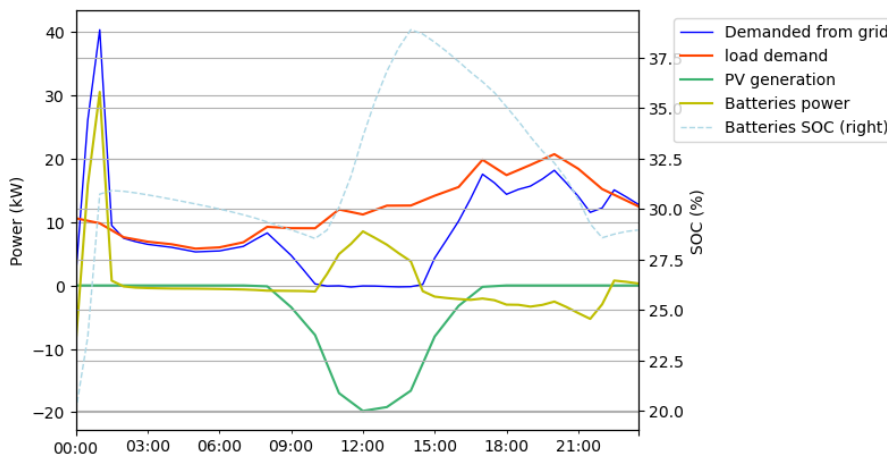


Figure 82: Preliminary test results of the CARTIF optimisation algorithm

Further performance tests are planned, including studies on the grid impact of different charging requests for electric vehicles. The current version of the software is available on the CARTIF private GitLab repository, as the definition of software requirements for the deployment phase remains an open task.

5.7. Outlook for the future expected release

This section outlines a plan for the anticipated developments in future releases of the local flexibility management for energy communities and self-consumption/cost optimisation service. These enhancements and upgrades are aimed at refining the service's functionality, increasing its efficiency, and ensuring it continues to meet the evolving needs of energy communities. The planned developments for future expected releases of this service include:

- Enhancements to the Pre-Aggregation Framework:
 - Extending user interface to enable community manager to manage the local flexibility.
 - Integration of nudging recommendations.
- Automatization of DR actions on the pre-aggregation platform and Universal Secure Gateway.
- Refinement of Optimisation Problem Formulation
 - Further refine the optimisation problem to better address the specific requirements of both the Austrian and Spanish pilots.



- Advancements in Virtual Balancing Group Management on OurPower P2P Marketplace:
 - Finalize the development of the virtual balancing group management system.
 - Complete the remuneration and reward concept for flexibility management, incorporating insights from ongoing cost analyses.
- Enhancements to the OurPower Community Manager:
 - Complete the integration of data from assets.
 - Enhance the community dashboard further by incorporating additional metrics and interactive tools for better visualization and decision-making by community managers.
 - Implement flexibility management/trading capabilities for pre-aggregated flexibility.
 - Upgrade of community billing and accounting.
- Deployment and demonstration.



6. Optimal Aggregation of flexible energy resources for heat vs electrical flexibility valorisation

The aim of this chapter is to provide updated information about the software expected to be developed as outcome of Task 4.5. This development can correspond to more than one services and must be linked to the TE-14 Aggregation of flexible energy resources for district heating management.

The services developed under the headline Aggregation of flexible energy resources for district heating management will extended the NeoGrid Platform with new services for optimal aggregation of thermal flexibility at individual buildings and at district level, where flexibility in how an energy-efficient building is heated can be used towards the district heating system, so either a building or a pool of buildings can move their energy consumption in time to minimise the collective load on the energy system. Furthermore, the solution will be extended to include ventilations systems connected to the electrical grid and include the flexibility from these systems toward the electrical grid, i.e. having one solution which can handle flexibility towards district heating and electricity systems and at the same time maintain the required indoor climate. The aim is to extend the existing system from building level to apartment level via smart thermostats in cooperating feedback from tenants with respect to comfort and flexibility requirements i.e. having humans in the loop. Having control possibility (thermostats and ventilation system) in each apartment, we expect to extract more flexibility from the building and at the same time honouring that tenants in the building can be more or less interested in providing flexibility.

The service will be closely tied to and tested in Pilot 2 (The Danish pilot).

High-level services in focus:

- Preheat control of District Heating (DH) flow temperature w/valve feedback (TE-14-FR1).
- In-apartment radiator set-point control for increased flexibility (TE-14-FR2 and (TE-14-FR3)).
- Cost- & indoor climate-optimisation of ventilation systems (electricity) (TE-14-FR4).
- Flexibility forecast and aggregated control of district heating in apartment, building and pool of buildings (electricity and district heating) (TE-14-FR5)

To comply with and deliver the above services, general functionality in the platform is required going from focusing on the building as a whole, to being able to interact with tenants in apartments. Here backend and GUI functionality are developed, so tenants can see own apartment data and furthermore provide input on set points and flexibility preferences.

The development process involves several steps, which in the following are presented as high-level steps, which will be carried out in relation to deliver the proposed services:

1) Preheat v2 - Valve feedback



- a) Forecasts models of building and apartments energy use, indoor temp., valve opening, etc.
 - b) Develop & deploy Preheat with valve opening feedback, collect data.
 - c) Visualisation / statistics on DH peak shaving for building managers (dashboard).
- 2) General functionality required by all T4.5 services (UC14, UC16, UC17)
- a) Implement hierarchy within building model and backend so the frontend can view devices in the right information context.
 - b) Implement a permission structure for zones, restricting access so that users can only view and control their designated zones and related sub-zones.
 - c) Implement push notifications for e.g. behavioural nudges within the system targeting each apartment/tenant related GUI's.

The development activities in the reported timeframe of the project have focused primarily on the above-mentioned development steps, which will be elaborated in the next sections.

The functional requirements for the service are listed in Table 29 below:

Table 29: Task4.5 - Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
Control of heat in building via central control with apartment temperature feedback	TE-14-FR1	UC14	TE-3,8,9	Pilot 2 - DK	Ongoing
Control of heat in building via central control and using smart thermostat valves as feedback signal to controller	TE-14-FR2	UC14, UC17	TE-3,8,9	Pilot 2 - DK	Ongoing
Monitor and Control of heat in apartment via central control and via Smart thermostats in	TE-14-FR3	UC14, UC17	TE-3,8,9	Pilot 2 - DK	Started



Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
apartments to minimize heat consumption and enable district heating flexibility					
Monitor and Control of ventilation in apartment via central control to minimize heat loss via air change and enable to enable electrical flexibility	TE-14-FR4	UC14, UC17	TE-3,8,9	Pilot 2 - DK	Not Started
Reduction and shifting of heat consumption for building or cluster of buildings	TE-14-FR5	UC16	TE-1, 3,4,9	Pilot 2 - DK	Not Started

Figure 83 shows the high-level architectural diagram of how the developed services are interacting in the DEDALUS setup having the NeoGrid commercial platform as a main part of that.

There are no relevant updates with respect to the high-level architecture of the service provided in D4.1 [5] taking in account the DEDALUS Software Architecture reported in D2.4.



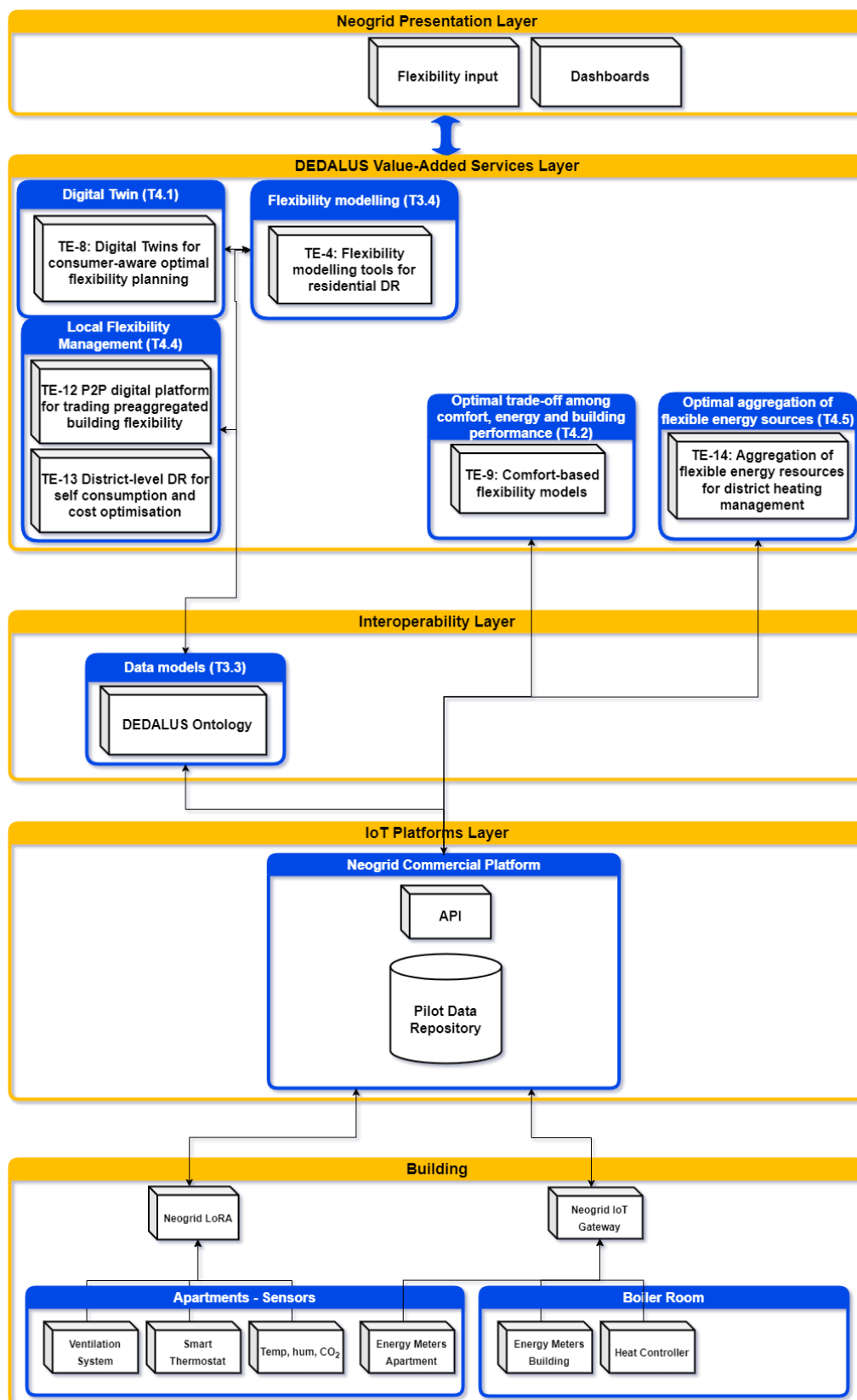


Figure 83: Pilot 2 - High level Architectural Diagram



6.1. Service TE-14-FR1

6.1.1. Overview of the service

This service is about the control of heat in building via central control with apartment temperature feedback. This controller is based on one of Neogrid's commercial controllers, which will be refined during this project. The aim is to have a control running so the building manager from the beginning can start getting a feeling about the system so they can be involved and consulted during the project.

Required data for running this service is time series data from central energy meter, metrological whether data (historical and forecasts) and indoor climate data from the apartments.

Service output will be set point schedules which will be applied for the district heating water circulated around the building.

The measured indoor climate data and energy data will be used as base line for the pilot site.

6.1.2. Architectural diagram

Figure 84 shows an architectural diagram of the services TE14-FR1-3. These three services are layered where TE14-FR1 is the most basic, and TE14-FR2 and TE14-FR3 brings additional features and complexity while still including the features of the previous layer.



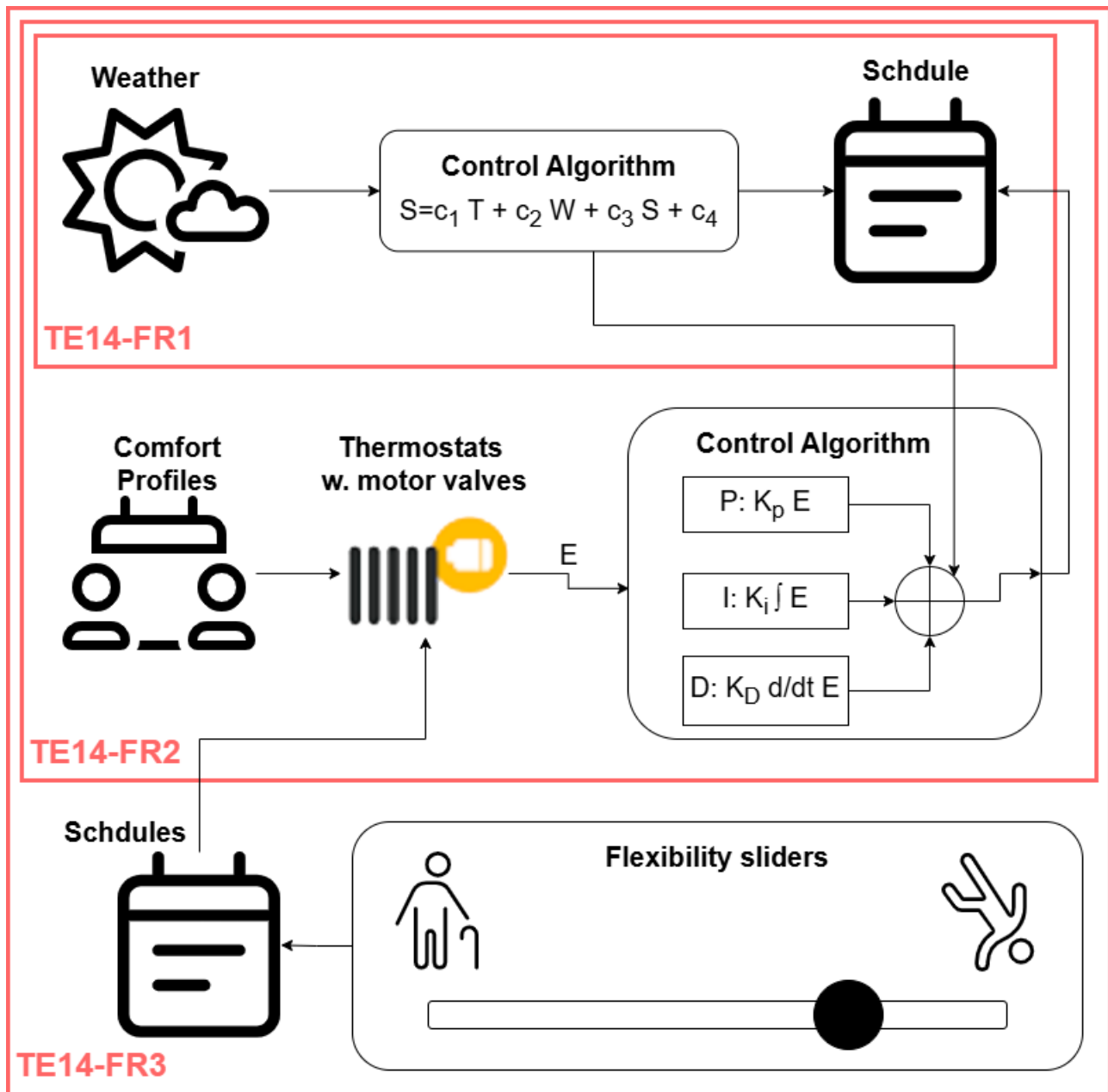


Figure 84: Pilot 2 - Architectural diagram of the services TE14-FR1 to TE14-FR3.

6.1.3. Development activity

In this phase an existing, thoroughly tested, control algorithm is used for the optimization of the heating. In the algorithm the supply temperature is a linear function for the outdoor temperature, sun intensity and wind at the location. The coefficients are chosen based on extensive usage on similar buildings and are periodically updated based on data from the building.



6.2. Service TE14-FR2

6.2.1. Overview of the service

This service is the control of heat in building via central control and using smart thermostat valves as feedback signal to controller. The main goal is to extend the TE14-FR1 controller with feedback from radiator valves, to have a better view on how the heat is distributed around the building and to get information about which apartments are the limiting factor or not using all the radiators in the building.

Required data besides what is specified for Service TE14-FR1, valve opening signals from the thermostats are required.

Service output will be set point schedules which will be applied for the district heating water and domestic hot water circulated around the building.

6.2.2. Development activity

In this phase feedback from the smart thermostats is used to extend TE14-FR1 to a PID controller.

The error signal E for the PID controller is coming from the motor valves in the smart thermostats that has been installed on all radiators in the building. The thermostats are controlled by the residents through comfort profiles on the NEOGRID app.

The valve opening is calculated using the mean value of the motor position and motor range for the last hour. In the first iteration of this service the mean value of the valve opening across all smart thermostats are used. When the mean valve opening for the building is known it is compared to an ideal valve opening set by an expert, this gives the error needed to calculate the P, I and D terms. The P, I and D terms are added to the supply temperature as calculated in TE14-FR1 and then send as the schedule for the building, see Figure 84.

The development of service TE14-FR2 is almost complete, it is expected to start running on locations in 2025 Q1. The development of the services TE14-FR2-5 have warranted an update to the NEOGRID framework used for processing and analysing data. The update includes optimization of data loading and caching making it feasible to include all the smart thermostats installed at the Pilot 2-DK demonstration site. Additionally, the update includes a reorganization of the building model in the python package, to make it faster and easier to develop new features.

6.3. Service TE14-FR3

6.3.1. Overview of the service

Besides the goals for Service TE14-FR1 and Service TE14-FR2, this service aims to optimise and control the set points for the individual radiators in the apartments to estimate and activate



flexibility for district heating across apartments and buildings and at the same time honouring the set points and tenant requirements for indoor comfort.

Required data besides what is specified for Service TE14-FR2, this service will require flexibility preferences (translated from requirement for indoor climate) and requested setpoints from apartments.

Service output will be a setpoint schedules for the individual radiators and for the central district heating water and domestic hot water production.

6.3.2. Development activity

In this phase flexibility is added to the service TE14-FR2 by allowing the residents to indicate a flexibility level using a slider. The flexibility level is translated into a schedule for manipulating the setpoint temperature of the smart thermostats to optimise the use of district heating. In addition to the input needed for TE14-FR2, TE14-FR3 needs flexibility slider levels set by the residents as input. This service outputs schedules for the smart thermostats in addition to the supply temperature schedule from TE14-FR2.

The translation of the flexibility slider input to the schedule for manipulating the smart thermostats is an important point of research in this project and it is anticipated that experiments must be conducted with several different schemes to optimise the use of district heating in the building.

It has been agreed upon between frontend, backend and analytics how the input from the flexibility slider will be implemented internally. The implementation for passing this information internally in the NEOGRID structure is a necessary step in developing and implementing a control algorithm based on the flexibility of residents.

6.4. Service TE14-FR4

6.4.1. Overview of the service

This service aims to optimise and control the air change rate in each apartment to estimate and activate flexibility for electricity (ventilation system) across apartments and buildings and at the same time honouring the tenant requirements for indoor comfort.

Requested data besides what is specified for previous services, this service will require data from ventilation systems in the apartments (time series data) and electricity price data.

Service output will be a setpoint schedules for the individual ventilation systems.



6.5. Service TE14-FR5

Based on the output from previous specified services, the aim for this service is to provide a tool to monitor, administer and execute in district heating domain for single building or cluster of buildings with focus on reduce and shifting the heat consumption (heating and domestic hot water production), to reduce the overall peak load of the building or a cluster of buildings.

The tool should aim to provide services to district heating grid like:

- Peak reduction.
- Reduce return temperature, maximizing cooling.
- Optimise heat consumption with respect to constraints from district heating grid/company.
- Optimise according to prices.

Requested data from the previous services and configuration inputs from building manager. Furthermore, requirements from the district heating grid are required. Here this will be modelled and configured via the building manager inputs.

Service output will be set point schedules which will provide the requested flexibility.

6.6. General functionality required by all T4.5 services

In order to comply with and deliver on the services developed in T4.5, it is necessary to further develop and extend the functionality in the platform. The platform general framework has to be able to extend the present setup focusing on the building as a whole, to being able to interact with tenants in apartments. Here backend and GUI functionality are developed, so tenants are able to see own apartment data and furthermore provide input on set points and flexibility preferences. This functionality is described in the following sections.

6.6.1. Hierarchical zone support

Figure 85 illustrates a hierarchical structure for managing permissions and access within an organization, specifically tailored to residential heating and energy control systems. At the top level, the Organization represents the overarching entity, such as a housing association like "Fællesbo." Beneath this, the Location(s) level encompasses physical addresses or clusters of buildings, for instance, "Thyravej 2-8," which is also the level where the energy management system is implemented. Moving further down, the diagram highlights nested Zones that reflect the breakdown within the building. For instance, a Stairwell is a shared space, followed by individual Apartments as distinct zones. At the most granular level, individual Rooms within Apartments are controlled, facilitating specific device-level management such as smart thermostats or sensors. Each zone is depicted with a key symbol, emphasizing the addition of role permissions, this newly implemented system ensures that users can only access and control their assigned zone and connected subzones.



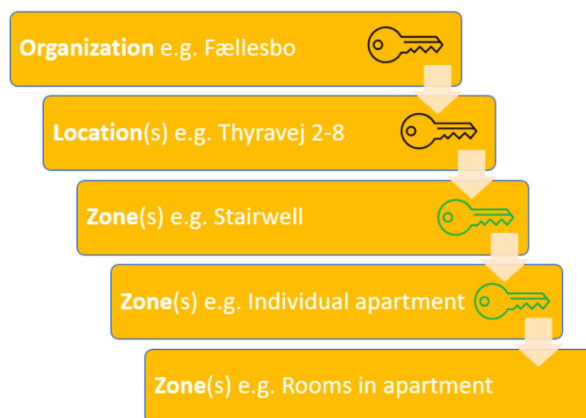


Figure 85: Hierarchical structure for managing permissions and access within an organization.

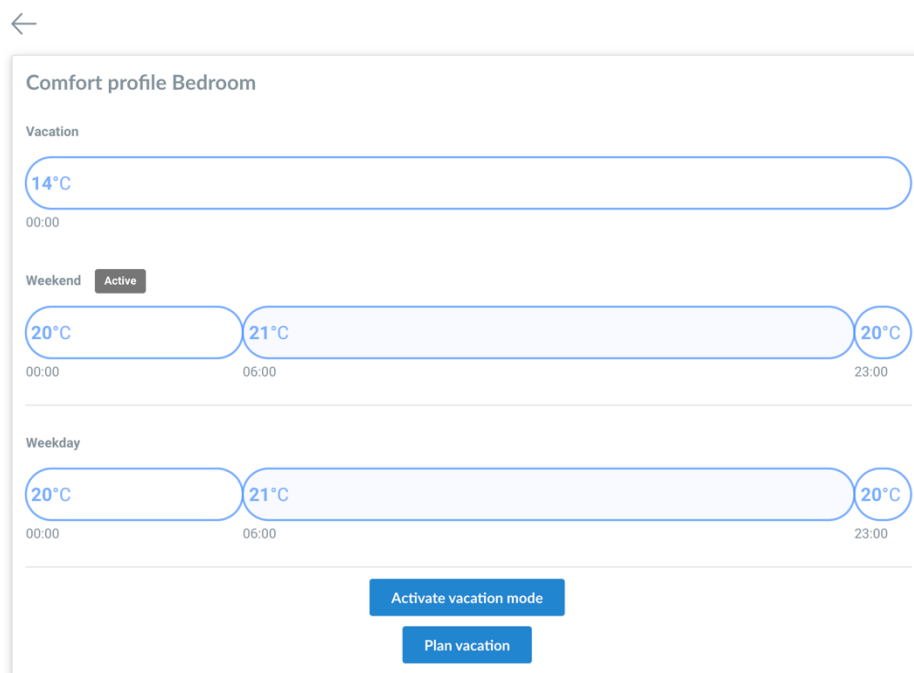
In practical scenarios, this system provides flexibility and oversight to various stakeholders. For example, an administrator from the housing association can access a specific building and all subsequent zones and subzones within it. This enables the administrator to address specific tenant issues, such as adjusting the indoor climate settings at the apartment level for tenants who may not have the application.

Since the most granular level, individual zones that represent Rooms within Apartments are controlled, we are able facilitate device-level management such as smart thermostats or sensors. For all edge zones (zones placed at the most bottom in the hierarchical structure) tenants can create a comfort profile.

6.6.2. GUIs for end user visualization on Zones

The comfort profile is depicted on Figure 86. The comfort profile enables tenants to create setpoints for all hours of the weekdays, weekend, and vacation mode. This setpoint will be used by the smart thermostats. A tenant can on a comfort profile select a flexibility on a scale of one to five, with five being the lowest. The selected flexibility will indicate to the building control how much the control can move the indoor temperature away from the current comfort profile.





←

Comfort profile Bedroom

Vacation

14°C

00:00

Weekend **Active**

20°C 21°C 20°C

00:00 06:00 23:00

Weekday

20°C 21°C 20°C

00:00 06:00 23:00

Activate vacation mode

Plan vacation

Figure 86: Comfort profile settings on zones for role: tenant

The zones in the building can be seen on the zone page depicted on Figure 87.

The zone system also incorporates colour-based nudging to provide users with intuitive feedback on whether specific values fall within expected thresholds. For instance, values like temperature, humidity, or air quality are visually marked with colour codes to indicate their status (e.g., green for acceptable, yellow for borderline, and red for out-of-range). colour-based nudging ensures that users can quickly identify potential issues or deviations immediately to promote user engagement and better energy habits.



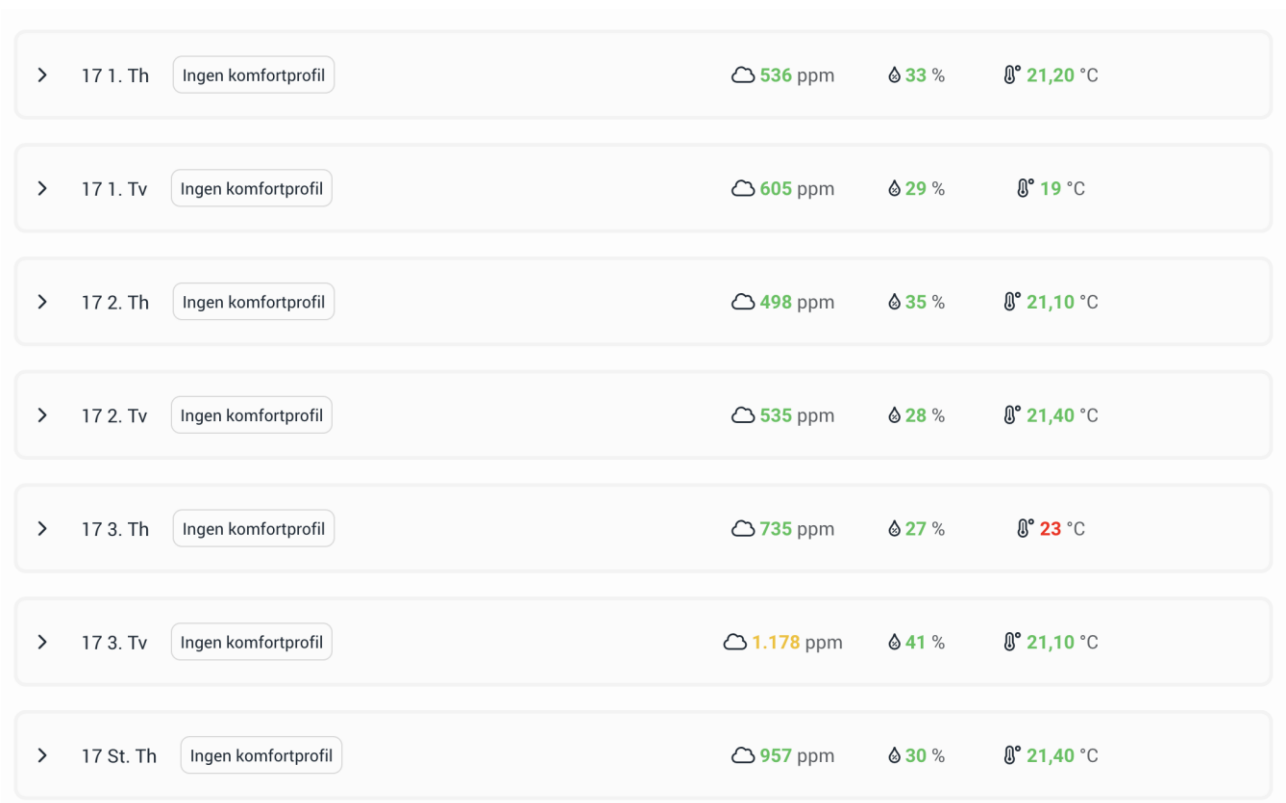


Figure 87: Apartments/Zones in a building

Additionally, the interface includes expandable zones, denoted by a chevron icon, which users can click to reveal subzones within a zone. For example, subzones for an apartment will be individual rooms, such as a bedroom depicted on Figure 88. In this scenario, the weekend comfort profile is active, setting a setpoint temperature of 21 °C. The system provides real-time data, showing the actual temperature at 19.82 °C, relative humidity at 40%, and air quality at 404 PPM. This granular feedback enables tenants or administrators to assess the indoor climate in detail and adjust as necessary to optimize comfort and energy efficiency.

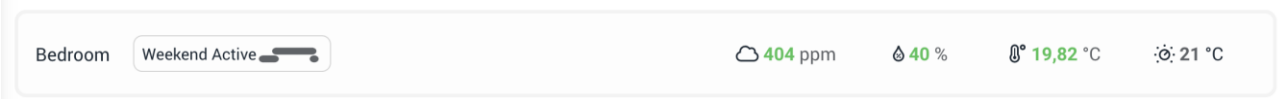


Figure 88: Subzones within a zone

6.6.3. Development activity

Development and implementation finished.

6.6.4. Notification hub

Other nudging initiatives include just-in-time notifications that have been introduced as part of a new notification centre, which is currently under design and implementation. The just-in-time notifications will be created based on the Collaboratory design workshop in the Danish pilot at Fællesbo in Herning. This system aims to enhance user engagement and promote proactive actions by delivering relevant notifications or alarms. Notifications are system-generated messages that provide actionable insights or recommendations to improve energy efficiency and indoor comfort. For example, a user might receive a message such as: “The temperature is set to 24°C in Bedroom. You can save up to 643 DKK annually by changing the temperature to 20°C.” Another example could be: “The CO2 level is high in Living room” These messages are designed to nudge users towards more energy-efficient behaviours and better indoor environmental management. Alarms, on the other hand, are user-configurable and provide real-time alarms based on personalized thresholds. Users can set specific alarms on the graph page for conditions they want to monitor closely. For instance, a parent might configure an alarm to trigger if the temperature in the children’s room drops below 20 °C or if the CO2 levels exceed 1000 PPM. These alarms ensure users are notified immediately about critical conditions that require their attention, adding a layer of customization and control to the system. Together, the notification and alarm features enhance the system’s ability to guide users in maintaining an optimal indoor environment and achieving energy efficiency goals.

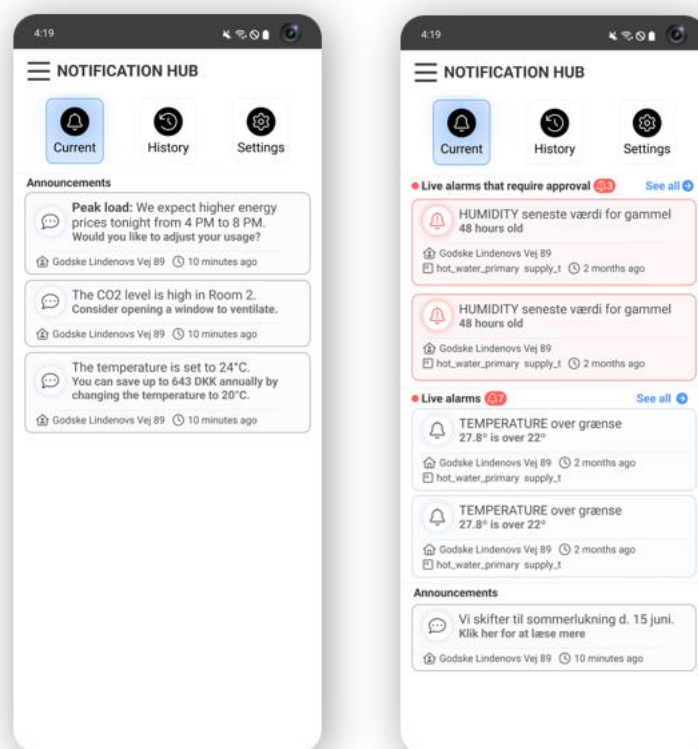


Figure 89: Notification hub.



The newly designed notification centre depicted on Figure 89 will serve as a centralized hub for managing both notifications and alarms. Within the notification centre, users will have access to all active alarms and notifications, as well as a complete history of past alarms and notifications. This interface will allow users to customize their experience by modifying settings for individual notifications or alarms to suit their preferences and schedules. For alarms, users will have the ability to interact with and acknowledge alarms directly within the notification centre. By tapping on a specific alarm, users can review the triggering event in detail, including visualizing the corresponding data on the system's graph page. In the alarm settings, users can further refine their configurations, such as setting specific timespans or days for the alarm to trigger and selecting their preferred medium for receiving alarms. Available mediums will include email, in-app notifications, and push notifications, the latter of which is currently under development. Customization allows users to stay informed while maintaining full control over how and when they receive alarms, thereby improving the system's usability and preventing notification fatigue.

6.6.5. Development activity

Notification hub is currently under development, with push notification support coming up.

6.6.6. Conceptual model for thermostats

The smart thermostat implementation has been designed to closely resemble the conceptual model of a traditional thermostat, ensuring familiarity for end-users. This implementation allows users to temporarily override the setpoint temperature defined in the comfort profile, replicating the behaviour of conventional thermostats. After a period of eight hours, the setpoint automatically reverts to the value specified in the active comfort profile, maintaining balance between user convenience and energy efficiency. Fully integrated into the hierarchical building structure, each thermostat is assigned to a specific zone, such as a room within an apartment, ensuring its data and settings are contextually follow the structure of the building. In zones where multiple thermostats are present, they are synchronized from the user's perspective, providing a seamless and unified control experience. When a user makes adjustments on the physical thermostat, the zone view reflects this with an 'overwritten' badge on the comfort profile badge, clearly indicating the temporary change as seen on Figure 90. This cohesive integration between the physical thermostat and digital interface ensures users can effectively monitor and manage their indoor climate.



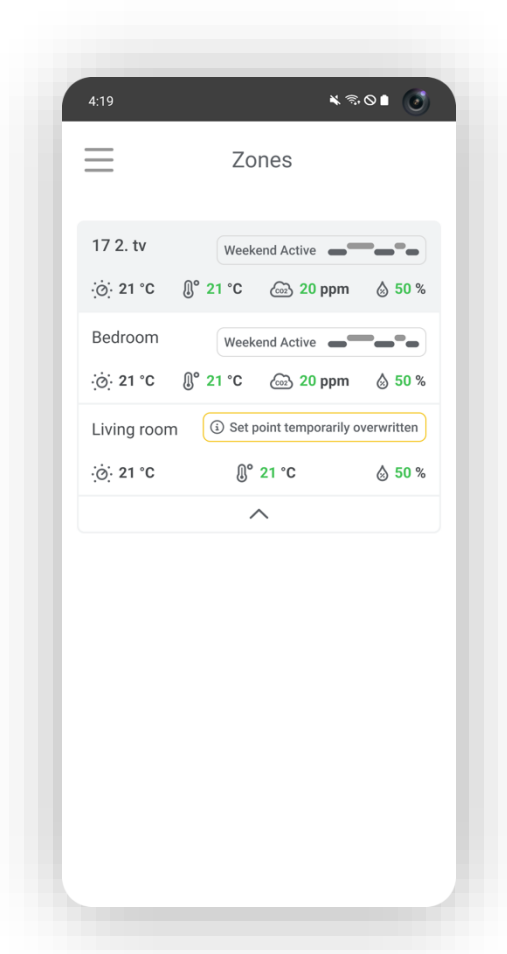


Figure 90: Room / Zone settings.

6.7. Main benefits and innovation

The main benefit and innovation steps are to extend an existing heating control system from focusing on central control at building level to apartment level via smart thermostats in cooperating feedback from tenants with respect to comfort and flexibility requirements i.e. having humans in the loop.

Having control possibility (thermostats and ventilation system) in each apartment, we expect to extract more flexibility from the building and at the same time honouring that tenant in the building can be more or less interested in providing flexibility. Furthermore, tenants are actively involved in the control loop, since they can indicate via the provided App the level of comfort and level of flexibility. By moving to the apartment level with heating control, we expect to improve the energy savings compared to only controlling the building as a whole from the basement.



6.8. Outlook for the future expected release

This section outlines a plan for the anticipated developments in future releases of the aggregation and control services.

Activities where development are either ongoing or done:

- 1) Initial phase:
 - a) Complete physical installations – DONE.
 - b) Set-up building model / data collection / get meter keys from Lejerbo and placement of meters – DONE.
 - c) Start collecting baseline data – DONE.
 - d) Deploy NeoGrid Preheat, collect “baseline” data (DEADLINE: SUMMER 24) – DONE.
- 2) Preheat v2 - Valve feedback:
 - d) Forecasts models of building and apartments energy use, indoor temp., valve opening, etc. – STARTED.
 - e) Develop & deploy Preheat with valve opening feedback, collect data. – ONGOING.
 - f) Visualisation / statistics on DH peak shaving for building managers (dashboard) – STARTED.
- 3) Preheat v2.5 - Individual apartment activation with aggregated activation:
 - a) Develop UI with flexibility slider for tenants – based on input from co-creation workshops – DONE.
- 4) General functionality required by all T4.5 services (UC14, UC16, UC17):
 - a) Implement hierarchy within building model and backend so the frontend can view devices in the right information context.
 - b) Implement a permission structure for zones, restricting access so that users can only view and control their designated zones and related sub-zones.
 - c) Implement push notifications for e.g. Behavioural nudges within the system targeting each apartment/tenant Related GUI's.
 - d) Extend backend framework for Notifications - ONGOING.

The planned developments for future expected releases of this service include:

- 5) Preheat v2.5 - Individual apartment activation with aggregated activation:
 - a) Develop forecast models for flexibility in building thermodynamics.
 - b) Estimate thermal capacity of apartment and building to determine how long we can close heating without affecting the room temperature.
 - c) Develop UI with flexibility slider for tenants – based on input from co-creation workshops (Nudging work) – DONE.
 - d) UI for building manager to list flexibility settings and adjust them (people that calls in and complains about lack of heating, etc.) and expand dashboard with graph of building flexibility (how much of building thermo-capacity is used?).
 - e) Develop & deploy Preheat with valve opening control, flexibility slide & maybe comfort profiles, collect data.



- 6) Ventilation Control:
 - a) Develop & deploy demand-based ventilation control based on air change, CO₂, and humidity, collect data.
 - b) Visualisation / statistics on ventilation, so building manager can confirm that QoS is maintained.
- 7) Parallel steps to support and interact to DEDALUS:
 - a) Data API for digital twin, 1st step REST based interface as delivered today from the platform. 2nd step via the DEDALUS interoperability layer.
 - b) Support for DEDALUS ontology.
 - c) Develop interactions between flexibility components in DEDALUS which we are interfacing to.
- 8) General functionality required by all T4.5 services (UC14, UC16, UC17):
 - a) Extend backend framework for Notifications.
 - b) Framework for Notification HUB enabling the web app to send push notifications to users.
 - c) Finalize implementation of generalized functionality in the platform.



7. Conclusions

The progress made between Deliverables D4.1 and D4.2 marks a significant milestone in the DEDALUS project, reflecting the collective effort to bridge conceptual frameworks with practical implementation across the pilots. The work presented in this chapter underscores the development of initial versions of digital twins. In the following is outlined the specific advancements achieved in all tasks detailing the outcomes realized.

Significant progress was made in Task 4.1: Digital Twins for optimal assets vs consumers flexibility planning. Notably, for Pilots 2 and 4, the project transitioned from a theoretical definition of interfaces and basic requirements to the development of an initial version of the digital twin. This first iteration provides a simplified representation of data and service outputs, marking a crucial step forward. For Pilot 3, a highly detailed mock-up of the digital twin interface was developed, establishing the foundation for significant advancements in the next phase. Substantial progress was also achieved in building simulation models designed for building managers and defining their integration into the digital twin. Additionally, for Pilot 2, a navigable 3D model of the apartments was created. All developed interfaces were made available in the project repository. These interfaces were implemented using React TypeScript with Bootstrap 5, resulting in improved usability and an intuitive user experience. The unified technology stack simplified development and enhanced modularity and integration among the various digital twins. The Apartment building digital twin for blockchain integration has entered development stage. It will be integrated with the DLT/Blockchain/Smart contracts solution (TE-5) to facilitate the simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts. The DT has been refined in terms of components and models and two algorithms have been integrated: an Electricity Load Disaggregation Algorithm and a DR Action Generation algorithm. At the same time the mock-ups for the DT dashboards have been designed. In the next iteration of the deliverable the mock-ups will be implemented, the optimization algorithms will be finalized, and the DT will be integrated with the smart contracts developed in TE-5.

Significant progress was also made in Task 4.2: Optimal trade-off among comfort, energy and building performance. Efforts were primarily focused on defining and implementing the architecture of the comfort flexibility model. For the Italian and Danish pilots, data collection and processing from installed sensors were initiated, including the calculation of results such as thermal comfort. Specific questionnaires on thermal sensation votes were distributed to tenants in the Danish and Italian pilots, with particular emphasis on simplifying the questionnaire for the Italian pilot. Lastly, a draft demand response strategy was developed to support end users in selecting the optimal trade-off between comfort and energy efficiency.

In the Task 4.3, Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers, we develop optimal DR strategies to be applied in the Greek, the Austrian and the Irish pilots. 8 different services have been defined, that are being developed by the involved partners and soon will be applied in the respective pilots. Significant progress has already been made in all services.

DOMX prepares a service for nudging the end-users of the Greek pilot towards improving their energy-behaviour. To do so, it implements specific algorithms for simulating the energy-



performance of each heating system and a clustering algorithm for creating groups of households with similar behaviour to apply different strategies to every set of households. NTUA, exploits the same data provided by the DOMX smart heating systems in order to cluster the different households by employing different clustering techniques. Both clustering results will be used in the Greek pilot for selecting the DR strategies in the pilot. Moreover, the role of the NTUA is to develop energy disaggregation mechanisms. To this end, NTUA is exploring data collected by the HERON's energy meters already deployed in several households. A mature version of the algorithm is already developed while some first results are available. As a next step, the developed method will be applied in all involved Greek's pilot households.

Heron is responsible for developing a tariff system that announces to the Greek pilot end-users the mix of the energy sources, in an effort to promote an energy-efficient operation. Specifically, this feature will inform the end-users to use their power-hungry appliances during hours with high-RES production. This service has been already integrated in the testing version of the new HERON's mobile application and will be soon released to the pilot users. In addition, HERON develops a multi-value DR approach for identifying the energy-savings potential of household's electrical appliances. Some significant results are already available, while the method will be applied in all Greek pilot's households. ENG is leading another service, aiming to integrate the result of the aforementioned multi-value DR service into the digital twin implemented in the Task 4.1. In the same domain of energy-consumption minimization, OURPOWER is in close collaboration with BLUEPRINT having already released a new version of their mobile application that enhances user engagement and optimizes energy consumption within communities. The application focuses on personalized messages towards promoting energy-efficient operation. It will be released to the Austrian pilot end-users. Lastly, CENTRICA is in the process of defining the exact implementation path to apply optimal control of the heat-pump systems in the Irish pilot.

In Task 4.4, Local flexibility management for energy communities and self-consumption/cost optimisation, transitioning from foundational conceptualization to the practical implementation and integration of key components. This included advancements in flexibility management, optimization algorithms, secure data transactions, and enhanced system interfaces across the Austrian and Spanish pilots. These developments reflect a robust evolution toward operational readiness.

Finally, in Task 4.5: Optimal aggregation of flexible energy resources for heat vs electrical flexibility valorisation, significant progress has been made in scoping, developing and implementing the required functionality. The platform general framework has been extending from the present setup focusing on the building as a whole, to being able to interact with tenants in apartments. Here backend and GUI functionality have been developed, so tenants are able to see their own apartment data and furthermore provide input on set points and flexibility preferences. Progress has been made on developing and implementing the controller taking feedback from the smart thermostats to extend TE14-FR1 to a PID controller, required for extending the flexibility towards apartment requirements. This work is almost completed.



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